

TENDER DOCUMENT SUBMISSION PACKAGE

Tender ID:	T-2026-0417
Tender Title:	Supply of IT Equipment
Procuring Entity:	Example Directorate
Bidder Name:	Example Company Ltd.
Submission Deadline:	2026-10-20
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INCLUDED DOCUMENTS (IN ORDER):

1. Trade License	[File: Wk_8_Intellectual_Property.pdf 58 pages]
2. TIN Certificate	[File: Wk_11_Evaluating and Controlling Technolog.pdf 50 pages]
3. VAT Registration Certificate	[File: Wk_9_Errors_Failures_Risks.pdf 29 pages]
4. Bank Solvency Certificate	[File: SPIC lecture slides_merged.pdf 141 pages]
5. Experience Certificate	[File: CSE 498-Course Outline- Spring 2022-converted.pdf 9 pages]
6. Technical Proposal	[File: Lecture 08(I&C).pptx.pdf 9 pages]
7. Financial Proposal	[File: Lecture 09(I&C).pptx.pdf 19 pages]

Intellectual Property

'Who owns my polio vaccine? The people! Could you patent the sun?'

Salk (1914-1995), who developed the first effective anti-polio vaccine



Topics

- Intellectual Property and Changing Technology
- Copyright law
- Copying Music, Movies, Software, and Books
- Solutions (Good and Bad)
- Free-Speech issues
- Free Software
- Software Patents



IP and Changing Technology

- Intellectual Property relates to intangible creative work
 - Not necessarily the physical form on which it is stored or delivered.
- Compare:
 - Physically owning a compact disc versus...
 - ... owning the copyrights for the music on the CD.
- Four types of protection:
 - Copyright: the right to copy a work
 - Patent: the right to use/recreate/manufacture an invention
 - Trademark: the right to use a word, phrase, sound
 - Trade secret: you don't tell anybody, nobody should know



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Wednesday, February 2, 2011



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News | Canada

Feds split on digital locks

By JESSICA MURPHY, QMI AGENCY

Last Updated: January 31, 2011 7:37pm

Story

Comments

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OTTAWA – Copyright reform is shaping up to be one of the most hotly debated bills in Parliament's winter session — and one with wide-ranging implications for Canadian consumers.

Among the most contentious issues in Bill C-32 is the ban on breaking so-called "digital locks," which can prevent people from doing things like shifting music from CDs to mp3 players or making

backup copies of DVDs.

The entertainment industry supports digital locks as a powerful tool to protect copyrighted material. Under the new legislation, the locks will trump other provisions that loosen the current reigns on legally purchased digital products.

In announcing the new copyright bill last summer, Industry Minister Tony Clement cheered the provisions for equipping businesses with necessary mechanisms to defend intellectual property.

But as University of Ottawa law professor Michael Geist sees it: "You think you've bought something but control doesn't rest with you at all."

Geist, Canada's research chair in Internet and e-commerce law, is an outspoken critic of the digital lock provisions and calls them too hard line for the average consumer.

"It is one of the most inflexible and consumer unfriendly approaches on digital locks in the world," he said. "In a bill designed to hurt (digital) pirates it's actually hurting consumers most."

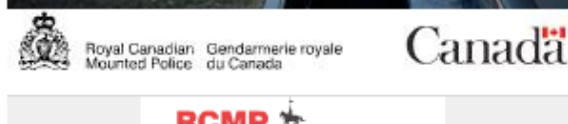
The government's hard line stance on the protective tool has also drawn the ire of opposition MPs tasked with helping refine Canada's copyright legislation.

"They're offering citizens a whole number of rights they won't be able to exercise," said NDP copyright critic Charlie Angus.

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Latest News Stories

- Cop threatened to Taser prisoners
- New trial, new beginning after Smith scandal, mom hopes
- Sally Ann has other plans for building city rejected
- City boosts development fees
- Kormos wants Welland 'poor' school stopped
- Ombudsman finds bad city service
- TTC approves Ashbridges Bay streetcar yard
- Picture released of man wanted in January assault



- What are some things the entertainment industry has done to protect copyright?
- Do you think these are justified?



Copyright

- (In Canada) Copyrights expires after ...
 - 50 years after death of a person
 - 50 years in case of a company
- With some exceptions, copyright owners have the exclusive right to:
 - Make copies of the work
 - Produce derivative works
 - Distribute copies
 - Perform the work in public
 - Display the work in public



Examples of copyrighted work

- Literary works
 - books, pamphlets, poems
 - other works consisting of text
 - computer programs (!)
- Dramatic works
 - films, videos, plays, screenplays, scripts
- Music works
 - Compositions (words & music, or music only)
- Artistic works
 - Paintings, drawings, maps, photographs, sculptures, architectural works



Copyright vs. Patent

- Copyright:
 - Protects works that owe their origin to the **expressive efforts** of an individual.
 - Key is that if the work is “original” (i.e., created independently) then it can be “copyrighted”.
 - Even if there is a closely similar work in existence.
 - Copyright owner has rights only against those who use their work without permission (i.e., defense against plagiarism).
- Patents:
 - Protect the **discoveries of inventors**.
 - For as long as patent lasts, holder has a monopoly right.
 - This right prevents anyone from producing implementations of their invention, even if same discovery was made independently.



IP and changing technology

- Problems for IP owners with newly available IT:
 - High-quality copying
 - High-quantity distribution
 - Faster / easier copying
 - Less expensive
- But note!
 - Intellectual property law has always evolved...
 - ... and has done so because of technological change.



Copyright law: History

- “Copyright” as a concept was not invented until after advent of the printing press
 - Prior to printing press, manuscripts were copied by hand.
 - Movable type introduced an economic incentive for piracy
- England: mid-16th to mid-17th century
 - Crown exercised authority over printing.
 - “Stationers’ Company” guild chartered to have a monopoly on printing.
 - Original charter established by Mary I...
- England: 1710, “Statute of Anne”
 - Created a copyright system that applied to the public in general (and not just a chartered company)
 - Copyright was now vested in the author, and not the printing-guild member who published the work.
 - Introduced the concept of a “time limitation” (set at 28 years)
- Europe: Similar model of “sovereign grants”
- International copyright protection: non-existent for centuries



First victim of music piracy?



Georg Phillip Telemann
1681 to 1767

- An important Baroque composer from Germany
- Prolific composer
 - Estimates range from 800 to 3000 works
 - Publishing his works was one of his sources of income
- In 1737:
 - Discovered his works were being pirated in France.
 - Went to Paris for six months
 - Cultivated royal favour
 - Works were then published in France.



International copyright

- 1852:
 - France extended protection to all works of authorship...
 - ... regardless of their national origin.
- Berne, Switzerland
 - September 9, 1886
 - First multilateral copyright convention in history.
 - A country agrees to give foreign authors the same protection it accords its own authors.
- This set up an international copyright union of member states (originally 14 members)
 - UK: one of the original group
 - Canada: joined in 1928
 - US: joined in 1989! Why?
 - Would have required major modifications to US law



US copyright law: history

- 1790: First US copyright law - covered printed material
 - newer technologies added later (photography, sound recordings)
- 1909: Definition of “unauthorized copy” was formed.
 - Established a compulsory licensing system
 - This meant permission for broadcast was not needed from the copyright owner...
 - ... but payment of a royalty (set out in statute) was required.
- 1960s: Some software and databases receive copyright protection.
- 1992: Making copies for personal gain becomes a felony.
- 1997: Illegal to make copies regardless of financial gain.
- 1998: Illegal to circumvent copy-protection schemes. (DMCA)



Canadian copyright law: history

- 1924: First Copyright Act (based on UK Copyright law of 1911)
- 1988: Phase 1 of copyright reform (Bill C-60)
 - Introduced protection of computer software (recognized as a “literary work”)
 - Copyright Board introduced
- 1989: Canada-US Free Trade Agreement
 - Provisions for satellite transmissions, radio, televisions
- Further changes also includes provision for royalties for performers and producers of sound recordings
- Last several years – repeated attempts to update copyright laws to be more in line with US
 - Latest: Bill C-32 (changes to fair dealing, ban on breaking digital locks)
 - will be discussed in upcoming session of Parliament



Copyright Law: exceptions

- Fair Dealing doctrine (= US “Fair Use”)
 - For some uses of copyright work, permission is not required.
 - Allows uses of copyright material contributing to the creation of new work.
 - ... and which do not significantly affect sales of the material.
 - Allows some research and educational uses.
 - Also allows for news reporting and critiquing.
- Guidelines for determining fair dealing are found in legal precedent.



Fair dealing: Six Factors

1. Purpose of the Dealing
 - Research? Private study? Criticism?
2. Character of the Dealing
 - Single copies made? Multiple copies?
 - Distributed widely? Or to a limited group?
 - Were copies destroyed after its purpose accomplished?
3. Amount of the Dealing
 - How much of the work is used?
 - What is the importance of the infringed work?
4. Alternatives to the Dealing
 - Are non-copyrighted versions available to the user?
 - Could the work be properly criticized without being copied?
5. Nature of the Work
 - Was the work published? Or unpublished? Secret?
6. Effect of Dealing on the Work
 - Is it likely to affect the market of the original work?



Example: Who should win?

- Political group organizes a web site where people post copies of news articles and then comment on them, other readers also add comments.
- Newspapers who originally published the articles sue for copyright infringement.



Example 2: Who should win?

- Online service for checking essays for plagiarism
 - Uploaded essays are stored in database and used for future checking
- Some students sue the company for copyright infringement on their essay



Purpose of copyright?

- Should it protect the author in something that is rightly theirs?
 - Berne Convention
 - Protection should be as long and broad as possible.
 - Should only provide those exceptions and limitations essential in the public interest.
- Should it benefit the public by stimulating creation of artistic work?
 - Universal Copyright Convention (Geneva, 1952)
 - The law should only give as much protection as is necessary to induce authors to create and disseminate their works.
- Should it do both?
 - Middle ground
 - Purpose of copyright is to stimulate the creation and public dissemination of works
 - And also to give their authors a generous reward for their contributions to society.
- A delicate balance.



Fair Use Cases: US

- 1984: Sony v. Universal City Studios
 - Supreme Court ruled that non-commercial copying (recording) of a movie for viewing at a later time was fair use.
 - Court ruled that copying devices should not be banned if they have significant legal uses.
- 1992: Sega Enterprises, Ltd. v. Accolade, Inc.
 - Reverse engineering a complete program in order to produce new, creative work was ruled fair use.
 - In essence: Sega did not have a monopoly on selling games for its machines.



Copying: Music

- Improved technology allows easy, fast, cheap and widespread copying of music on Web.
- Entrepreneurs create businesses to facilitate storing and sharing of music files.
 - Many individuals set up free sites for music sharing, too.
- US: RIAA (Recording Industry Association of America) continues to fight unauthorized copying of music.
- Canada: CRIA (Canadian Recording Industry Association)
 - Launched a lawsuit in 2004 against 29 unnamed alleged music file sharers.
 - By May 2005: case heard by trial court and the appeals court
 - Both courts ruled that CRIA's case was not strong enough
 - "The court concluded that the evidence created the risk that innocent persons might have their privacy invaded and also be named as defendants where it is not warranted."
 - Courts instructed the CRIA to come back with stronger evidence.



Copying: Music (Canada)

- Before 1998, copying any sound recording almost always infringed copyright
- 1998: Part VIII of the Copyright Act
 - You can legally make a copy of musical recordings for **personal use**
 - Does not legalize
 - Copies made for someone else
 - Copies of anything other than music recordings



Copying: Music (Canada)

- Section VIII of the Copyright Act leads to some odd situations
- It is legal to:
 - Lend a commercial CD to a friend, give him a blank CD-R, let him use your computer, and help him burn the CD-R which he can keep for private use.
 - Copy a commercial CD, keep the copy, and give your friend the original.
- BUT, you cannot legally make the copy yourself and give your friend the copy.



- What are some fair reasons to copy music / movies?
- When does this cross the line?



Copying: Movies & TV Programs

- Improved digital technologies + greater bandwidth = copying and transfer of movies and TV programs
- Some businesses provide(d) free services to facilitate copying of broadcast material
 - RecordTV.com
 - Scour.com (bankrupt)
- US MPAA (Motion Picture Association of America)
 - Fighting unauthorized copying of their IP
 - Attempting to chip away at “Sony v. Universal Studios”



Copying: Software

- Same technologies contributed to unauthorized copying of software
 - commercial (i.e., within or by companies)
 - non-commercial (individuals)
- Many individuals & whole businesses continue to do this
 - Reproducing, transporting, & selling copies of software, manuals and supporting materials.
 - Also covers those who give away such material for free.
- US SIIA (Software Information Industry Association)
 - Lobby for increased protections for software
 - Raising awareness



Which should be fair use?

- Copying a friend's spreadsheet software to try out for 2 weeks, then either delete or buy your own copy?
- Copying a computer game to play for 2 weeks, with no intention to buy later?
- Others?



Copying: Books

- Again, improved technology enables copying of books that:
 - is simple to perform
 - quick
 - cheap
- Counterfeiters
 - Textbooks, novels, other printed matter
 - Profit is in not paying publishers or authors for their IP
- E-books use encryption to reduce copying
 - But some of these schemes have been cracked.



Napster

- Benefits (besides from being free)
 - Share music with other users
 - Obtain individual songs from a CD
 - Sample song from a CD
 - Access more songs (especially those unavailable commercially)
- Legal Issues
 - Was copying and distributing music through Napster within fair-use guidelines?
 - If not, was Napster responsible for user actions?
- US Supreme Court decision
 - Napster was guilty of encouraging and assisting copyright infringement



Solutions to IT's impact on copyright

- There are three groups of solutions
 - Technological
 - Markets and Management
 - Regulations and Enforcement
- Technological
 - Expiration date encoded.
 - Hardware dongle required.
 - “Activation” features.
 - Digital-rights management (DRM) – encryption and other schemes
 - Can prevent saving, printing, making more than N copies, distribution, extracting excerpts, ffw over commercials



Solutions... contd.

- Markets and Management
 - Subscribe to services
 - Collect fees from users and large organizations.
 - Meter usage of copyrighted material on a network.
 - Offer discounts to educational users.
 - Educate the public about the value of copyrighted material produced by creators and publishers.
- Regulations and Enforcement
 - US: The Digital Millennium Copyright Act (DMCA) and other laws.
 - Canada: Will there be a Canadian DMCA?
 - Enforce current laws and punish abusers.



Technology: Restrictions & Bans

- In the past, lawsuits have delayed, restricted, or banned the release of new technologies
 - CD-recording devices
 - Digital Audio Tape (DAT) systems
 - DVD recorders
 - DVD players
 - MP3 players
- Some governments or industry groups have created taxes or levies...
 - Goal: Reduce or prevent unauthorized copying and distribution of copyrighted material.
 - Further goal: Assist affected creators of this material.
 - Examples
 - Audio tapes, Blank CD-ROMs
 - CD recorders
 - Personal computers
 - Printers, Scanners

2011: The NDP wants a tax on all devices with a hard drive. Do you agree?



Technology: DRM?

- Digital Rights Management
 - When combined with laws such as the US's DMCA...
 - ... can result in heavy fines or imprisonment (or both) for violators.
- Legal & monetary consequences can be applied to:
 - those who pirate intellectual work
 - scientists and researchers of technology



Circumventing Digital Locks

- Note: Canada does not have the equivalent of the DMCA
 - But we have come very, very close with Bill C-60 (tabled in 2005)
 - Now on the table again with C-32
- Fair Dealing
 - Prohibiting the use of circumvention tools may block exercise of Fair Dealing rights
- Free Speech
 - Prohibiting the sharing of information about circumvention of DRM or copy-protection schemes may violate freedom of speech.



Future of Copyright

- Challenges to the principles of copyright
 - Methods to circumvent copy-protection schemes.
 - Peer-to-peer (P2P) file transfer.
 - Ethical position among some that if copying is either easy, or cheap online access is absent, then copying is okay.
- Challenges to Fair Dealing
 - Technological (DRM) and legal (DMCA) restrictions.
 - Conflicting outcomes (e.g., reverse engineering case) in the courts.
 - Non-traditional uses (e.g., online teaching materials)



Free-Speech Issues

- IP protection?
- Or violation of free speech?
- Copyright:
 - Unauthorized posting of copyrighted documents for purpose of criticizing and organization.
- Trademark:
 - Domain names that infringe upon trademark claims.
- Trade Secret
 - Posting internal documents to expose unfair labour or business practices.



Free Software

- Notion of “free software” has been advocated by Richard Stallman
 - “Free” means “free from certain copyright restrictions”
- Examples:
 - GNU project
 - Emacs
 - “free” compilers and utilities
 - Linux
 - Apache webserver tools
 - GNOME



Patents

- Q: Should software be Copyrighted or Patented?
- Copyrights:
 - Protect the expression of an idea in a fixed or tangible form.
 - Are cheap, easy to obtain, and last a long time.
 - Allow fair dealing of the intellectual property.
- Patents:
 - Protect new, non-obvious, and useful processes.
 - Are expensive, difficult to obtain, and last for short periods of time (20 years in Canada & USA)
 - Allow licensing to other developers.



Copyright & software forms

- Recall that software can take several forms:
 - source code
 - object code
 - microcode (i.e., within processors)
- Each of these can be copyrighted
 - For instance, copyrighting object code deals with software piracy.
- But what happens if:
 - The copyrighted source code is in C...
 - ... yet someone rewrites it into Java?
 - Is copyright infringed?
- US Courts have drawn no distinction between source and object code
 - Sculpture analogy: doesn't matter if sculpture is in bronze or in marble



Copyright and software forms

- But what about software versus hardware?
- Software is copyrightable
 - Yet we know that the same software could be turned into hardware...
 - ... and hardware falls under the purview of patent.
 - (Note: the hardwired version could not be copyrighted.)
- Is software patentable?
 - 1970s: US Supreme Court initially ruled software as unpatentable
 - That is: algorithms were deemed to constitute abstract ideas or methods
 - This placed software outside of patentable subject matter.



Change...

- 1981: US Supreme Court revisited their earlier ruling
- “Diamond v. Diehr”
 - New position amended earlier one.
 - Software by itself is still unpatentable...
 - ... but if tied to some physical component or substrate, it could be patentable.
- This particular case covered a technique for curing rubber.
 - The device included some element that performed computerized calculations.
 - These determined when curing was done.



Patents

- Possible patentable material:
 - machines
 - industrial processes
 - compositions of matter
 - articles of manufacture that are novel, useful, and non-obvious
- Patents are obtained in many jurisdiction by submitting an application to a Patent Office
 - US: Patent and Trademark Office (PTO)
 - Canada: Patent Office
- Patent must describe:
 - how to make the claimed invention
 - how to use it
 - claims delineating the inventions covered by the patent



Software Patents

- US law since 1981:
 - chipping away at restrictions on software patents
 - “stretching” patent law to accommodate software
- Examples: physical substrate
 - Software needs to produce some physical change in a machine.
 - Programs certainly do this
 - stored data in the form of computer memory (i.e., hardware with varying electrical or magnetic states)
 - moving from one such state to another is a physical change
 - Storing data or software code on a magnetic disk
 - Recognized as producing a novel and patentable article of manufacture.
 - That is, the disk with the stored data differs from the disk without it.



Algorithms & data structures

- US 1998: State Street Bank & Trust vs. Signature Financial Group
 - Signature Financial obtained a patent on a “hub-and-spoke” method of running mutual funds.
 - This depended heavily on computer technology.
- Idea:
 - Several mutual funds are the “spokes”; The single investment portfolio is the “hub”
 - Software determines the values of each fund based on its percentage of assets in the “hub”
 - Information is tracked on a daily basis (fund share pricing, tax accounting).
- State Street Bank asked that this business process be judged unpatentable as it was just a mathematical algorithm.
- A high-level court disagreed:
 - Court explicitly stated that business processes can form patentable subject matter
 - Consequence: algorithms and data structures are now considered by many to be “patentable subject matter”





US005193056A

United States Patent [19][11] **Patent Number:** 5,193,056**Boes**[45] **Date of Patent:** Mar. 9, 1993[54] **DATA PROCESSING SYSTEM FOR HUB AND SPOKE FINANCIAL SERVICES CONFIGURATION**[75] Inventor: **R. Todd Boes**, Boston, Mass.[73] Assignee: **Signature Financial Group Inc.**, Boston, Mass.

[21] Appl. No.: 667,777

[22] Filed: **Mar. 11, 1991**[51] Int. Cl.⁵ G06F 15/21; G06F 15/30

[52] U.S. Cl. 364/408

[58] Field of Search 364/401, 408

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Primary Examiner—Roy N. Envall, Jr.

Assistant Examiner—David Huntley

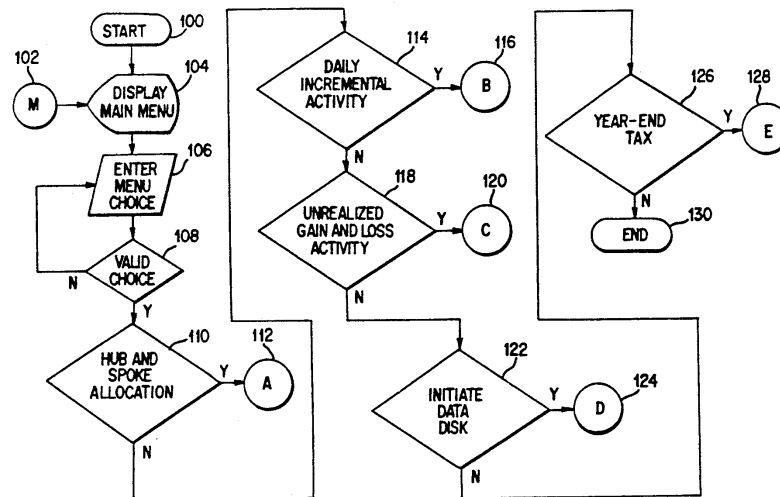
Attorney, Agent, or Firm—Pennie & Edmonds

[57]

ABSTRACT

A data processing system is provided for monitoring and recording the information flow and data, and making all calculations, necessary for maintaining a partnership portfolio and partner fund (Hub and Spoke) financial services configuration. In particular, the data processing system makes a daily allocation of assets of two or more funds (Spokes) that are invested in a portfolio (Hub). The data processing system determines the percentage share (allocation ratio) that each fund has in the portfolio, while taking into consideration daily changes both in the value of the portfolio's investment securities and in the amount of each fund's assets. The system also calculates each fund's total investments based on the concept of a book capital account, which enables determination of a true asset value of each fund and accurate calculation of allocation ratios between the funds. The data processing system also tracks all the relevant data, determined on a daily basis for the portfolio and each fund, so that aggregate year-end data can be determined for accounting and for tax purposes for the portfolio and for each fund.

6 Claims, 18 Drawing Sheets



Cover page from
Signature Financial Patent

Controversy

- Software may be protected under various forms of IP:
 - trade secret
 - assumes source code is unavailable ...
 - ... or at least closely held by a small group
 - copyright
 - covers expressions but not ideas
 - patent:
 - covers processes
- Economic advantage comes from patents
 - Many companies now have large “patent portfolios”
 - IBM, HP, Sun, Oracle, SAP, Novell
 - Patent infringement is avoided by extensive cross-licensing between companies.
- However, some claim software patents will devastate the computer industry
 - i.e., “absurd” patents



United States Patent [19]
Welch

[11] **Patent Number:** 4,558,302
[45] **Date of Patent:** Dec. 10, 1985

- [54] **HIGH SPEED DATA COMPRESSION AND DECOMPRESSION APPARATUS AND METHOD**
[75] **Inventor:** Terry A. Welch, Concord, Mass.
[73] **Assignee:** Sperry Corporation, New York, N.Y.
[21] **Appl. No.:** 505,638
[22] **Filed:** Jun. 20, 1983
[51] **Int. Cl.:** G06F 5/00
[52] **U.S. Cl.:** 340/347 DD; 235/310
[58] **Field of Search:** 340/347 DD; 235/310, 235/311; 364/200, 900

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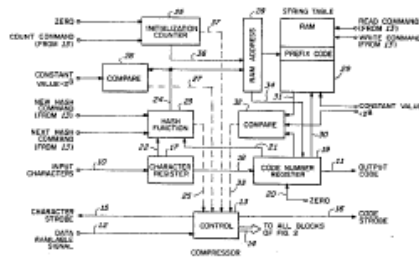
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Ziv, "IEEE Transactions on Information Theory", IT-24-5, Sep. 1977, pp. 530-537.
Ziv, "IEEE Transactions on Information Theory", IT-23-3, May 1977, pp. 337-343.

Primary Examiner: Charles D. Miller
Attorney, Agent, or Firm: Howard P. Terry; Albert B. Cooper

[57] **ABSTRACT**
A data compressor compresses an input stream of data character signals by storing in a string table strings of data character signals encountered in the input stream. The compressor searches the input stream to determine

the longest match to a stored string. Each stored string comprises a prefix string and an extension character where the extension character is the last character in the string and the prefix string comprises all but the extension character. Each string has a code signal associated therewith and a string is stored in the string table by, at least implicitly, storing the code signal for the string, the code signal for the string prefix and the extension character. When the longest match between the input data character stream and the stored strings is determined, the code signal for the longest match is transmitted as the compressed code signal for the encountered string of characters and an extension string is stored in the string table. The prefix of the extended string is the longest match and the extension character of the extended string is the next input data character signal following the longest match. Searching through the string table and entering extended strings therein is effected by a limited search hashing procedure. Decompression is effected by a decompressor that receives the compressed code signals and generates a string table similar to that constructed by the compressor to effect lookup of received code signals so as to recover the data character signals comprising a stored string. The decompressor string table is updated by storing a string having a prefix in accordance with a prior received code signal and an extension character in accordance with the first character of the currently recovered string.

181 Claims, 9 Drawing Figures



Consumer and
Corporate Affairs Canada

Consommation
et Corporations Canada

(11) (A) No. 1 223 965

(45) ISSUED 870707

(52) CLASS 354-68

(51) INT. CL. H03M 7/30, 7/50

(19) (CA) **CANADIAN PATENT** (12)

(54) High Speed Data Compression and Decompression Apparatus and Method

(72) Welch, Terry A.,
U.S.A.

(73) Granted to Sperry Corporation
U.S.A.

(21) APPLICATION No. 455,994

(22) FILED 840606

(30) PRIORITY DATE U.S.A. (505,638) 830620

No. OF CLAIMS 38

Canada

DEPOSITED BY THE PATENT OFFICE, OTTAWA
CDA-274 (11-82)

Cover pages from infamous LZW patent
(US on right, Canadian on left)
Compression scheme used in GIF file format



US005443036A

United States Patent [19]

Amiss et al.

[11] Patent Number: 5,443,036

[45] Date of Patent: Aug. 22, 1995

[54] METHOD OF EXERCISING A CAT

[76] Inventors: Kevin T. Amiss, 255 S. Pickett St., #301, Alexandria, Va. 22304; Martin H. Abbott, 10549 Assembly Dr., Fairfax, Va. 22030

[21] Appl. No.: 144,473

[22] Filed: Nov. 2, 1993

[51] Int. Cl.⁶ A01K 29/00

[52] U.S. Cl. 119/707

[58] Field of Search 119/702, 707, 174, 905; 446/485

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Primary Examiner—Todd E. Manahan

[57]

ABSTRACT

A method for inducing cats to exercise consists of directing a beam of invisible light produced by a hand-held laser apparatus onto the floor or wall or other opaque surface in the vicinity of the cat, then moving the laser so as to cause the bright pattern of light to move in an irregular way fascinating to cats, and to any other animal with a chase instinct.

4 Claims, 1 Drawing Sheet

U.S. Patent

Aug. 22, 1995

5,443,036

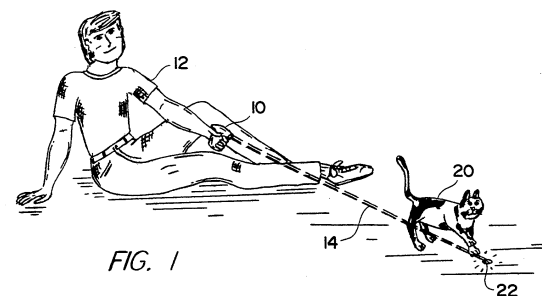


FIG. 1

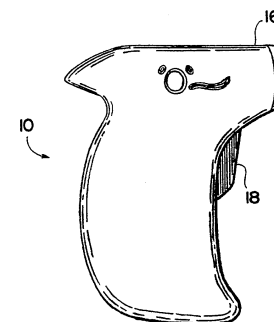


FIG. 2

Cover page and diagram from
"Method for Exercising a Cat"

Progress bar patent - Sony

- A method of providing information about recorded media content having a beginning and end time, said method comprising: displaying a progress bar having a first portion, said first portion graphically representing the duration of said recorded media content and having a first color; and displaying a second portion having a second color in said progress bar, said second portion graphically representing a section of said recorded media content that is viewed during a viewing session; wherein said second color is distinct from said first color; and wherein said second portion graphically tracks a navigation through said recorded media content being viewed in a trick mode.
- Filed: **August 25, 2004**
- **United States Patent 7,290,698**



How would you protect your software?

- Patent, Copyright, or Trade secret?
- Is it worth the time & \$ to patent your algorithm?
 - Keep in mind – your patent is only as good as your ability to legally defend it
 - A patent is only effective in the country it is filed in – so you may need to file in many countries



Licensing

- Microsoft is probably infringing on Sony's progress bar patent
- But Sony is probably infringing on some Microsoft patents
- They sign licensing deals to allow each other to use the IP without risk of being sued
- Holding lots of patents is a defensive strategy for big companies



Free software & Open Source Licenses

- Berkeley software distribution license (BSD)
 - Code is free & can be modified, but must include a copyright notice
- Gnu public license (GPL)
 - New works made from software under the GPL must also be released under the GPL and have source code available (=“copyleft”)



Open source software

- What are the benefits of open source software?
- Drawbacks?
- As a for-profit company, would you contribute?



Perspective on controversy

- Computer Scientists and Software Engineers generally take stands against software patents.
 - This is perhaps to be expected.
 - Therefore we must beware of our use of “special pleading” arguments.
- Example of such an argument:
 - “Software is unique, therefore we must find a unique mechanisms other than patents and copyright to protect it.”
 - Only to be followed by “Digital media should not be treated any differently with respect to fair dealing under copyright.”
- IT professionals are (usually) not patent lawyers
- Even legal scholars doing research on software patents may not have first-hand experience practicing patent law.
- We should acknowledge our dislike of “unwanted” complexity.



Questions

1. Since patents are “property rights”, do they promote innovation?
 - Monopoly right with strict time limit
 - Some arguments against software patents are actually arguments against any patents.
2. Have patents evolved to address concerns raised by those who suspect software patents?
 - Recall that patents have been in existence for over 500 years.
 - Software has been around for a tenth of that time.
 - The infrastructure around software patents is being assembled, including better handling of infringement, examination, and identification of prior art.



Questions (continued)

3. Must the patent system be perfect?

- Rhetorical question; but if there are too many mistakes in granting patents, the patent system will fail.
- Patents may inhibit standards if holders do not agree to standard licensing agreements.
- QWERTY keyboard; automobile controls arrangement – both were patented, but did not inhibit standards creation.

4. Is software different enough from other technologies that it is suitable for special treatment?

- Other complex systems benefit from patents (i.e., airplane design, chip fabs, oil refineries)
- Other small innovative companies benefit from patents (i.e., the “cottage industry” argument)



Questions (continued)

5. Is the Open-Source model superior to patent systems?
 - Open-source licenses are based on the copyright model.
 - However, some open-source projects (such as Linux) are based on innovative ideas developed by companies using patents (AT&T and UNIX).

6. Should difficulties in identifying “prior art” motivate alternate models for software IP?
 - Many horror stories of software patents are derived from the early days of such patents.
 - The US PTO has steadily improved its search for prior art.
 - Political dimension: How well do we fund our patent offices?



Questions (continued)

7. Is it fair to pose the question, “Is software patentable?”

- This may now be a moot point.
- Software has been patented now for twenty years or so.
- There is now a lot of motivation for society to preserve the concept of such patents.
- There also exists a huge pool of experts who know patent law as it applies to software.



7

EVALUATING AND CONTROLLING TECHNOLOGY

7.1 Evaluating Information

7.2 The “Digital Divide”

7.3 Neo-Luddite Views of Computers, Technology, and Quality of Life

7.4 Making Decisions About Technology

Exercises



In this chapter we consider such questions as these: Does the openness and “democracy” of the Web increase distribution of useful information or inaccurate, foolish, and biased information? How can we evaluate complex computer models of physical and social phenomena? How does access to digital technology differ among different populations? Is computing technology evil? Why do some people think it is? How should we control technology to ensure positive uses and consequences? How soon will robots be more intelligent than people? What will happen after that?

Whole books focus on these topics. The presentations here are necessarily brief. They introduce some of the issues and arguments.

7.1 Evaluating Information

*A little learning is a dang'rous thing;
 Drink deep, or taste not the Pierian spring;
 There shallow draughts intoxicate the brain,
 And drinking largely sobers us again.*

—Alexander Pope, 1709¹

7.1.1 THE NEED FOR RESPONSIBLE JUDGMENT

Expert information or the “wisdom of the crowd”?

*We can get the wrong answer to a question quicker than our fathers
 and mothers could find a pencil.*

—Robert McHenry²

There is a daunting amount of information on the Web—and much of it is wrong. Quack medical cures abound. Distorted history, errors, outdated information, bad financial advice—it is all there. Marketers and public relations firms spread unlabeled advertisements through blogs, social media, and video sites. Search engines have largely replaced librarians for finding information, but search engines rank Web pages by popularity (at least partly) and give prominent display to content providers who pay them; librarians do not. Wikipedia, the biggest online encyclopedia, is immensely popular, but can we rely on its accuracy and objectivity when anyone can edit any article at any time? On social journalism sites, readers submit and vote on news stories. Is this a good way to get news? The nature of the Internet encourages people to post their immediate thoughts and reactions without taking time for contemplation or for checking facts. How do we know what is worth reading in contexts where there are no editors selecting the well-written and well-researched?

When we see a video of a currently popular performer singing with Elvis Presley, we know we are watching creative entertainment—digital magic at work. But the same technology can deceive. Video-manipulation tools (and increased bandwidth) provide the opportunity for “forging” people. A company developed an animation system that modifies video images of a real person to produce a new video in which the person is speaking whatever words the user of the system provides. Another system analyzes recordings of a person’s voice and synthesizes speech with the voice, inflections, and tones of that person. Combined, these systems will likely have many uses, including entertainment and advertising, but clearly people can also use them to mislead in highly unethical ways.³ How do we know when someone is manipulating us?

Example: Wikipedia

To explore some issues of information quality, we consider Wikipedia. Wikipedia is a collaborative project among large numbers of strangers worldwide. It is huge, free, participatory, noncommercial, ad-free, and written by volunteers. The English edition has almost four million articles, more than 10 times as many as the long-respected *Encyclopaedia Britannica*, first published in 1768 and online since 1994.⁴ Wikipedia is one of the Internet’s most-used reference sites. But are its entries true, honest, and reliable?

We expect encyclopedias to be accurate and objective. Traditionally, expert scholars selected by editorial boards write encyclopedias. Volunteers, not carefully selected scholars, write and continually edit and update Wikipedia articles. Anyone who chooses to participate can do so. People worry that the lack of editorial control means no accountability, no standards of quality, no way for the ordinary person to judge the value of the information. They argue that because hundreds of millions of people—anyone at all—can write or edit articles, accuracy and quality are impossible. Truth does not come from populist free-for-alls. Members of the staffs of political candidates have distorted the Wikipedia biographies of their candidates to make their bosses look better. Opponents and enemies regularly vandalize profiles of prominent people. The staff of a federal agency removed criticisms of the agency from its Wikipedia article. Discredited theories about historic events such as the terrorist attacks on September 11, 2001, and the assassination of John F. Kennedy reappear regularly. A lawyer reported that one party in a case edited Wikipedia entries to make information appear more favorable to that party. (Jurors are not supposed to consult online sources about a trial, but some do.) Removing false information, hoaxes, and the like requires constant effort, according to Wikipedia volunteers. The *Encyclopaedia Britannica* has had errors and oddities, but the nature of Wikipedia makes it prone to more. Anonymity of writers encourages dishonesty. Open, volunteer, instant-publishing systems cannot prevent errors and vandalism as easily as publishers of printed books or closed, proprietary online information sources.

In spite of the errors, sloppiness, bad writing, and intentional distortions, most of Wikipedia is, perhaps surprisingly, of high quality and extraordinary value. Why? What

protects quality in large, open, volunteer projects? First, although anyone *can* write and edit Wikipedia articles, most people do not. Thousands write and edit regularly, not millions. Most are educated and have expertise in the subjects they write about. They correct articles promptly. (Wikipedia saves old versions, so it can restore an article someone has vandalized.) After well-publicized incidents of manipulation of articles, Wikipedia's managers developed procedures and policies to reduce the likelihood of such incidents. For example, they lock articles on some controversial topics or people; the public cannot directly edit them.

We, as users, can (and must) learn to deal appropriately with side effects or weaknesses of new paradigms. Even though so much of Wikipedia is excellent and useful, we learn that someone might have wrecked the accuracy and objectivity of any individual article at any hour. We learn that articles on technology, basic science, history, and literature are more likely to be reliable than those on politics, controversial topics and people, and current events. We learn to use Wikipedia for background, but to check facts and alternative points of view. Should we judge Wikipedia (and, by extension, the mass of information on the Web) by the excellent material it provides or by the poor-quality material it includes?

Written by fools for the reading of imbeciles.

—An evaluation of newspapers, not websites, by a character in Joseph Conrad's novel *The Secret Agent* (1907)

The “wisdom of the crowd”

People ask all sorts of questions on Yahoo! Answers (and other sites like it) about dating, make-up, food, college (“Are online college classes as good as classroom classes?”), and wide-ranging technical, social, economic, and political issues (“If we can produce enough food to feed everyone in the world, why don't we?”) Of course, a lot of answers are ill-informed. The questioner designates the posted answer he or she deems the best. What qualifies the questioner, presumably a person who does not know the answer, to judge the worthiness of the replies? To what extent does the ease of posting a question reduce the likelihood that a person will seek out well-researched or expert information on the subject? There are obviously questions for which this kind of forum might not provide the best results. (An example might be: Is it safe to drink alcohol while using an acne medicine?) However, the first two sample questions I quoted above are likely to generate a lot of ideas and perspectives. Sometimes that is exactly what the questioner wants. Without the Web, if someone asked questions like those of only a few friends, the answers might be less varied and less useful.

Some health sites on the Web encourage the public to rate doctors, hospitals, and medical treatments. Are such ratings valuable or dangerous? Will they motivate doctors and hospitals to change their practices to achieve higher ratings at the expense of good medical care? Steve Case, co-founder of AOL and founder of a health site that emphasizes ratings by the public, argues that if millions of people participate, the results will be very useful. Others are extremely suspicious of “the wisdom of the crowd.” And there is always

concern about manipulation. Websites have sprung up to buy and sell votes to get prominent display for articles on social media sites. What are the implications of such practices for sites where the public rates medical care? Will providers of new or questionable medical treatments generate fake favorable reviews and votes? Will responsible operators of sites that display material based on rankings or votes anticipate manipulation and protect against it?

Let's pause briefly to put the problems of incorrect, distorted, and manipulated information in perspective. Quack medical cures and manipulative marketing are hardly new. Product promotions not labeled as advertising date back hundreds of years. Eighteenth-century opera stars paid people to attend performances and cheer for them or boo their rivals. "Hatchet jobs" in the form of news articles, books, ads, and campaign flyers have dishonestly attacked politicians long before the Web existed. There are plenty of poorly written and inaccurate books. Historical movies merge truth and fiction, some for dramatic purposes, some for ideological purposes. They leave us with a distorted idea of what really happened. Two hundred years ago, cities had many more newspapers than they do today. Most were opinionated and partisan. At supermarket counters, we can buy newspapers with stories as outlandish as any online. The *New York Times* is a prime example of a respected newspaper, staffed by trained journalists, with an editorial board in charge. Yet one of its reporters fabricated many stories. Numerous other incidents of plagiarism, fabrication, and insufficient fact-checking have embarrassed newspapers and television networks.

OK, the problems of unreliable information are not new. But they are problems, and the Web magnifies them. So we consider two questions: How good is the wisdom of the crowd? And how can we distinguish good sources of information on the Web?

Researchers find that crowds do, in fact, generate good answers to certain kinds of questions. When a large number of people respond, they produce a lot of answers, but the average, or median, or most common answer is often a good one. This works well when the people are isolated from each other and express independent opinions. Some researchers think a large (independent) group is likely to be more accurate than a committee of experts for a variety of questions such as estimating economic growth or how well a new product or movie will do. (A Canadian mining company, perhaps hoping for such a phenomenon, posted a large set of geological data on the Web and held a contest to choose areas to look for gold.) However, when people see the responses provided by others, some undesirable things happen. People modify their responses so that the set of responses becomes less diverse, and the best answer may no longer stand out. People become more confident from reinforcement even though accuracy has not improved. The wisdom of crowds depends on diversity and independence. In social networks (as well as in-person teams working on projects in businesses, organizations, and government agencies), peer pressure and dominant personalities can reduce the wisdom of the group.⁵

How can we distinguish good sources of information on the Web? Search engines and other services at first ranked sites by the number of people who visit them. Some developed more sophisticated algorithms to consider the quality of information on sites

where users provide content. (In response, some sites added editors and fact-checking to improve quality.) A variety of people and services review and rate sites and blogs. Critics of the quality of information on the Web and the lack of editorial control disdain such ratings as merely popularity contests, contending, for example, that the Internet gratifies the “mediocrity of the masses.”⁶ For blogs, as for Wikipedia or health care sites, they argue that popularity, voting, and consensus do not determine truth. That is correct, but there is no magic formula that tells us what is true and reliable either on the Web or off the Web. That a large number of people visit a website does not guarantee quality, but it provides some information. (Why have newspapers long published “best seller” lists for books?) We can choose to read only blogs written by Nobel Prize winners and college professors, if we wish, or only those recommended by friends and others we trust. We can choose to read only product reviews written by professionals, or we can read reviews posted by the public and get an overview of different points of view.

Over time, the distinction between the online equivalents of responsible journalism and supermarket tabloids becomes clear. Good reputations develop, just as they have for decades offline. Many university libraries provide guides for evaluating websites and the information on them. (I list some at the end of this chapter.) One good step is to determine who sponsors the site. If you cannot determine the sponsor of a site, you can consider its information as reliable as the information on a flyer you might find under your car’s windshield wiper when you park in a busy parking lot. Ultimately, we must find sites, reviewers, ratings, editors, experts, and other sources we trust. Good judgment and skepticism are always useful.

The only way to preserve the wisdom of the crowd is to protect the independence of the individual.

—Jonah Lehrer⁷

Vulnerable viewers

Since you are reading this book, you probably are a student, a reasonably well-educated person who is learning how to analyze arguments and make good judgments. You can develop skills to evaluate material you read on the Web. But what about people who have less education or ability? For example, what risks does bad information pose to children who find it on the Web? Some critics of the Web worry most about the impact of inaccurate information on such vulnerable people. The fears of some seem to edge toward a belief that we (or experts, or the government) should somehow prevent such information from appearing. The many strong arguments for freedom of speech in general are arguments against any centralized or legally mandated way of accomplishing this. What can we do to improve the quality of information? Basic social and legal forces help (to a degree): freedom of speech (to provide responses, corrections, alternative viewpoints, and so on), teachers and parents, competition, fraud and libel laws—and people who care,

who volunteer to write, review, and correct online information. What else can we do to reduce access to dangerously wrong information by vulnerable people?

Narrowing the information stream

All the problems of junk and nonsense on the Web notwithstanding, the Web now gives us access to more high-quality, up-to-date information than libraries did in the past. Consider current events, politics, and controversial issues. We can read and listen to thousands of news sources on the Web from our own and other countries, getting different cultural and political perspectives on events. We can read the full text of government documents—bills, budgets, investigative reports, congressional testimony and debate—instead of relying on a few sentences quoted from an official news release or a sound bite from a biased spokesperson. We can search archives of millions of news articles from the past 200 years. We can follow websites, blogs, tweets, and social media news of conservatives, liberals, libertarians, tea party activists, environmentalists, evangelical Christians, animal rights activists, and so on, far more easily and cheaply than when we had to seek out and subscribe to their print newsletters and magazines. But what do people actually do? Some get all their news and interpretation of events from a small number of sites that reflect a specific political point of view. Online tools make it easy: you just set up your bookmarks and feeds and never look anywhere else, except at other sites recommended by the ones you frequent. Some critics see the Web as significantly encouraging political narrowness and political extremes by making it easy for people to avoid seeing alternative opinions.

The phenomenon of using the information that is easy to get applies to other fields besides politics, of course. I hear sad complaints from librarians and experienced researchers: Too many students and professionals believe whatever they read in Wikipedia. Researchers “reinvent the wheel” (or apply for grants to do so) because they do not read relevant work in their field published in the past and available only in paper journals. Because there is so much on the Web, it is too easy to ignore what is not there. How serious is the problem of ignoring older, undigitized work? Is it a temporary problem that will go away when virtually all research is available electronically?

A researcher analyzed millions of academic articles published over 50 years and found that as journals moved online, authors tend to cite fewer articles, more recent ones, and articles from a narrower set. The speculation is that researchers using search engines to find articles related to their work select from among the ones that appear high in search results—the ones that are already cited frequently. Those articles might indeed be the most important, but this approach reinforces previous choices and can lead researchers to miss less popular but very relevant work. Researchers have far more (and easier) access to articles and journals online than they had in the stacks of libraries. However, as the author of the study says, searching online “puts researchers in touch with prevailing opinions, but this may accelerate consensus and narrow the range of findings and ideas built upon.”⁸ The effect of accelerating consensus and narrowing results is similar to what researchers saw with the wisdom of the crowd when crowd members were not

Idiots and dunderheads

A fool and his money are soon parted.
—Old English proverb

New technologies can have the unintended side effect of diminishing older skills. Computing technology has reduced the use of cursive writing, for example. Microsoft made a conscious decision with the effect of diminishing language skills. The thesaurus in Microsoft Word 2000 (and some later versions) lists the verb “trick” as the only meaning for “fool.” It omits noun synonyms “clown,” “blockhead,” “idiot,” “ninny,” “dunderhead,” “ignoramus,” and others—all present in earlier versions. Standard

references such as dictionaries and Roget’s Thesaurus contain some of these and more choices.

Microsoft said it eliminated words “that may have offensive uses.”^{9*} Was this a dunderheaded decision that dulls the language and reduces literacy? Do producers of widely used reference works have an ethical responsibility to report the substance of their field accurately, or a social responsibility to remove potentially offensive words from the language?

* Microsoft restored some synonyms meaning a foolish person but continues to omit the more colorful and more offensive terms.

independent, though the mechanism is different. Clearly, it is good for researchers to be aware of this phenomenon and to broaden their searches when appropriate. The number of scholarly papers published each year has grown enormously (to, very roughly, a million yearly). Is it the tendency to use search tools in a somewhat lazy way—or the sheer number of papers—that causes some valuable work to be missed?

If we receive too much information that does not interest us, we stop reading it. To counter this problem, Facebook implemented algorithms to filter news feed updates from friends based on how recently a member communicated with them. But of course, sometimes we want to hear from those people we have not heard from in a long time. What better methods could Facebook use? And is this relevant to social issues beyond personal relationships? It is. Eli Pariser, president of (liberal) MoveOn.org, includes conservatives among his Facebook friends because he wants to be aware of views different from his own. Over time, he realized he was no longer receiving updates from them (because he did not communicate with them regularly). Although Facebook members can turn off the filtering of news feeds, most people are not aware of it. Pariser considered the problem of filtered information so disturbing that he wrote a book about it.¹⁰ What lessons can we learn from Facebook’s filtering? It is not ideal to use Facebook as our main source of access to political discussion. Facebook’s choice of a default setting (filtering turned on) might not be best (but, then again, most people might prefer it). More fundamentally, the problem of determining what information is relevant and desired does not have an easy or obvious solution. We observed (in the box in Section 3.2.4) that in some situations filtering out too much is better than filtering out too little, while in other situations the opposite is true. Any solution that a search engine or social media service adopts will not

be perfect. However, when people do not know that they are seeing filtered information, they do not know to turn off the filter or to look elsewhere for more information. There should be a clear indication when filters are active and an explanation of what they do.

How else does the Web narrow information streams? In Chapter 2 we saw that search engines personalize results for users based on their location, past searches, profile information, and other criteria. Given the huge amount of information on the Web, this fine tuning helps us find what we want quickly. It is very valuable. However, it does mean that when we are searching for something outside our usual context, including perhaps information on controversial subjects, we might have to make an effort to look a little harder.

Do the various aspects of the Internet that narrow our information stream significantly diminish access to different points of view on controversial social and political topics? Members of radical political groups (left and right) and cults got information and opinions from narrow sources well before the Web. Does the Web encourage or increase ideological isolation? Does it simply reflect the choices that some people have often made in the past? Does it make it more likely that we will see a variety of points of view? When we criticize aspects of the Web, it is helpful to look to human nature and the past for perspective. It is also helpful to look toward an ideal to suggest improvements.

Abdicating responsibility

*I have a spelling checker.
It came with my PC.
It plainly marks four my revue,
Miss steaks aye can knot sea.
Eye ran this poem threw it,
I'm sure your pleased too no.
It's letter perfect in it's weigh,
My checker tolled me sew.*

—Jerrold H. Zar, “Candidate for a Pullet Surprise”¹¹

The tools and technologies we use encourage certain practices and consequences by making them easier. The spelling-checker verse above humorously illustrates the problem of doing what the tool makes easy and ignoring other important tasks. Software can check the spelling of all the words in a document faster than a person can find the first one by flipping through the pages of a printed dictionary. But a simple spell checker looks up each word only to discover whether it is in its dictionary. It does not check whether the writer uses the word properly.*

* Grammar checkers were rudimentary when the poem first circulated on the Internet. They would now catch some of the errors.



The convenience of using a computer system and abdication of responsibility to exercise judgment can encourage a mental laziness with serious consequences. A trucker in Britain got his truck stuck on a small farm road after ignoring a sign saying the road was not suitable for large vehicles. He was unquestioningly following the directions of a navigation system. A newspaper editor in Pakistan received a letter to the editor by email and inserted it into the newspaper without reading beyond the title. The letter was an attack on the prophet Muhammad. Angry Muslims set fires in the newspaper office. Several editors were arrested and charged with blasphemy, sometimes punishable by death.¹² Back when newspaper content was still typeset and copyedited, such an accident would have been unlikely.

Businesses make decisions about loan and insurance applications with the help of software that analyzes risks. School districts make decisions about the progress of students and the careers of administrators on the basis of computer-graded and -calibrated tests. They sometimes make bad decisions because of ignorance of the kinds of errors that limitations of the system can cause. Law enforcement agents arrested people when a check of an FBI database showed an arrest warrant for someone with a similar name. Do officers think that because the computer displayed the warrant, the system has decided that the person they are checking is the wanted person? Or does an officer understand that the system simply displays any close matches and that the responsibility for the arrest decision lies with the officer?

Sometimes reliance on a computer system rather than human judgment becomes “institutionalized” in the sense that an organization’s management and the legal system can exert strong pressure on individual professionals or employees to do what the computer says. In bureaucracies, a decision maker might feel that there is less personal risk (and less bother) in just accepting what the software produces rather than doing additional checking or making a decision the software does not support. Computer programs advise doctors on treatments for patients. It is critical to remember that, in complex fields, the computer systems might provide valuable information and ideas but might not be good enough to substitute for an experienced professional’s judgment. In some institutions, when something goes wrong, “I did what the program recommended” is a stronger defense (to superiors or against a lawsuit) than “I did what my professional judgment and

experience recommended.” Such institutions are encouraging abdication of personal responsibility, with potentially harmful results.

7.1.2 COMPUTER MODELS

Likeness to truth is not the same thing as truth.

—Peter L. Bernstein¹³

Evaluating models

Computer-generated predictions based on mathematical models of subjects with important social impact frequently appear in the news. Figure 7.1 shows a few examples. A mathematical model is a collection of data and equations describing, or simulating, characteristics and behavior of the thing studied. The models and simulations of interest to us here require so much data and/or computation that they must be run on computers. Researchers and engineers do extensive modeling to simulate both physical systems, such as the design for a new car or the flow of water in a river, and intangible systems, such as parts of the economy. Models allow us to simulate and investigate the possible effects of different designs, scenarios, and policies. They have obvious social and economic benefits: They help train operators of power plants, submarines, and airplanes. They enable us to consider alternatives and make better decisions, reducing waste, cost, and risk. They enable us to project trends and plan better for the future.

Although the models we consider are abstract (i.e., mathematical), the meaning of the word “model” here is similar to its meaning in “model airplane.” Models are simplifications. Model airplanes generally do not have an engine, and the wing flaps might not move. In a chemistry class, we could use sticks and balls to build models of molecules to help us understand their properties. The molecule models might not show the components of the individual atoms. Similarly, mathematical models do not include equations for every factor that could influence the outcome. They often include simplified

-
- Population growth
 - The cost of a proposed government program
 - The effects of second-hand smoke
 - When we will run out of a critical natural resource
 - The effects of a tax cut on the economy
 - The threat of global warming
 - When a big earthquake is likely to occur
-

Figure 7.1 Some problems studied with computer models.

equations because the correct ones are unknown or too complicated. For example, we use a constant known as the acceleration of gravity in a simple equation to determine when an object dropped from a high place will hit the ground. We ignore the effect of wind in the equation, but, on some days, wind could make a difference.

Physical models are usually not the same size as the real thing. Model planes are smaller; the molecule model is larger. In mathematical models, it is time rather than physical size that often differs from reality. Computations done on a computer to model a complex physical process in detail often take more time than the actual process takes. For models of long-range phenomena, such as population growth and climate change, the computation must take less time than the real phenomenon for the results to be useful.

Predictions from expensive computers and complex computer programs impress people, but models vary enormously in quality. Some are worthless. Others are very reliable. Politicians and special interest groups use model predictions to justify multibillion-dollar programs and laws with significant impact on the economy and the standard of living and choices of millions of people. It is important for both computer professionals and the general public to have some idea of what is in such computer programs, where their uncertainties and weaknesses might lie, and how to evaluate their claims. It is the professional and ethical responsibility of those who design and develop models for public issues to describe honestly and accurately the results, assumptions, and limitations of their models.

The following questions help us determine the accuracy and usefulness of a model.

1. How well do the modelers understand the underlying science or theory (be it physics, chemistry, economics, or whatever) of the system they are studying? How well understood are the relevant properties of the materials involved? How accurate and complete are the data?
2. Models necessarily involve assumptions and simplifications of reality. What are the assumptions and simplifications in the model?
3. How closely do the results or predictions of the model correspond with results from physical experiments or real experience?

Among three models developed to predict the change in health care costs that would result if the United States adopted a national health system, the predictions varied by hundreds of billions of dollars. Two of the models predicted large increases and one predicted a drastic decrease.¹⁴ Why was there such a difference? There are both political and technical reasons why models might not be accurate. Political reasons, especially for this example, are probably obvious. In addition to technical reasons that the questions above suggest (incomplete knowledge of the system being modeled, incomplete or inaccurate data, and faulty assumptions or oversimplification), other reasons are that computing power could be inadequate for the number of computations needed to model the full complexity of the system, and the difficulty, if not impossibility, of numerically quantifying variables that represent human values and choices.

-
- How many times do parents reuse a cloth diaper before discarding it? (Values ranged from 90 to 167.)
 - Should the model give credit for energy recovered from incineration of waste? Or does pollution from incineration counterbalance the benefit?
 - How many cloth diapers do parents use each time they change a baby? (Many parents use two at once for increased protection.) Numbers in the models ranged from 1.72 to 1.9.
 - How should the model count pesticides used in growing cotton?
-

Figure 7.2 Factors in diaper life cycle modeling.

Are reusable (washable cloth) diapers better for the environment than disposable diapers? When environmentalists proposed bans and taxes on disposable diapers, this controversy consumed almost as much energy as diaper manufacturing. Several modelers developed computer models to study the question. We call this particular kind of model a life cycle analysis. It attempts to consider the resource use and environmental effects of all aspects of the product, including manufacture, use, and disposal. To illustrate how difficult such a study might be, Figure 7.2 lists a few of the questions about which the modelers made assumptions. Depending on the assumptions, the conclusions differed.¹⁵ It is worth noting also that the models focused on one quality—environmental impact. To make a personal decision, we might consider the results of such a model (if we think it reliable), and we might also consider other factors such as cost, aesthetics, convenience, comfort, and health risks.

The U.S. Army Corps of Engineers uses mathematical models to predict how long an artificially constructed or replenished beach will last before waves wash it away. Two geologists have explained weaknesses in these models.¹⁶ Among other simplifying assumptions, the models assume that all waves have the same wavelength, that all waves come from the same direction, and that all grains of sand are the same size. A model uses only 6 of 49 parameters that might affect the amount of sand washed away. Even if these 6 are the most important (or if the model included all 49), the appropriate values for a particular beach are uncertain. Often, say the critics, the beaches do not last as long as the models predict, partly because the models do not accurately provide for relevant but irregular natural phenomena such as big storms.

Example: Modeling car crashes*

Car crash analysis programs use a technique called the finite-element method. They superimpose a grid on the frame of a car, dividing the car into a finite number of small

* An earlier version of this section appeared in my chapter, “Social and Legal Issues,” in *An Invitation to Computer Science* by G. Michael Schneider and Judith L. Gersting, West Publishing Co., 1995. (Used with permission.)

pieces, or elements. The grid is entered into the program, along with data describing the specifications of the materials making up each element (e.g., density, strength, and elasticity). Suppose we are studying the effects on the structure of the car from a head-on collision. Engineers initialize data to represent a crash into a wall at a specified speed. The program computes the force, acceleration, and displacement at each grid point and the stress and strain within each element. It repeats these calculations to show what happens as time passes in small increments. These programs require intensive computation to simulate 40–100 milliseconds of real time from the impact.

A real crash test can cost several hundred thousand dollars. It includes building and testing a unique prototype for a new car design. The crash analysis programs allow engineers to consider alternatives—for example, to vary the thickness of steel for selected components, or change materials altogether—and discover the effect without building another prototype for each alternative. But how good are the programs?

How well is the physics of car crashes understood? How accurate and complete are the data? Force and acceleration are basic principles. The physics involved in these programs is straightforward. Engineers know the relevant properties of steel, plastics, aluminum, glass, and other materials in a car fairly well. However, although they understand the behavior of the materials when force is applied gradually, they know less about the behavior of some materials under abrupt acceleration, as in a high-speed impact, and their behavior near or at breaking point. There are good data on the density, elasticity, and other characteristics of materials used in the model.

What simplifications do the programs make? The grid pattern is the most obvious. A car is smooth, not made up of little blocks. Also, time is continuous. It does not pass in discrete steps. The accuracy of a simulation depends in part on how fine the grid is and how small the time intervals are. Current computer speeds allow updating the calculations on fine grids with small time intervals (e.g., one millionth of a second).

How do the computed results compare to actual crash tests on real cars? High-speed cameras record real crash tests. Engineers attach sensors to the car and mark reference points on the frame. They compare the values the sensors record with values the program computes. They physically measure the distortion or displacement of the reference points, then compare these measurements to the computed positions of the points. Starting with the results of the physical crash, the engineers use elementary physics to calculate backward and determine the deceleration and other forces acting on the car. They compare these to the values computed in the simulation. The conclusion? Crash analysis programs do an extremely good job. In part because of the confidence that has developed over time in the validity of the results, engineers use variations of the same crash analysis modeling programs in a large variety of other impact applications, including those in Figure 7.3.

Engineers who work with the crash analysis programs do not believe that they will or should eliminate physical crash testing. The computer program is an implementation of theory. Results could be poor if something happens that the program designers simply did

-
- Predict damage to a hazardous waste container if dropped.
 - Predict damage to an airplane windshield or nacelle (engine covering) if hit by a bird.
 - Determine whether beer cans would get dented if an assembly line were speeded up.
 - Simulate a medical procedure called balloon angioplasty, where doctors insert a balloon in a blocked artery and inflate it to open the artery. The computer program helps researchers determine how to perform the procedure with less damage to the arterial wall.
 - Predict the action of airbags and the proper location for sensors that inflate them.
 - Design interior parts of cars to reduce injuries during crashes (e.g., from the impact of a steering wheel on a human chest).
 - Design bicycle and motorcycle helmets to reduce head injuries.
 - Design cameras to reduce damage if dropped.
 - Forecast effects of earthquakes on bridges and buildings.
-

Figure 7.3 Other uses of crash analysis programs.

not consider. The crash analysis programs are excellent design tools that enable increases in safety with far less development cost. The physical crash test is confirmation.

Example: Modeling climate

The earth has Ice Ages and warm interglacial periods. We are now in an interglacial period that is more than 11,000 years old. Within such periods, climate varies over time and in different parts of the world. For example, the Northern Hemisphere experienced both a medieval warm period about a thousand years ago and a later colder period (roughly 1550–1850), sometimes called the Little Ice Age.

Climate change is a very complex phenomenon. Solar radiation warms the earth. Some of the heat is reflected back, and gases trap some in the atmosphere. The latter phenomenon is known as the greenhouse effect. Without it, the temperature on the earth would be too cold to support life. Water vapor is the main greenhouse gas, but there are several other significant greenhouse gases as well. Among those whose concentration has been increased by human activity (in particular, burning of fossil fuels), carbon dioxide (CO₂) is most important. An upward trend in CO₂ concentration began roughly 16,000 years ago. However, since the beginning of the Industrial Revolution, CO₂ concentration has been increasing at a faster rate.* Between the period 1850–1899 and the period 2001–

* CO₂ concentration has increased by almost 40% since 1750.¹⁷ The older data come from measurements of gases trapped in ice cores drilled in Antarctica and Greenland.

2005, average global temperature rose roughly 0.76°C .^{*18} The increase has been steeper since roughly 1980. The global temperature increase raised concern about the threat of excess global warming, possibly caused by human-induced increase of CO_2 and other greenhouse gases in the atmosphere. Global warming predictions are based on computer models of climate. We consider those models. Since 1990, the Intergovernmental Panel on Climate Change (IPCC), sponsored by the United Nations and the World Meteorological Organization, has published comprehensive reports on the science of climate change and the quality and projections of climate models roughly every five years. Much of the information in this section comes from those reports.¹⁹

Climate models, like the car crash analysis models, calculate relevant variables for grid points and elements (grid boxes) for specified simulated time intervals. The grid circles the earth, rises through the atmosphere, and goes down into the ocean. The models contain information about the sun's energy output; the orbit, inclination, and rotation of the earth; geography (a map of land masses); topography (mountains, etc.); clouds; sea and polar ice; soil and air moisture; and a large number of other factors. Equations simulate atmospheric pressure, temperature, wind speed and direction, moisture, precipitation, ocean currents, and so forth. Researchers use climate models to study several aspects of future climate. They try, for example, to determine the effect of doubling CO_2 concentration in the atmosphere. (Current trends suggest the concentration will have doubled, from its approximate level at the beginning of the 20th century, by some time in the 21st century.[†]) Models also project the likely increase in global temperature, sea level, and other climate characteristics in various scenarios with assumptions about population, industrial and economic activity, energy use, and so on, for the rest of this century. Another task for the models is to distinguish how much warming is caused by human activity and how much is natural. The IPCC has concluded that it is "extremely likely" that human activity has had a substantial warming effect on climate since 1750.²⁰

Climate models have improved over the few decades that scientists have been developing and working with them. The models used in the 1980s and 1990s were quite limited. Here is a brief sampling of simplifications, assumptions, and factors modelers did not fully understand: The models did not distinguish day and night.²¹ They used a fairly coarse grid (with points roughly 500 kilometers apart). They did not include the El Niño phenomenon. They made assumptions about methane (a greenhouse gas) that scientists later determined were incorrect. They did not include aerosols (small particles in the air) that have a cooling effect. Clouds are extremely important to climate, but many processes involved with the formation, effects, and dissipation of clouds were not particularly well understood. The IPCC summarized in 2001: "As has been the case since the first IPCC Assessment Report in 1990, probably the greatest uncertainty in future projections of

* With an error range of ± 0.19 .

† Other greenhouse gases are included too by converting their amount and effect to the equivalent number of units of CO_2 .

climate arises from clouds and their interactions with radiation. . . . Clouds represent a significant source of potential error in climate simulations.”²² The extremely simplified representations of the oceans in these models was another very significant weakness. Computing power was insufficient to do the many calculations to simulate ocean behavior. When run on past data, some of the early climate models predicted temperature increases three to five times as high as what actually occurred over the previous century. The 1990 IPCC report predicted that temperature would increase 0.3°C per decade (with an error range of 0.2°–0.5°C). The actual temperature increase over the next two decades was lower than that.²³ Thus, it should not be surprising that there was much skepticism about the climate models and their projections.

Current models are more detailed and complex. Increased computer power allows the use of finer grids (with points spaced roughly 100–300 kilometers apart) and more experiments with the models. Increased data collection and basic science research have been improving the understanding of the behavior and interactions of climate system components. The models project that doubling the concentration of greenhouse gases in the atmosphere will cause a global temperature increase within the range 2°–4.5°C. The models project warming of 0.2°C per decade for the next few decades and a sea level rise of between 8 and 23 inches by the end of the 21st century.²⁴

How well is the science understood? How accurate are the data? Climatologists know an enormous amount about climate. The models incorporate a huge amount of good science and data. But the amount not known is also large.

Much of the variation in model results comes from the still troublesome lack of full understanding of clouds. When the earth warms, water evaporates, and the additional water vapor in the atmosphere absorbs more thermal energy, warming the atmosphere farther. On the other hand, water vapor forms clouds, which reflect incoming solar radiation with a cooling effect. Thus, clouds have positive (destabilizing) and negative (stabilizing) feedback effects. The basic science of the mechanisms is fairly well understood, but not the complexity and magnitude of the feedbacks.²⁵

A related area of uncertainty has been the impact of variations in output from the sun. Recent research on the interactions between solar activity, cosmic rays (radiation from space), and cloud formation suggests that solar activity might have an impact on cloud formation, and thus on warming, that the climate models do not include. The research is at an early stage, the theory is controversial, and the magnitude of the impact is unknown. Experiments are continuing.²⁶

There is insufficient data on many phenomena for the period before satellites collected data. For example, the IPCC lists among “key uncertainties” insufficient data to draw conclusions about trends in thickness of Antarctic sea ice.

The temperature data sets that models use for temperature over the past century have been a source of some contention. They include many kinds of limitations (for example, few monitoring stations in the oceans and remote land areas) and errors. In 2011, the Berkeley Earth Surface Temperature project completed a multiyear effort analyzing

temperature data from 15 different data sets. It reviewed algorithms and statistical methods used to develop the data sets, and it developed new statistical methods to try to overcome problems in the previous methods. It developed and published a new global surface temperature record and provided an uncertainty analysis.²⁷ Climate science researchers had much praise for the quality of the work in this project and for Berkeley Earth's decision to openly publish its methodology.

What are the assumptions and simplifications in the models? Ideally, equations derived from the underlying science (generally, physics and chemistry) would model all the processes that affect climate. This is not possible, because it would require too much computation time and because all the underlying science is not known. Simplified equations, called parametrizations, represent many processes; they seem to give realistic results but are not derived from scientific theory. The specific parametrizations vary among the models, reflecting the choices of the modelers.

The IPCC acknowledges that the underlying complexity of the problem still hampers the accuracy of projections for future climate change. That is, even the extremely powerful computers of today are not sufficient to achieve an ideal level of resolution (grid size) and to include simulation of more processes that affect climate.²⁸

The model projections based on scenarios (rather than a specific increase in greenhouse gas concentration) include numerous assumptions about technological development, political control of emissions, population, economic development, energy use, and so on, throughout a century.

Science and fiction

Why do science fiction movies about global warming show the buildings of cities underwater? The entertainment industry exaggerates and dramatizes, of course. Why does an exhibit in a science museum show water up to the middle of the Statue of Liberty (about 200 feet above sea level)? A climate scientist once said: “[T]o capture the public’s imagination,” “we have to offer up scary scenarios, make simplified dramatic statements, and make little mention of any doubts we may have. . . . Each of us has to decide what the right balance is between being effective and being honest.”²⁹ Although he said he hoped climate scientists

could be both effective and honest, there is clearly an ethical problem when we trade honesty for something else. Is it a good idea? A 20-inch rise in sea level would be a very significant problem, but one we can tackle. Tens or hundreds of feet of sea level rise would be an enormous disaster. Exaggeration might lead people to take constructive action. Or exaggeration might lead to overreaction and counterproductive, expensive actions, draining resources from effective approaches. If we hope to solve real potential problems (such as flooding in low-lying areas), we must first identify them accurately.

How well do predictions of the models correspond with actual experience? The models predict seasonal variations and other actual broad-scale phenomena. The general patterns of predictions by different models are similar. For example, they all predict warming, and they all predict that more of the warming would take place near the poles and in winter. Many models now do a good job predicting air temperature near the surface of the earth (that is, close to observed temperatures) for the recent past. The models do well enough that the IPCC expresses many of its projections as very likely or likely.

For more than a decade at the beginning of the 21st century, global temperature fluctuated but did not rise overall. The models did not indicate that this would happen. Scientists are devising and testing theories to account for it. The amount of water vapor in the stratosphere is a key suspect.³⁰ Models designed to project long-term trends might not predict short-term variations well. Thus, we do not know yet if the first decade of the century was a short-term variation or whether it will require revisions in the models.

7.2 The “Digital Divide”³¹

The term *digital divide* refers to the fact that some groups of people (the “haves”) enjoy access to and regularly use the various forms of modern information technology, while others (the “have-nots”) do not. The focus of the discussion about “the digital divide” has shifted over time. In the 1990s, the focus was on access to computers and the Internet for poor people, people in rural areas, and certain demographic groups within the United States (and other developed countries). As more people acquired digital technology and Internet access, focus shifted to a divide among those who have broadband and those who do not. There is also more focus now on the digital divide between developed countries and poor countries.

7.2.1 TRENDS IN COMPUTER ACCESS

Once upon a time, everyone in the world had equal access to personal computers and the Internet. They did not exist, and we all had none. Later, a small, elite minority enjoyed these new, expensive tools. As the technology began to spread and its value became clearer, people became more concerned about the gap in access. Poor children and children of some ethnic minorities had less access to computers both in schools and at home. In the early 1990s, only about 10% of Net users were women. By 1997, the gender gap had vanished,³² but other gaps remained.* Black and Hispanic households were about half as likely as the general population to own a computer. Access in rural and remote regions lagged behind the cities.

* A gender gap remains among those who work in information technology fields. Only about one-quarter are women, and the percentage of women undergraduates interested in the computer science major dropped drastically between 2000 and 2009.

Cost is one factor that affects access by the general population. Ease of use is another. At first, personal computers and the Internet were difficult to use. Software innovations, such as point-and-click graphical user interfaces, Web browsers, and search engines made computer use significantly more comfortable for ordinary people. With lower prices, more useful applications, and ease of use, ownership and access spread quickly. The data I found about the extent of computer ownership and Web access differ in specific numbers, but all showed the same trends: In 1990, 22% of households in the United States owned a computer. In 2001, 84% of homes with children in middle and high school had Internet access. That was a significantly higher percentage of households overall, suggesting that families perceived access to be important for their children and allocated their spending accordingly.³³

Individuals, businesses, community organizations, foundations, and government programs contributed to the spread of computers and Internet access. Internet cafés sprang up from Alaska to Cairo in the 1990s when Net access from home was relatively uncommon. The federal government and local governments spent billions of dollars on technology for schools. By the end of the 1990s, most public libraries provided Internet access for the public for free. By 2000, 98% of high schools had Internet access. At about the same time, African Americans, people 65 and older, and Hispanics increased their use of the Internet significantly. Groups with low access in earlier years began to catch up. The gaps among Hispanic, black, and white people almost completely disappeared among those with the same education levels.³⁴ By 2011, there were more than 300 million cellphone subscriptions in the United States. Computing and new communication technology reached more households much more quickly than earlier technologies such as telephone, television, electricity, and automobiles.

Virtually all technological innovation is first available to the rich (or others willing to pay the initially high price). The early purchases finance improvements in design and production techniques that bring the price down. Prices of many consumer products follow this pattern. Telephones and televisions were originally luxuries of the rich. Now, almost everyone in developed countries has them. (By 2006, there were more televisions than people in the United States.) When first introduced in the 1980s, compact disk music players cost \$1000. Now we play music and video on our phones. Computer prices plunged more dramatically than prices of most other products, even while the memory, speed, and variety of input/output devices and software increased enormously.* The vast resources of the Internet are available for about what home telephone service used to cost. The phenomenon that new technologies and inventions first are expensive luxuries, then become cheaper and spread throughout the population, has led some observers to conclude that it is more accurate to think of people as “haves” and “have-laters” rather than “haves” and “have-nots.”³⁵

* For example, the cost of disk storage fell from hundreds of dollars per megabyte in the 1980s to about \$100 a terabyte (one million times as much space) by 2012.

Access to broadband connection is a newer version of a digital divide. The same demographic groups that were the have-laters with respect to Internet access in the 1990s had less broadband access in the first decade of the 2000s, with similar disadvantages. Without broadband access, it is more difficult to find employment opportunities, access news and information, and make use of some online health information and tools. Children are less likely to graduate from high school. People without broadband are less likely to create Internet content. According to Connected Nation, in 2011 only 46% of low-income households with children and 37% of low-income minority households with children had broadband at home, compared to 66% of households nationally. (About 20% of all U.S. households had broadband in 2003.) The largest barrier to adopting broadband, according to people surveyed, was cost, followed by digital literacy and not believing that the Internet was relevant to their lives. In 2011, the FCC approved a program to extend broadband service to rural areas of the United States. Businesses and nonprofit organizations started a program to address cost and digital literacy. Under this initiative, major cable carriers offer broadband service at a low price to low-income customers. Best Buy, Microsoft, America’s Promise Alliance, and United Way, among many others, provide digital literacy training.³⁶

A related digital divide exists among content consumers and content producers on the Internet. Internet users create blogs, Web pages, videos, and product reviews. Being a content creator empowers a user to communicate his or her message to a large number of people. The Internet can be a strong agent for change for those who have the skills, education, and tools to create content. Content creators tend to be people who access the Internet frequently from multiple places using multiple gadgets. They also are more educated.³⁷ The content-production divide shows a gap among users based on socioeconomic status. How should we view the Internet content-production divide? Before the Internet, a very small percent of people wrote books and articles and produced movies and television shows. The vast majority of people were content consumers only. Is the current divide less of a social problem than the pre-Internet divide because so many more people can now create content, or is it more of a problem than before because it isolates a smaller part of the population that cannot?

7.2.2 THE GLOBAL DIVIDE AND THE NEXT BILLION USERS

Approximately two billion people worldwide use the Internet, a fivefold increase over roughly a decade.³⁸ From one perspective, that is an extraordinary accomplishment in a very short time. From another perspective, it means that about five billion people do not use the Internet. Lack of access to the Internet in much of the world has the same causes as lack of health care, education, and so on: poverty, isolation, poor economies, and politics.

Both nonprofit organizations and huge computer companies have ongoing projects to spread computer access to more people in developing countries. Some companies use

the catchphrase “the next billion users” to describe the people their programs address. For the companies, these programs create good will and—if successful in improving the standard of living and economies of the target countries—a large future customer base. Companies have trained hundreds of thousands of teachers to use technology effectively in classrooms in China, India, and other countries.

One Laptop per Child is a nonprofit organization that supplies an inexpensive laptop computer specially designed for elementary school children in developing countries. The laptop works in extreme heat or cold, extremes of humidity, and dusty or rainy environments. The power requirements are very low. The program provided an important lesson: giving out computers and walking away will not close the digital divide. The success of the program in implementing the technology into school curricula depends on the presence of supporting social and technical infrastructures such as electricity, networks, tech support, parental support, teacher attitudes towards technology, and administrative school support. Purely financial resources can be less important than these factors.³⁹

Some people active in movements to shrink the “digital divide” emphasize the need to provide access in ways appropriate to the local culture. For example, one website argues that access can hurt the poor “by loosening the bonds of tradition.” In many countries, access “is one-way, entertainment-oriented, commercial.” Access might accelerate the exodus of untrained, unprepared young people from rural areas into cities.⁴⁰ How significant are these concerns? What can be done to alleviate them?

Only a few years ago, most people in the world had never made a telephone call. By the end of 2010, there were five billion cellphone subscriptions.*⁴¹ Almost every time I have read about a program to bring Internet access or cellphones to rural, third-world adults over the past several years, the most immediate uses are similar. Farmers use the Internet to learn about better farming techniques and to get up-to-date pricing information for their crops. Fishermen use their cellphones to find a nearby village where they will get a good price for their catch. As the technology spreads, food production and economic well-being improve. Some see each new digital divide as a serious social problem. What is perhaps most surprising is how quickly most of these divides shrink and how much more quickly they shrink than did previous technological divides between rich and poor, men and women, black and white, or developed and undeveloped countries.

7.3 Neo-Luddite Views of Computers, Technology, and Quality of Life

The microchip is . . . made of silicon, or sand—a natural resource that is in great abundance and has virtually no monetary value. Yet the

* That does not mean that five billion people had cellphones. In wealthier areas, some people have more than one.

combination of a few grains of this sand and the infinite inventiveness of the human mind has led to the creation of a machine that will both create trillions of dollars of added wealth for the inhabitants of the earth in the next century and will do so with incomprehensibly vast savings in physical labor and natural resources.

—Stephen Moore⁴²

Quite apart from the environmental and medical evils associated with them being produced and used, there are two moral judgments against computers. One is that computerization enables the large forces of our civilization to operate more swiftly and efficiently in their pernicious goals of making money and producing things . . . And secondly, in the course of using these, these forces are destroying nature with more speed and efficiency than ever before.

—Kirkpatrick Sale⁴³

7.3.1 CRITICISMS OF COMPUTING TECHNOLOGIES

The quotations above, both from 1995, illustrate the extreme divergence of views about the anticipated value of computer technology. Evaluations cover the spectrum from “miracle” to “catastrophe.” Although most of this book discusses problems that arise with the use of computers, the Internet, and other digital communications technologies, the implicit (and sometimes explicit) view has been that these technologies are a positive development bringing us many benefits. The potential for loss of freedom and privacy via government surveillance and the building of consumer dossiers is a serious danger. Computer crime is expensive, and changes in employment are disruptive. Our discussion of systems failures in the next chapter warns us that some potential applications can have horrifying risks. We might urgently try to prevent implementation of some applications and urgently advocate for increased protection from risks, yet not consider the threats and risks as reasons for condemning the technology as a whole. For the most part, we have looked at new risks and negative side effects as problems that occur in the natural process of change, either problems we need to solve or the price we pay for the benefits, part of a trade-off. Many people with quite different political views share this attitude, although they disagree about the significance of specific computer-related problems and about exactly how to solve them.

On the other hand, there are people who utterly reject the view that computing technology is a positive development with many important benefits. They see the benefits as few and overwhelmingly outweighed by the damage done. Neil Postman says that

voting, shopping, banking, and getting information at home online is a “catastrophe.” There are fewer opportunities for people to be “co-present,” resulting in isolation from neighbors. Richard Sclove and Jeffrey Scheuer argue that electronic communication will erode family and community life to the point that people will mourn the loss of depth and meaning in their lives.⁴⁴ A comment made by one reviewer of this book illustrates the difference in perspective. He objected to the “gift of fire” analogy I use to suggest that computers can be very useful and also very dangerous. The reviewer thought “Pandora’s box” was more appropriate. Pandora’s box held “all the ills of mankind.” Kirkpatrick Sale, author of *Rebels Against the Future*, used to demonstrate his opinion of computers by smashing one with a sledgehammer at public appearances.

In England in 1811–1812, people burned factories and mills in efforts to stop the technologies and social changes that were eliminating their jobs. Many were weavers who had worked at home on small machines. They were called Luddites.* For 200 years, the memory of the violent Luddite uprising has endured as the most dramatic symbol of opposition to the Industrial Revolution. The term “Luddite” has long been a derisive description for people who oppose technological progress. More recently, critics of technology have adopted it as an honorable term. Kirkpatrick Sale and many others who share his viewpoint call themselves neo-Luddites, or simply Luddites.

What do the neo-Luddites find so reprehensible about computers? Some of their criticisms are problems that also trouble people whose view of computing technology is generally positive, problems we discussed in earlier chapters. One of the differentiating characteristics of the neo-Luddites is that they focus on these problems, seeing no solutions or trade-offs, and conclude that computers are a terribly bad development for humankind. Among their specific criticisms are the following:

- Computers cause massive unemployment and de-skilling of jobs. “Sweatshop labor is involved in their manufacture.”⁴⁵
- Computers “manufacture needs”; that is, we use them just because they are there, not because they satisfy real needs.
- Computers cause social inequity.
- Computers cause social disintegration; they are dehumanizing. They weaken communities and lead to isolation of people from each other.
- Computers separate humans from nature and destroy the environment.
- Computers benefit big business and big government most.
- Use of computers in schools thwarts development of social skills, human values, and intellectual skills in children. They create an “ominous uniformity of knowledge” consistent with corporate values.⁴⁶

* The name Luddite comes from General Ned Ludd, the fictitious, symbolic leader of the movement.

- Computers do little or nothing to solve real human problems. For example, Neil Postman, in response to claims of the benefits of access to information, argues that “if families break up, children are mistreated, crime terrorizes a city, education is impotent, it does not happen because of inadequate information.”⁴⁷

Some of these criticisms might seem unfair. The conditions in computer factories hardly compare to conditions in the sweatshop factories of the early Industrial Revolution. In Chapter 6, we saw that computers eliminate some jobs, and that the pace of computerization causes disruptions, but the case that computers, and technology in general, cause massive unemployment is not convincing. Blaming computers for social inequity in the world ignores thousands of years of history. Postman is right that inadequate information is not the source of most social problems. A computer in the classroom does not replace good parents in the home. But should this be a criticism of computers and information systems? Access to information and communication can assist in solving problems and is not likely to hurt. The main problem for ordinary people, Postman says, is how to find meaning in life. We need answers to questions like “Why are we here?” and “How are we supposed to behave?”⁴⁸ Is it a valid criticism of computing technology that it does not solve fundamental social and philosophical problems that have engaged us for thousands of years?

To the neo-Luddites, the view that computers are fundamentally malevolent is part of a wider view that almost all of technology is malevolent. To the modern-day Luddites, computer technology is just the latest, but in many ways the worst, stage in the decline of what was good in human society. Computers are worse than earlier technologies because of their enormous speed and flexibility. Computers increase the negative trends that technology causes. Thus, if one points out that a particular problem blamed on computers already existed because of an earlier technology, Luddites consider the distinction to be a minor one.

The depth of the antipathy to technology in the Luddite view is perhaps made clearer by attitudes toward common devices most of us use daily. For example, Sale has said, “I find talking on the phone a physical pain, as well as a mental anguish.” Sven Birkerts, another critic of computers, says that if he lived in 1900, he would probably have opposed the telephone.* Speaking of the invention of the printing press, Sale laments that “literacy . . . destroys orality.” He regards not only computers but civilization as a catastrophe. Some of us see modern medicine as a life-saving and life-enhancing boon to humanity; some Luddites point out that it gave us the population explosion and extended senility.⁴⁹

* Critics of telephones complained that they replaced true human interaction with disembodied, remote voices. They actually expanded and deepened social relationships for isolated people—for example, women in general (farm wives, in particular) and the elderly.⁵⁰

Having read and listened to the arguments of technology enthusiasts and technology critics, I find it striking that different people look at the same history, the same society, the same products and services, the same jobs—and come to diametrically opposed conclusions about what they see. There is a fundamental difference between the world views of supporters and opponents of technology. It is more than the difference between seeing a glass as half full or half empty. The difference seems to be one of contrasting views about what should be in the glass. Supporters of technology see an upward trend in quality of life, beginning with people living at the mercy of nature with an empty glass that technology has been gradually filling. Neo-Luddites view the glass as originally full when people lived in small communities with little impact on nature; they see technology as draining the glass.

The neo-Luddite view is associated with a particular view of the appropriate way of life for human beings. For example, Sale's first point, in the quotation at the beginning of this section, makes the moral judgment that making money and producing things is pernicious. His introductory remark and his second point barely hint at the unusually high valuation he places on not disturbing nature (unusually high even in the contemporary context, where there is much awareness of the importance of protecting the environment). We explore these views further.

7.3.2 VIEWS OF ECONOMICS, NATURE, AND HUMAN NEEDS

Luddites generally have a negative view of capitalism, business, markets, consumer products, factories, and modern forms of work. They see the profit-seeking goals of businesses as in fundamental conflict with the well-being of workers and the natural environment. They see work in factories, large offices, and business in general as dehumanizing, dreary, and bad for the health of the workers. Hence, for example, the Luddite criticisms of the clock. Neil Postman describes the invention of the clock as “the technology of greatest use to men who wished to devote themselves to the accumulation of money.”⁵¹

Choice of words, making subtle differences in a statement, sometimes illustrate the difference in perspective between Luddites and non-Luddites. What is the purpose of technology? To the Luddites, it is to eliminate jobs to reduce the costs of production. To proponents of technology, it is to reduce the effort needed to produce goods and services. The two statements say nearly the same thing, but the first suggests massive unemployment, profits for capitalists, and a poorer life for most workers. The second suggests improvements in wealth and the standard of living.

The Luddite view combines a negative attitude toward business with a high estimation of the power of corporations to manipulate and control workers and consumers. For example, Richard Sclove describes telecommuting as being “imposed by business.” (Interestingly, one of the common criticisms of the Industrial Revolution was that working in factories instead of at home weakened families and local community.)

Luddites make particularly strong criticisms of automobiles, of cities, and of most technologies involved in communications and transportation. Thus, it is worth noting that most of us get both personal and social benefits from them. Cities are centers of culture, wealth production, education, and job opportunities.⁵² Modern transportation and communication reduce the price of products and increase their variety and availability. For example, we can eat fresh fruits and vegetables all year. We can look up menus and movie schedules on our smartphone to find what we want. We can shop worldwide on the Web. We can commute a long distance to take a better job without having to sell our house and move. If we move to a new city for college or a job, modern conveniences such as airplanes, telephones, and the Internet make the separations less unpleasant. We can visit more often in person. We can share greetings and activities with friends and family members via social media. Luddites and other critics of technology do not value these advantages highly. In some cases, in their point of view, the advantages are merely ameliorating other problems technology causes. For example, Postman quotes Sigmund Freud's comment, "If there had been no railway to conquer distances, my child would never have left his native town and I should need no telephone to hear his voice."⁵³

Does the technology create the need for itself?

A common criticism of capitalism is that it survives by convincing us to buy products we do not need. Luddites argue, similarly, that technology causes production of things we do not need. This contrasts with the market-oriented view that sees consumer choices as determining which products, services, and businesses succeed or fail (in the absence of government favoritism, subsidies, and restrictions). We examine the issue of created needs.

Sale argued that small, portable computers do not "meet any known or expressed need," but companies produced them simply because miniaturization of computing components made it possible. People have bought many millions of laptops, tablet computers, and cellphones. The number of uses is phenomenal. So, does a mobile computer meet a need? It depends on what we mean by "need." Do we need to do homework in the backyard or listen to music on an iPod? Does an architect or contractor need a laptop at a construction site? Those who emphasize the value of individual action and choices argue that needs are relative to goals, and goals are held by individuals. Thus, should we ask whether "we," as a society, need portable computers? Or should this be an individual decision with different responses? Many people demonstrate, by their purchases, that they want portable computers. Anyone who does not feel a desire or need for one does not have to buy one. The Luddites, who believe that advertising, work pressure, or other external forces manipulate buyers, reject this individual-oriented approach.

Environmental and anti-technology groups use computers and the Web. The editor of *Wild Earth*, who considers himself a neo-Luddite, said he "inclines toward the view

Wal-Mart and e-commerce versus downtown and community⁵⁴

Does electronic commerce force changes on communities that no one wants? Richard Sclove and Jeffrey Scheuer think so.⁵⁵ They use the analogy of a Wal-Mart store draining business from downtown shops, resulting in the decline of the downtown community, a “result that no consumers wanted or intended.” They generalize from the Wal-Mart scenario to cyberspace. As we conduct more economic transactions electronically, we lose more local stores, local professional and social services, and convivial public spaces like the downtowns of small towns. Consumers are “compelled” to use electronic services, “like it or not.” Other strong critics of technology share the underlying point of view of Sclove and Scheuer, so it is worth examining their argument.

The Wal-Mart analogy is a good one. The scenario is useful for illustrating and clarifying some issues about the impact of e-commerce on communities. Suppose, say Sclove and Scheuer, that a new Wal-Mart store has opened just outside of town and about half the town residents begin to do about a third of their shopping there, while the others continue to do all their shopping downtown. Everyone shops downtown, and everyone wants the downtown stores to remain. But downtown stores have lost about 16.5% of their sales, and many will not survive. Sclove and Scheuer describe this as an “involuntary transformation” that no consumer wanted or intended. It occurs, they say, because of a “perverse market dynamic.” The changes, however, are not involuntary or perverse. The core of the problem with Sclove’s and Scheuer’s interpretation is their failure to make two important distinctions: the distinction between wanting something and the willingness to pay for it, and the distinction between something being coerced or involuntary, on the one hand, and being unwanted, unintended, or unexpected on the other.

Consider a simpler situation for a moment. Suppose we poll the adult residents of a small town with a population of, say, 3000 and ask if they would like to have a fine French restaurant in town. Almost everyone says yes. Will a French restaurant open in the town? Probably not. Almost everyone wants it, yet there is not enough potential business for it to survive. There is a market dynamic at work, but it is not perverse. The fact that consumers want a particular service, store, or product is irrelevant if not enough people are willing to pay the prices that make the business viable. In Sclove’s

and Scheuer’s Wal-Mart scenario, the downtown stores could stay in business if the people were willing to pay higher prices to make up for the 16.5% of revenue lost to Wal-Mart. But we know that if the stores raise prices, they will almost certainly lose even more customers. The town residents are not willing to pay what it costs to keep the downtown stores in business. You might object: The townspeople did not have to pay the higher prices before. Why now? Because now the people who shop at Wal-Mart—or online—*have another choice*. Whatever price advantage or convenience lured them, they were not getting that benefit before. Again, a market dynamic is at work, but not a perverse one: competition.

The second issue about the Wal-Mart/e-commerce scenario is whether the change is an “involuntary” transformation. Sclove and Scheuer say that, as local businesses decline, people will be compelled to use electronic services, like it or not. Is this accurate? No more so than Wal-Mart shoppers or cyberspace enthusiasts were compelled to shop downtown (or from other offline stores), like it or not, before they had the new option. The new status quo is no more involuntary than the previous one. Although no one wants to see the downtown decline, the actions that could lead to that result are all voluntary. When a new store opens (online or offline), no one is forced to shop there. The impact on the downtown stores might not have been obvious to all the townspeople at the beginning (although now it is common enough that they might anticipate it), but an unexpected or unintended result is not the same as a coerced result. In a free society, individuals make millions of decisions based on their knowledge and preferences. This decentralized, individualized decision making produces a constantly changing pattern of stores, services, and investments (not to mention social and cultural patterns). No one can predict exactly what the result will be, and no one intends a particular picture of the economy or society, but (apart from government subsidies, prohibitions, and regulations) the actions of the consumers and merchants are voluntary. No one person can expect to have exactly the mix of shopping options (or other community characteristics) that he or she wants. If the result flows from the myriad decisions that consumers and producers make, it is not coerced. It is the process, not the result, that tells us whether an outside force is coercing people.

Do we need cellphones?

Hundreds of thousands of people have heart attacks in the United States each year. Treatment received in the first few minutes can be critical to their survival. A fire department in California helped develop a smartphone app that alerts people trained in CPR if they are near the location where a person is having a

heart attack, perhaps in the same office building, shopping center, or neighborhood. The app provides the location of the victim and the locations of any nearby emergency defibrillator devices, so a trained person can get to the scene quickly and has the tools to save a life.

that technology is inherently evil,” but he “disseminates this view via E-mail, computer, and laser printer.”⁵⁶ An interviewer reported that in 2007, after a long career attacking computers, Kirkpatrick Sale was using a laptop. The question is: Are Sale and the editor of *Wild Earth* using computer equipment because of an artificial need or because it is useful and helpful to them? Sale sees the use of computers as an uncomfortable compromise. The use of computers, he says, insidiously embeds into the user the values and thought processes of the society that makes the technology.⁵⁷

The argument that capitalists or technologies manipulate people to buy things they do not really want, like the argument that use of computers has an insidiously corrupting effect on computer users, displays a low view of the judgment and autonomy of ordinary people. It is one thing to differ with another person's values and choices. It is another to conclude that, because of the difference, the other person is weak and incapable of making his or her own decisions. The Luddite view of the appropriate way of life puts little value on modern comforts and conveniences or on the availability of a large variety of goods and services. Perhaps most people value these things more highly than the Luddites do. To get a clearer understanding of the Luddite view of a proper life style, we consider some of their comments on the relationship of humans and nature.

Nature and human life styles

Luddites argue that technology has made no improvement in life, or at best improvements of little importance. Sale's list of benefits includes speed, ease, and mass access—all of which he disdains. Sale says that although individuals might feel their lives are better because of computers, the perceived benefits are “industrial virtues that may not be virtues in another morality.” He defines moral judgment as “the capacity to decide that a thing is right when it enhances the integrity, stability, and beauty of nature and is wrong when it does otherwise.”⁵⁸ Jerry Mander, founder of the Center for Deep Ecology and author of books critical of technology and globalization, points out that thousands of generations of humans got along without computers, suggesting that we could do just fine without them too. While some people evaluate trade-offs between negative side effects of pesticides

and the benefits of reducing diseases or protecting food crops, Mander's objections to technology lead him to the conclusion that there can be no "good" pesticide. While many people work on technological, legal, and educational approaches to reducing the gasoline usage of automobiles, Mander says there can be no "good" automobile.⁵⁹

What are the underlying premises behind these comments by Sale and Mander? We consider Sale's comment on moral judgment first. Many debates about the environment set up a humans-versus-nature dichotomy.⁶⁰ This is not the true conflict. Nature, biodiversity, forests, a hospitable climate, clean air and water, open space away from cities—these are all important and valuable to humans. So is shelter from the rain, cold, and heat. So are life-saving medicines and medical techniques. Conflicts about the environment are not conflicts between humans and nature. They are conflicts between people with different views about how to meet human needs. In contrast to Sale's statement, moral judgment, to many people, and for many centuries, has meant the capacity to choose that which enhances human life, reduces misery, and increases freedom and happiness. Sale's comment chooses nature, not humanity, as the primary standard of moral value.

Whether an automobile (or computing device) is "good," by a human-centered standard, depends on whether it meets our needs, how well it does so, at what cost (to the environment and society, as well as to our bank account), and how well it compares to alternatives. Critics of modern technologies point out their weaknesses but often ignore the weaknesses of alternatives—for example, the millions of acres once needed to grow feed for horses and the hundreds of tons of horse manure dropped on the streets of cities each day, a century ago.⁶¹ Mander's comment about automobiles again raises the issues of our standard of value and our need for a product or service. Candles, gas lamps, and kerosene lamps filled homes with fumes and soot. Do we need electricity? Do we need hot water on tap, movies, and symphony orchestras? Or do we need nothing more than food and shelter? Do we need an average life expectancy of more than 25 years? Do we want to merely exist—do we *need* even that?—or do we want long, happy, comfortable lives filled with time for love, interesting activities, and an opportunity to use our marvelously inventive brains?

*The Web is alive, and filled with life, nearly as complex and, well,
natural as a primordial swamp.*

—John Perry Barlow⁶²

Accomplishments of technology

It is easy to miss the extreme changes in quality of life that have taken place over the past few centuries. We mention here a scattering of examples.

Technology and the Industrial Revolution have had a dramatic impact on life expectancy. A study in 1662 estimated that only 25% of people in London lived to age

Environmental impacts of computing technology

I had thought of including a section in this book on environmental impacts of computers, mobile devices, and the Internet. As I looked for data, I concluded that attempts to quantify environmental benefits and costs would be subject to the same weaknesses and criticisms of models that we discussed in Section 7.1.2. It is extremely difficult to measure impacts and to determine how to compare to impacts of technologies and activities that computing technology replaces. However, we can make some observations.

Production of computers is energy intensive and uses hazardous materials. Because of these materials, disposal is an issue, as it is for fluorescent light bulbs. Running and cooling the millions of servers on the Internet in the United States accounts for about 2% of U.S. electric power usage,⁶³ more than the U.S. auto industry and less than the chemical industry. There are estimates that production of computers uses roughly twice as much energy as operating them.

On the other hand, digitally controlled machinery uses less power than older electromechanical controls. Digital sensors and controls for regulating lighting, heating, air conditioning, and farm irrigation (among many other examples) save resources by determining just what is needed and thus reduce waste. Microprocessors control hybrid cars, reducing gasoline use. Telecommuting, e-commerce, and online libraries and information sites significantly reduce the need for driving and flying and thus, the need for fuel. One fiber-optic cable, with about 150 pounds of silica, carries more messages than a ton of copper wire.⁶⁴

Digital storage of documents, data, photos, and so on, reduce the need for paper (and the

amount of trash produced.) Specific examples suggest the reductions: A large insurance company reduced its use of paper by 100 million pages in a nine-month period by storing its manuals digitally instead of printing them. A computerized system for recording insurance claims replaced more than 30 million index cards. We use email and texting instead of sending letters and cards on paper. Electronic payments eliminate paper bills and checks. We read books, newspapers, magazines, and so on, on tablets, e-readers, and smartphones, reducing paper use. The decline in business for the U.S. Postal Service and printed newspapers, while population and economic activity grow, are indications of these reductions. But do we actually use less paper than we did before? I could not find clear data for total paper use. However, between 2001 and 2011, annual consumption of newsprint for daily newspapers in the United States dropped by an estimated 61%, and the number of pieces of first class mail dropped by about 24%.⁶⁵

We take, post, and share far more photos (billions per month) than we did when we made prints and slides. This is one example of a phenomenon that occurs in many fields: as a product or service becomes more efficient and cheaper, we use more of it. It seems that people have certain levels of cost that they are willing to accept. We do more and use more when the cost goes down. Perhaps we have increased our use of resources as we have increased our use of computing technology. Certainly, we shift resources from areas where there are savings to other uses, including improved medical technology, more music and video, easier access to education, and other products and services that bring us benefits.

26. Records from 18th-century French villages showed that the median age of death was lower than the median age of marriage. Until recent generations, parents had to endure the deaths of most of their children. Starvation was common. In the United States, life expectancy at birth increased from 47.3 years in 1900 to 77.9 in 2007. Worldwide average life expectancy increased from approximately 30 in 1900 to approximately 64 in 2006. Science and technology (along with other factors such as education) reduced or almost eliminated typhoid, smallpox, dysentery, plagues, and malaria in most of the world. Deaths at work, during travel, and by accidents declined dramatically.⁶⁶

In the early 2000s, Americans spent less than 10% of family income on food, compared to approximately 47% in 1901. Agronomist Norman Borlaug, who won a Nobel Peace Prize for his work in improving agricultural productivity, reported that when new forms of wheat and crop management were introduced in India, yields rose from 12.3 million tons in 1965 to 73.5 million tons in 1999. In about the same timeframe, U.S. production of its 17 most important crops increased from 252 million tons to 596 million tons, but used 25 million fewer acres. Nicholas Eberstadt, an expert on population, reported that food supplies and gross domestic product have been growing faster than population for decades in most areas of the world, in both developing and developed countries.⁶⁷

The benefits of telecommunications and information technology are enormous in developing countries. A report of a United Nations Conference on Trade and Development, for example, observes that developing economies can make productivity gains worth billions of dollars by encouraging the growth of electronic commerce. The report said that “it is because the internet revolution is relevant not just to the high-tech, information-intensive sectors but also to the whole organisation of economic life that . . . developing countries stand a better chance of sharing in its benefits earlier than in previous technological revolutions.”⁶⁸

Technology is certainly not the only factor in improving quality of life. Progress against disease, discomfort, and early death depends on the stability, freedom, and flexibility of political and economic systems as well. Measuring quality of life is subjective, and some find other measures more important than the few we cited above. But, for many people, these data suggest that technology has contributed much to human well-being.

7.4 Making Decisions About Technology

No one voted for this technology or any of the various machines and processes that make it up.

—Kirkpatrick Sale⁶⁹

7.4.1 QUESTIONS

We saw, in Section 7.3, that the determination of what are true needs depends on our choice of values. Throughout this book, we saw controversies about specific products, services, and applications of computer technology (for example, personalized advertising, anonymous Web surfing, and face recognition systems). How should we make decisions about the basic question of whether to use a whole technology, or major segments of it, at all? Who would make such decisions?

Most people in science, engineering, and business accept, almost without question, the view that people can choose to use a technology for good or ill. Some critics of technology disagree. They argue that technologies are not “neutral.” Neil Postman says, “Once a technology is admitted [to our culture], it plays out its hand; it does what it is designed to do.”⁷⁰ This view sees the technologies themselves as being in control.

In the view of some critics of computing technology, big corporations and governments make decisions about uses of the technology without sufficient input or control by ordinary people. Kirkpatrick Sale’s lament at the beginning of this section expresses this view: there was never a vote on whether we should have computers and the Internet. Some people argue that we should not use a new technology at all until we have studied it, figured out its consequences, and made a determination that the consequences are acceptable. The idea is that if the technology does not meet certain criteria, we would not permit its development and use.

This view leads to a few basic questions. Can a society choose to have certain specific desirable modern inventions while prohibiting others or prohibiting whole technologies? How well can we predict the consequences of a new technology or application? Who would make the decisions? We consider the first question here and the others in the next few sections.

How finely can we make decisions about acceptable and unacceptable technologies? In response to a criticism that the tribal life he extolled would have no pianos, no violins, no telescope, no Mozart, Sale replied, “[I]f your clan thought that the violin was a useful and nonharmful tool, you could choose to invent that.”⁷¹ Perhaps critics of computing technology who recognize its value to disabled people would permit development of applications for them. The question is whether it is possible for a clan or society to choose to invent a violin or a book reader for blind people without the technological and economic base on which development of these products depends. That base includes the freedom to innovate, a large enough economy to get materials from distant sources, and a large number of potential applications that make the research, development, and production of the basic ingredients of these products economically feasible. It is unlikely that anyone would even think of developing a book reader for the blind if some of the components did not already exist in prior products (for example, perhaps, a photocopy machine).

Telemedicine: A bad application of technology?

In Chapter 1, we described long-distance medicine, or telemedicine, as a benefit of computer technology. Computer and communications networks make possible remote examination of patients and medical test results, and they make possible remotely controlled medical procedures. You should be able to think of potential privacy and safety problems with such systems. You might think of other objections as well. Should we ban telemedicine?

Several states passed laws prohibiting the practice of telemedicine by doctors who are not licensed in that state. The main argument they give for the laws is safety, or concern about out-of-state “quacks.” The laws will “keep out the charlatans and snake-oil salesmen,” according to one supporter.⁷² Also, telemedicine could increase the influence of large, well-financed

medical centers—to the detriment of local physicians in private practice. Large hospitals might become the “Wal-Marts of medicine,” says one writer. Telemedicine might make medical care even more impersonal than it is already.

Is concern for patients the real reason for the laws? The arguments about charlatans and quacks seem weak, considering that the laws target doctors who are licensed, but in another state. Many doctors who support the bans see telemedicine as a significant competitive threat. As the director of one state medical board put it, “They’re worried about protecting their turf.”⁷³ The laws restrict competition and protect established special interests—a risk of any mechanism designed to prohibit a new technology or product.

7.4.2 THE DIFFICULTY OF PREDICTION

A brief look at the development of communications and computer technology suggests the difficulty of evaluating the consequences and future applications of a new technology. Early computers were developed to calculate ballistics trajectories for the military. The PC was originally a tool for doing computation and writing documents. No one but a few visionaries imagined most of their current uses. Each new technology finds new and unexpected uses. When physicists began developing the World Wide Web, who would have predicted online auctions, social networking, or sharing home video? Would anyone have predicted even a small fraction of the ways we use smartphones? Postman’s statement that a technology does “what it is designed to do” ignores human responsibility and choice, innovation, discoveries of new uses, unexpected consequences, and social action to encourage or discourage specific applications. Computer scientist Peter Denning takes a different view: “Although a technology does not drive human beings to adopt new practices, it shapes the space of possibilities in which they can act: people are drawn to technologies that expand the space of their actions and relationships.”⁷⁴ Denning says people adopt technologies that give them more choices. Note that he does not say more choices of consumer products, but more actions and relationships. Don Norman also suggests that society influences the role of a technology when he says, “The failure to predict the computer revolution was the failure to understand how society would modify the original notion of a computational device into a useful tool for everyday activities.”⁷⁵

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- *The telephone is so important, every city will need one!*
—Anonymous
 - *My personal desire would be to prohibit entirely the use of alternating currents. They are unnecessary as they are dangerous.*
—Thomas Edison, 1899
 - *I think there is a world market for maybe five computers.*
—Thomas J. Watson, chairman of IBM, 1943
 - *Computers in the future may . . . only weigh 1.5 tons.*
—*Popular Mechanics*, 1949
 - *There is no reason for any individual to have a computer in their home.*
—Ken Olson, president of Digital Equipment Corp., 1977
 - *The U.S. will have 220,000 computers by the year 2000.*
—Official forecast by RCA Corporation, 1966. The actual number was close to 100 million.
-

Figure 7.4 Predictions.⁷⁶

How well can a government committee, a think tank, or a computer industry executive predict the consequences of a new technology? The history of technology is full of wildly wrong predictions—some overly optimistic, some overly pessimistic. Consider the quotations in Figure 7.4. Some scientists were skeptical of air travel, space travel, and even railroads. (They believed that passengers would not be able to breathe on high-speed trains.) The quotations in Figure 7.4 reflect a lack of imagination about the myriad uses people would find for each new technology, about what the public would like, and about what they would pay for. They demonstrate humorously that many experts can be utterly wrong. We examine the prediction problem more seriously and in more depth by considering arguments made by computer scientist Joseph Weizenbaum in 1975 against the development of a particular computer technology: speech recognition systems.⁷⁷ We now have more than 35 years of hindsight. However, many inexpensive applications of speech recognition had already appeared by the early 1990s. Here are Weizenbaum's objections, accompanied by comments from our perspective today.

- *“The problem is so enormous that only the largest possible computers will ever be able to manage it.”* Speech recognition software runs on smartphones.
- *“. . . a speech-recognition machine is bound to be enormously expensive, . . . only governments and possibly a very few very large corporations will therefore be able to afford it.”* Millions of people own smartphones and other devices that include speech recognition.

- “*What can it possibly be used for?*” Much more than I will mention here. (Speech recognition technology is a multibillion-dollar industry.) We can search the Web from a cellphone by speaking what we want instead of typing. We can call a business, speak the name of the person we want to reach, and automatically connect to that person’s extension. Other customer-service applications include checking airline flight schedules, getting stock quotes and weather information, conducting banking transactions, and buying movie tickets on the phone by speaking naturally instead of pushing buttons.

Recall some of the applications described in Sections 1.2.3 and 1.2.4: training systems (e.g., for air traffic controllers and for foreign languages) and tools that help disabled people use computers and control appliances in their homes. People who suffer from repetitive strain injury use speech recognition input instead of a keyboard. IBM advertised speech-input software for poets, so they can concentrate on poetry instead of typing. People with dyslexia use speech recognition software so they can write by dictation.

Speech translation systems recognize speech and translate it into other languages. Full translation is still a difficult problem, but tourists, business people, social service workers, hotel reservations clerks, and many others use specialized versions.

Voice-activated, hands-free operation of cellphones, car stereos, and other appliances in automobiles eliminates some of the safety hazard of using these devices while driving.

- *The military planned to control weapons by voice command, “a long step toward a fully automated battlefield.”* Some argue that we should have the best possible weapons to defend ourselves. Others argue that, if wars are easier to fight, governments fight more of them. If countries fight wars with remotely controlled automated weapons and no humans on the battlefield, is that an improvement over wars in which people are slaughtered? What if only one side has the high-tech weapons? Would that cause more wars of aggression? Is there any technology that the military cannot or does not use? Should we decline to develop strong fabrics because the military can use them for uniforms? Clearly, military use of high-tech tools raises serious ethical and policy questions. Are these questions sufficient reason to abandon or condemn a technology?
- *Governments can use speech recognition to increase the efficiency and effectiveness of wiretapping.* Abuses of wiretapping concerned Weizenbaum (e.g., tapping done by oppressive governments). He does not explicitly mention wiretapping of criminal suspects. One can argue that governments can use the same tool beneficially in legal wiretapping of suspected criminals and terrorists, but it is true that speech recognition, like many other technological tools, can be a danger in the hands of governments. Protection from such abuses depends in part on the recognition

of the importance of strictly controlling government power and in part on the appropriate laws and enforcement mechanisms to do so.

Discussion of Weizenbaum's objections is important for several reasons. (1) Although Weizenbaum was an expert in artificial intelligence, of which speech recognition is a subfield, he was mistaken in his expectations about the costs and benefits. (2) His objections about military and government use highlight the dilemma: Should we decline to develop technologies that people can misuse, or should we develop the tools because of their beneficial uses, and use other means, including our votes and our voices, to influence government and military policy? (3) Weizenbaum's argument against development of a technology because of its expected cost is similar to arguments expressed by others about current and future computer applications and other technologies. For example, a common objection to some new medical technologies is that they are so expensive that only the rich will be able to afford them. This shortsighted view can result in the denial of benefits to the whole population. For many new inventions, prices are high at first but quickly come down. A computer chip developed to float on the retina of the eye and send visual signals to the brain has the potential to restore sight to some blind people. The initial cost was \$500,000. Should we ban it because it would be available only to the very rich? The developer of the chip expected the cost to come down to \$50 with mass production.

Weizenbaum was not trying to evaluate computer technology as a whole but was focusing on one specific application area. If we are to permit the government, or experts, or the people via a majority vote to prohibit development of certain technologies, it is essential at least that we be able to estimate the consequences—both risks and benefits—of the technology fairly accurately. We cannot do this. The experts cannot do it.

But what if a technology might threaten the survival of the human race? We consider such an example in the next section.

7.4.3 INTELLIGENT MACHINES AND SUPERINTELLIGENT HUMANS— OR THE END OF THE HUMAN RACE?

Prominent technologists such as Hans Moravec, Ray Kurzweil, and Vernor Vinge describe a not-very-distant future in which intelligence-enhancing devices, artificial intelligence, and intelligent robots change our society and our selves in profound ways.* The more optimistic scenarios include human use of intelligent machines and services of many kinds. People might acquire advanced mental powers through brain implants and computer-brain interfaces. When someone has a stroke, doctors might remove the damaged part of a brain and replace it with a chip that performs the lost functions, perhaps with a large amount of extra memory or a chip to access the Web directly. Why wait for a stroke? Once the technology is available, healthy people will likely buy and install such implants.

* I include some references at the end of the chapter.

MIT robotics researcher Rodney Brooks, for example, suggests that by 2020 we might have wireless Internet interfaces that doctors can implant in our heads. He says people might be just as comfortable with them as they are now getting laser eye surgery at a mall.⁷⁸ Will such implants make someone less human than a heart transplant or pacemaker does? What social problems will intelligence enhancement cause in the next few decades? What philosophical and ethical problems arise when we combine human and machine intelligence in such intimate ways?

Going farther into the future, will we “download” our brains to long-lasting robot bodies? If we do, will we still be human?

The technological singularity

The term *technological singularity* refers to the point at which artificial intelligence or some combined human–machine intelligence advances so far that we cannot comprehend what lies on the other side. It is plausible, says computer scientist Vinge, that “we can, in the fairly near future, create or become creatures who surpass humans in every intellectual and creative dimension. Events beyond such a singular event are as unimaginable to us as opera is to a flatworm.”⁷⁹

Some technologists welcome the idea of the human race transforming into an unrecognizable race of superintelligent, genetically engineered creatures within this century. Others find it horrifying—and others unlikely. Some see potential threats to the survival of the human race. They see the possibility of the machines themselves achieving human-level intelligence, then rapidly improving themselves to a superhuman level. Once robots can improve their design and build better robots, will they “outcompete” humans? Will they replace the human race, just as various species of animals displace others? And will it happen soon, say within the next 20 years or so?

Two estimates support these scenarios. One is an estimate of the computing power of the human brain. The other is based on Moore’s Law, the observation that the computing power of new microprocessors doubles roughly every 18 to 24 months. If the progress of hardware power continues at this rate, then by roughly 2030 computer hardware will be about as powerful as a human brain, sufficiently powerful to support the computation requirements of intelligent robots.

Both those who think an extreme advance in machine intelligence or human–machine intelligence is likely in the near future and those who criticize these ideas provide several reasons why it might not happen. Here are some of them. First, hardware progress might slow down. Second, we might not be able to develop the necessary software in the next few decades, or at all. Developments in AI, particularly in the area of general intelligence, have been much slower than researchers expected when the field began. Third, the estimates of the “hardware” computing power of the human brain (the sophistication of the computing power of neurons) might be drastically too low. Finally, some philosophers argue that robots programmed with AI software cannot duplicate the full capability of the human mind.

Responding to the threats of intelligent machines

Whether the singularity occurs within a few decades, or later, or not at all, many in the relevant fields foresee general-purpose intelligent machines within your lifetime. By its definition, we cannot prepare for the aftermath of the singularity, but we can prepare for more gradual developments. Many of the issues we explored in previous chapters are relevant to enhanced intelligence. Will software bugs or other malfunctions kill thousands of people? Will hackers hack brains? Will a large division open up between the superintelligent and the merely humanly intelligent? We saw that protections for safety and privacy in computer systems are often weak because they were not designed in from the start. It is valuable to think about potential problems of superintelligent systems and intelligence enhancement for humans well before they confront us so that we can design the best protections.

Bill Joy is cofounder of Sun Microsystems and a key developer of Berkeley Unix and the Java programming language. In his article “Why the Future Doesn’t Need Us,”⁸⁰ Joy describes his worries about robotics, genetic engineering, and nanotechnology. He observes that these technologies will be more dangerous than technologies of the 20th century (such as nuclear weapons) because they will be self-replicating and will not require rare and expensive raw materials and huge factories or laboratories. Joy foresees profound threats, including possibly the extinction of the human race.

What protections do people who fear for the future of the human race recommend? Joy describes and criticizes some before suggesting his own. Space enthusiasts suggest creating colonies in space. Joy observes that it will not happen soon enough. If it does, it might save the human race but not the vast majority of humans on earth. If colonists take the current technologies with them, the threat goes too. A second solution is to develop protections that can stop the dangerous technologies from getting out of control. Futurist Virginia Postrel suggests “a portfolio of resilient responses.”⁸¹ Joy argues that we could not develop “shields” in time, and if we could, they would necessarily be at least as dangerous as the technologies they are supposed to protect us against. Joy recommends “relinquishment,” by which he means we must “limit development of the technologies that are too dangerous, by limiting our pursuit of certain kinds of knowledge.” He cites, as earlier examples, treaties to limit development of certain kinds of weapons and the United States’s unilateral decision to abandon development of biological weapons. However, relinquishment has the same kinds of weaknesses Joy attributes to the approaches he rejects: they are “either undesirable or unachievable or both.” Enforcing relinquishment would be extraordinarily difficult, if not impossible. As Joy recognizes, intelligent robots and the other technologies that concern him have huge numbers of potentially beneficial applications, many of which will save lives and improve quality of life. At what point should governments stop pursuit of knowledge and development? Ethical professionals will refuse to participate in development of some AI applications, but they too face the difficult problem of where to draw the line. Suppose we develop the technology to a point where we get useful applications with legal and technological safety controls. How will

we prevent visionary or insane scientists, hackers, teenagers, aggressive governments, or terrorists from circumventing the controls and going beyond the prohibited level? Joy sees a relinquishment verification program on an unprecedented scale, in cyberspace and in physical facilities, with privacy, civil liberties, business autonomy, and free markets seriously curtailed. Thus, relinquishment means not only that we might lose development of innovative, beneficial products and services. We would lose many basic liberties as well.

Although we can find flaws with all proposals to protect against the dangers of powerful technologies, that does not mean we should ignore the risks. We need to choose appropriate elements from the various proposals and develop the best protections we can.

*Prediction is difficult, especially about the future.*⁸²

7.4.4 A FEW OBSERVATIONS

We have presented arguments against the view that we should evaluate and perhaps ban new technologies at the start. Does this mean that no one should make decisions about whether it is good to develop a particular application of a new technology? No. The arguments and examples suggest two things: (1) that we limit the scope of decisions about development of new technology, perhaps to particular products, and (2) that we decentralize the decision-making process and make it noncoercive, to reduce the impact of mistakes, avoid manipulation by entrenched companies who fear competition, and prevent violations of liberty. We cannot often predict the decisions and the results of decisions made by individual engineers, researchers, programmers, entrepreneurs, venture capitalists, customers, and teenagers who tinker in their garages, but they have a valuable robustness. The fundamental problem is not *what* decision to make about a particular technology. Rather, it is to select a decision-making *process* that is most likely to produce what people want, to work well despite the difficulty of predicting consequences, to respect the diversity of personal opinions about what constitutes a desirable life style, and to be relatively free of political manipulation.

When we consider the most extreme potential developments, such as superintelligent robots, what level of certainty of dire consequences should we require before restricting the freedom to develop technologies and products that might have marvelous benefits?



EXERCISES

Review Exercises

- 7.1 What is one significant criticism of Wikipedia?
- 7.2 What questions do we use to evaluate computer models?
- 7.3 What is one common use of Internet access in rural areas of developing countries?
- 7.4 Give one of the neo-Luddite criticisms of electronic commerce.
- 7.5 Give an example of a mistaken prediction made about computers.

General Exercises

- 7.6 Consider a social media website on which display of news stories depends on the votes of readers. Is it an ethical obligation of the site operators to ensure that votes are not bought and sold, or is it merely a good business policy? Or is it both?
- 7.7 Describe a scenario in which biased or incorrect information a child finds on the Web might harm him or her. What, if anything, might have prevented the child from finding similar information before the Web existed? Suggest and evaluate one mechanism for preventing such harm (from the Web).
- 7.8 Describe an example (actual or hypothetical) of digital manipulation of an image or video for each of the following categories. Elaborate on the third one; give some arguments for each side.
 - (a) There is no ethical problem; it is clearly ethical.
 - (b) No complex argument is needed; it is clearly unethical.
 - (c) Deciding whether it is ethical or not is not simple. There are reasonable arguments on both sides, or the context might be important.
- 7.9 Some religion-based cellphone services charge less for calls to others in the same network of religion-based phones. Is this a positive way of reinforcing a community or an encouragement for insularity? Give reasons for your answer.
- 7.10 Rewrite the spelling-checker verse (in Section 7.1.1), correcting all the mistakes without using a computer or other electronic device. (There are more than a dozen mistakes.)
- 7.11 Give an example of a bad decision or poor work that you can attribute to mental laziness encouraged by computers or the Internet. (Try for one not described in the text.)
- 7.12 Approximately 6000 languages are spoken in the world. This number is declining rapidly as a result of increased communication and transportation, globalization of business and trade, and so on—all side effects of increased technology in general and of the Internet in particular. What are the advantages and disadvantages of losing languages? Overall, is it a significant problem?
- 7.13 Write three questions whose answers you would need for a life cycle analysis model comparing the environmental impact of juice boxes with the environmental impact of juice in bottles. (Use the questions for diapers in Figure 7.2 as a guide.) Consider manufacture, transportation, use, and disposal.
- 7.14 Suppose a computer program uses the following data to determine in how many years an important natural resource (say, copper) will run out.
 - The number of tons in the known reserves of the resource.
 - The average amount of the resource used per person (worldwide) per year.
 - The total population of the world.
 - An estimate of the rate of population increase for the next few decades.
 - (a) List all the reasons you can think of why this program is not a good predictor of when we will run out of the resource.
 - (b) In 1972, a group called the Club of Rome published a study using computer models that implied that the world would run out of several important natural resources in the 1980s. Today, even with the enormously increased demand from China and other developing countries, we have not run out. Why do you think many people accepted the predictions in the study?
- 7.15 What devices or technologies, available now or in the next few years, do you think schools might ban students from using for schoolwork to promote fairness among students who have them and students who do not?

- 7.16 By 2010, only about 45% of people over age 64 in the United States had broadband service, while about 67% of households overall did.⁸³ What are some likely reasons for this difference? What negative effects might it have? How is it likely to change in the near future? Is this digital divide a significant problem?
- 7.17 In the mid-1990s, approximately 70% of the computers connected to the Internet were in the United States. Did this suggest a growing gap between “have” and “have-not” nations? Give your reasons. (Try to find out what percentage of computers or websites are in the United States now.)
- 7.18 How do the opportunities for “co-present,” or in-person, social interactions today compare with those of 200–250 years ago? (If you think it is relevant, comment on the fact that Thomas Jefferson and Voltaire each wrote more than 20,000 letters in their lifetimes.⁸⁴)
- 7.19 A large number of college students enroll in online degree programs. Discuss some advantages and disadvantages (to the students and to society in general) of students getting degrees online instead of at traditional colleges where they are co-present with faculty and other students.
- 7.20 The number of small neighborhood bookstores declined because of competition from both large chain megabookstores and online stores like Amazon.com. Should a law have prohibited Amazon.com from opening? If not, should we prohibit it from selling used books, to help preserve small neighborhood used-book stores? Give reasons. Suppose you like to shop in your neighborhood bookstore and fear it might go out of business. What can you do?
- 7.21 Many games that children used to play on boards with dice, cards, and plastic pieces are now computer games. Is this an example of unnecessary use of the new technology just because it is there? Describe some advantages and disadvantages of replacing a boardgame by a computer version.
- 7.22 Analyze the following argument that we are forced to have a cellphone. Is it convincing?
Some people do not want to own or use a cellphone. Technology advocates say if you don’t want one, you don’t have to buy one. But this is not true. We have to have one. Before cellphones became popular, there were coin-operated telephones all over, on street corners, in and near stores, in restaurants, at gas stations, and so on. If we needed to make a call while away from home or work, we could use a pay phone. Now most pay phones are gone, so we have to have a cellphone whether we want to or not.
- 7.23 Which of the Luddite criticisms of computers listed in Section 7.3.1 do you consider the most valid and significant? Why?
- 7.24 Recall the discussion in the box “Wal-Mart and e-commerce versus downtown and community” (Section 7.3.2), and consider these questions: Do people have a right to shop in small neighborhood stores rather than online? Do people in a small town have a right to eat in a French restaurant? Distinguish between negative and positive rights (Section 1.4.2).
- 7.25 Sclove and Scheuer saw e-commerce as a major threat to local “brick and mortar” businesses. Give a few specific examples of ways in which online businesses and services contribute to the success of local businesses. What kinds of local businesses are least likely to be hurt by online competition?
- 7.26 Some governments adapted software originally developed to help parents and Internet service providers block access to material inappropriate for children to block access to political and religious discussions. In what way does this example illustrate the views that technology will inevitably have negative uses and that, as Neil Postman said, “once a technology is admitted, . . . it does what it is designed to do”?

- 7.27 Some neo-Luddites acknowledge that computing technology is beneficial to many people, but they see the main beneficiaries as government and big business. They say the key question is: Who benefits most? Consider the following questions and discuss the issue of who benefits most.
- When a drug company develops a new cancer drug and its executives make millions of dollars as the stock goes up, while people who had that cancer live 20 extra years, who benefits most? Who benefits most from social media: governments, businesses, or ordinary users?
- 7.28 A philosopher writing more than a decade ago argued against the use of speech synthesis. He found it unsettling and dangerous that a person might have a telephone conversation with a machine and think it was a real person. Describe a few uses of speech synthesis. What are the benefits? What are reasons for concern?
- 7.29 Speaker recognition software analyzes speech to determine who the speaker is (not what words the speaker is saying, as in speech recognition). Describe some potentially useful and some potentially threatening or risky applications.
- 7.30 Assume you are a professional working in your chosen field. Describe specific things you can do to reduce the impact of any two problems we discussed in this chapter. (If you cannot think of anything related to your professional field, choose another field that might interest you.)
- 7.31 Think ahead to the next few years and describe a new problem, related to issues in this chapter, likely to develop from digital technology or devices.

Assignments

These exercises require some research or activity.

- 7.32 Find an article in Wikipedia on a subject that you already know a lot about. Read and review the article. Is it accurate, well done, complete?
- 7.33 Find websites that provide recommendations about how much Vitamin C a person should consume each day. Find at least one site that is extreme in some way and at least one that you consider reasonably reliable. Describe the sites and explain the basis for your characterization of them.
- 7.34 This exercise explores whether the wisdom of the crowd can successfully run a soccer team. In 2008, thousands of soccer fans chipped in via a website, MyFootballClub.co.uk, to buy a British soccer team. The plan was to make management decisions by voting on the Web. Find out how well it worked and how well the team did.
- 7.35 Rumors spread quickly on the Internet and in social media. Some persist and show up repeatedly. Find a website that regularly reports on myths, rumors, and “urban legends”—and researches and reports on the facts about them.
- 7.36 Recent predictions for population growth in the 21st century have changed quite a bit from predictions made a few decades ago. Find reports of older population models (say, from the 1960s, 1970s, or 1980s), and find reports of recent population models. How do they differ? How have the assumptions in the models changed?
- 7.37 Three-dimensional “printers” create 3-D structures, layer by layer, using glues and resins, under direction of a computer file. Find out what people and businesses use these devices for. Suppose someone described these devices 10 years ago as a potential invention and asked: Will they fill any real needs? How do you think most people would have answered? What is your answer now?

Class Discussion Exercises

These exercises are for class discussion, perhaps with short presentations prepared in advance by small groups of students.

- 7.38 Some people who consider themselves capable of distinguishing reliable information from unreliable information on the Web are concerned that most ordinary people are not educated, experienced, or sophisticated enough to do so. They are likely to believe lies, they might follow dangerous medical or financial advice, and so on. How serious a problem do you believe this is? How might it be addressed?
- 7.39 Some critics of tools that enable us to specify the kinds of information we receive on the Web argue that the tools encourage fragmentation of society along political lines. Some argue that when most people got their news from three TV networks, there was a more cohesive, shared background of information. How serious is the problem of people seeing only one political point of view on the Web?
- 7.40 A number of people advocate a law requiring Google to make public the algorithms it uses to rank websites for display in response to search queries. Considering issues in this chapter, and any other relevant issues, discuss arguments in favor of such a requirement and arguments against it.
- 7.41 Google has been pursuing an ambitious goal of collecting and providing vast amounts of human knowledge. Google's projects have faced a variety of criticisms, some centered on the facts that it is a private company and that it is an American company. For example: Google's databases and the wide use of its search engines give it more power over information than a private corporation should have. Google's project of scanning millions of books contributes to American and English-language domination of world culture because it is scanning books in English. Give arguments in support of these criticisms and arguments against them.
- 7.42 What are some skills, traditions, and/or social conventions that have been or might soon be lost because of computer, phone, and Internet technology? Include at least one that you think will be a real loss and include at least one that you think is not a problem. Give reasons.
- 7.43 Which of the following models do you think would produce very accurate results? Which do you think would be less reliable? Give your reasons.
 - (a) A model developed by a team of mathematicians in 1895, using projections of population growth, economic growth, and traffic increase, to project the tonnage of horse droppings on the city streets in 1995
 - (b) A model to predict the effect of an income tax change on government revenue
 - (c) A model to predict the position of the moon in relation to the earth 30 years from now
 - (d) A model to predict how much optical fiber a major city will need 30 years from now
 - (e) A model to predict how much carbon dioxide the burning of fossil fuel for energy will emit worldwide 30 years from now
 - (f) A model to predict the speed of a new racing-boat hull design under specified wind conditions
 - (g) A model to project temperature increase and sea level rise over the next several centuries from a quadrupling of CO₂ concentration in the atmosphere (which the model assumes would happen in 140 years)⁸⁵

- 7.44 What form will the “digital divide” likely take 10 years from now? How do digital divides differ from social divisions that occurred with the introduction of earlier, nondigital information and communication technologies?
- 7.45 In the Prometheus myth, Zeus, the king of the gods, was furious at Prometheus for teaching science and technological skills to mankind because they made people more powerful. Zeus was jealous of his power and determined to withhold fire from mankind so that people would have to eat their food raw. Zeus and the Luddites represent different viewpoints on who benefits most from technology.
- (a) Give arguments in support of Zeus’ view that technology helps the less powerful, reducing the advantage of the more powerful.
 - (b) Give arguments in support of the Luddite view that technology helps the more powerful (e.g., governments and large corporations) most.
- 7.46 Read Bill Joy’s article and the reply by Virginia Postrel (see the references at the end of the chapter). Whose arguments are more convincing? Why?



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NOTES

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Errors, Failures, and Risks

*Understanding the risks and reasons
for computer failures.*



Trusting Computers

- What can go wrong?
- Case Studies
- Increasing Reliability and Safety
- Computer Models
- Perspectives on Failures, Dependence, Risk and Progress
- Learning from other technologies



What can go wrong?

- Describe a system failure that affected...
 - You or a friend
 - Many people



RISKS digest

- “Forum on Risks to the Public in Computers and Related systems”

BSoD forces students to retake standardized test (May 15, 2007)

2900 Virginia students will have to re-take standardized tests because the computer systems failed during the testing process. There are two descriptions of what went wrong: the testing vendor "reported that there was a problem with a connection between two servers" and students' "computer screens suddenly turned blue and displayed an error message" (i.e., a BSoD). Whether this is one problem or two is unclear... "State officials said there was an unrelated computer problem with online testing last week [where] 1,300 tests were interrupted and that the students will have to be retested."



RISKS digest

Scots Jail hi-tech door locking system broke (September 20, 2005)

Prison officers have been forced to abandon a new security system and return to the use of keys after the cutting-edge technology repeatedly failed. The system, which is thought to have cost over £3 million, used fingerprint recognition to activate the locking system at the high-security Glenochil Prison near Tullibody, Clackmannanshire. ... For more than a month, the 420 inmates - including some murderers and other high-risk inmates - had been able to wander around the high-security jail. Staff claim that the unlimited access to all parts of the prison had allowed some prisoners to settle old scores with rivals.



RISKS digest

Computer problem suspected in erratic Airbus flight (Jan 11, 2008)

An Air Canada flight from Victoria -> Toronto rolled suddenly from side to side and then plunged in the air. It may have suffered technical problems, according to passengers interviewed after the plane was diverted to Calgary. There had been a computer failure and pilots were flying the plane manually. Ten people were admitted to hospital.



What can go wrong?

- Facts about computer errors
 - Error-free software is very, very difficult to achieve.
 - Errors are often caused by several factors.
 - Errors can be reduced by following good software development procedures.
 - Confidence in code can be obtained by keeping it small enough to be understood.



What can go wrong?

- Roles
 - Computer user
 - Should understand limitations of computers
 - Should understand need for proper training and responsible use
 - Computer professional (buyer, developer, system manager...)
 - Needs to understand the sources and consequences of failures
 - Educated member of society
 - Understanding computer risks helps in discussing and deciding public policy.



What can go wrong?

- Categories of computer errors and failures
 - Problems for individuals
 - affects one or a few people
 - System Failures
 - affects large numbers of people or costs large amounts of money or both
 - classic example: telecommunications network
 - Safety-Critical Applications
 - where property may be damaged or destroyed
 - where people may be injured or killed



What can go wrong?

- Problems for Individuals
 - Billing errors
 - Lack of tests for inconsistencies and inappropriate amounts
 - Perhaps caused by data entry errors due to poor HCI
 - Database accuracy problems
 - Incorrect information resulting in wrongful treatment or acts
 - Causes
 - large, diverse population
 - human common-sense not part of automated processing
 - overconfidence in accuracy of data from a computer
 - information not updated or corrected
 - lack of accountability for errors
- Q: How accurate should computer systems be?



What can go wrong?

- System failures
 - Communications
 - Telephone, online, and broadcast services
 - Cellular, paging, long-distance network
 - Broadband outages
 - Business
 - Inventory and management software
 - Financial
 - Stock exchange, brokerages, banks, etc.
 - Transportation
 - Reservations, ticketing, and baggage handling
 - Causes
 - Insufficient testing and debugging time
 - Significant changes in specifications (during and after project start)
 - Overconfidence in system
 - Mismanagement of the project



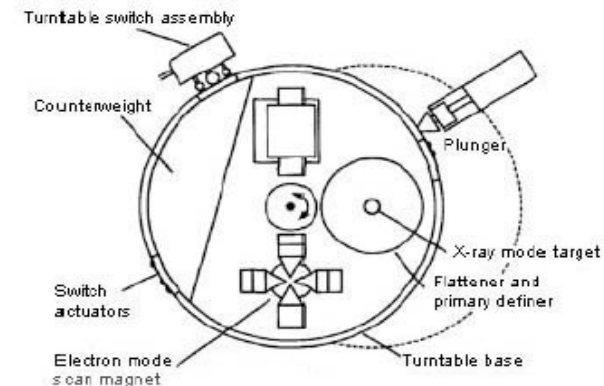
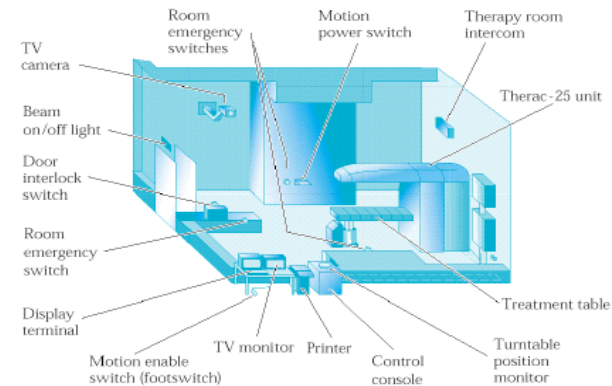
What can go wrong?

- Safety-Critical Applications
 - Uses
 - Medicine, health-services
 - Power plants
 - Aircraft
 - Trains
 - Automated Factories
 - Military applications
 - Causes of error:
 - Overconfidence
 - Lack of override features
 - Insufficient testing
 - Poor HCI
 - Sheer complexity of system (both computer and environment)



Case Study: Therac-25

- Produced by the Atomic Energy of Canada Limited (AECL)
- Introduced in 1983 as a successor model to the Therac-6 and Therac-20
- This new model provided full computer control, which itself was intended to provide several benefits
 - Faster setup by operators, hence more treatments per day.
 - Software checks replaced earlier hardware interlocks

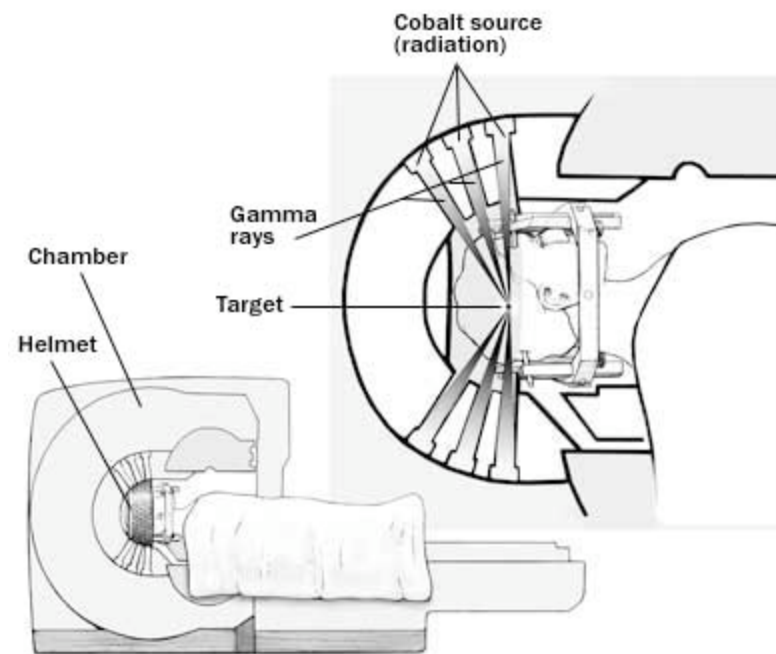


University of Victoria
Department of Computer Science

SENG 401: Social and Professional Issues
Trusting Computers: Slide 13

Detour: radiotherapy

- External Beam Therapy: idea
 - Destroy tumours by focusing energy at a specific region
 - Insight: to minimize damage to surrounding tissue, beam “rotates” around patient.
 - Simulation, treatment planning, treatment delivery
 - Radiation oncologist, radiation physicist, dosimetrist and radiation therapist are all involved
 - Radiation therapist controls the machine
 - Actual beams, radiation dosage, etc. are planned out
 - Linear accelerators or cobalt machines are used to generate radiation.



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Image of “gamma-knife radiosurgery”



University of Victoria
Department of Computer Science

SENG 401: Social and Professional Issues
Trusting Computers: Slide 14

Case Study: Therac-25

- Overdosed 6 patients over a two year period
 - First case occurred in Hamilton, Ontario
 - Normal radioactive dosages is 100-200 rads
 - To compare:
 - Dental x-ray: 0.02 rads
 - Chest x-ray (two views): 0.02 – 0.07 rads
 - CT scan of head or chest: 1 rad
 - Overdoses were estimated at 13,000 and 25,000 rads
 - Given to six people
 - Three of the six people died



Case Study: Therac-25

- Multiple causes of failure (many interrelated):
 - Poor safety design
 - Insufficient testing and debugging
 - Software errors
 - Lack of safety interlocks (hardware)
 - Overconfidence
 - Inadequate reporting / investigation of accidents
- Therac-25 machine continued to be used afterwards but after retrofitted with hardware interlocks



Case Study: Patriot Missile Failure

- February 1991, During first Gulf War
 - Patriot Missile battery in Dhahran, Saudi Arabia, failed to intercept incoming Iraqi Scud missile
 - Scud struck an American Army barracks
 - 28 soldiers killed
- Reported cause of failure
 - Inaccurate calculation of “time since boot”
 - Due to computer-arithmetic errors



Case Study: Patriot Missile Failure

- Technical details
 - System's internal clock measured time in units of 1/10th of a second
 - That is, a time of 1000 units corresponded to 100 seconds
 - To obtain time in seconds, units were multiplied by 1/10th.
 - Multiplication was performed in a 24-bit fixed-point register
 - Binary expansion of 1/10 is 0.00011001100110011 ...
 - Therefore some truncation is necessary
- System had been up for 100 hours
- Error in calculation was 0.34 seconds
- Scud missile travels at 1,676 meters per second
 - 0.34 seconds therefore equals about half a kilometer



Interface Design Perspective

- To the user “the interface is the system”
 - Understanding of the system operation is based on its interface
 - If this conceptual model is wrong, errors result
- Human attention is incomplete – will not notice every detail, every notification
- Good design will help prevent human error and help users recover from errors



Interface Design Perspective

- Beware of “response chaining”
- Example: Nurse accidentally turned off alarm on monitoring for critically ill patient
 - Normal operation – set alarm & then progress through a series of confirmation screens by pressing enter
 - Action became automatic (tap tap tap) – failed to notice warning that alarm was off



What goes wrong?

- Computer systems fail because:
 - The job they are doing is inherently difficult
 - The job is sometimes done poorly
- Compounding the reliability issue
 - Developers and users are overconfident in the system
 - Re-used system software may not work correctly in different environments
- Intellectual overload
 - Large bodies of code are more difficult to understand than smaller ones



Q: How can we increase reliability & safety?

- As developers?
- As system managers?
- As users?



Increasing reliability and safety

- Be professional and responsible:
 - Follow good software-engineering practices
 - What do you believe constitutes “good”?
 - Construct well-designed user interfaces + take human factors into account
 - Include built-in redundancy
 - Incorporate self-checking where appropriate
 - Follow good testing principles and techniques



Increasing reliability and safety

- Law and regulation
 - Civil and criminal penalties
 - to recover from loss due to faulty or unsafe systems
 - to provide incentives to produce reliable and safe systems
 - Warranties
 - to guarantee a certain level of quality
 - Provincial or Federal regulations
 - E.g. Database accuracy regulations - to protect the public from inaccurate information maintained by private companies and government
 - Mandatory licensing of software developers
 - to ensure proper training, competency, and continuing education
 - Q: Do you think this is a good idea?



Computer Models

- Points to consider
 - Models are simplifications of either physical or intangible systems
 - Those who design and develop models must be honest and accurate with results
 - Computer professionals and the general public must be able to evaluate the claims of the developers

Q: Give an example of a computer model. How accurate is it?



Computer models

- Evaluating models
 - Why models might not be accurate:
 - Developers have incomplete knowledge of the system being modeled
 - Data might be incomplete or inaccurate
 - Power of the computer might be inadequate
 - Variables are difficult to quantify (numerically)
 - Political and economic motivation to distort results
 - Two contrasting examples:
 - Car-crash models
 - Climate models



Computer models

- Car-crash models
 - How well do the modelers understand the system or material being studied (or both)?
 - How accurate and complete are the data?
 - What are the assumptions and simplifications of the model?
 - Do the results or predictions correspond with results in the real world?
- Climate models
 - (Same questions)



Perspectives

- Failures
 - What are “acceptable rates” of failures?
 - How accurate should software be?
- Dependence
 - How dependent on computer systems are our ordinary activities?
 - How useful are computer systems to our ordinary activities?
- Risks and progress
 - How do new technologies become safer?
 - Can progress in software safety keep up with the pace of change in computer technology?



Risk is not new

- Risks existed with older technologies
 - Cars, airplanes, trains...
 - Steam engines...
 - Elevators...
- Computers change the types and scope of risk and the allocation of blame
 - Larger, more complex systems
 - Computers make decisions



Unwrapping the Gift: Benefits of Computer Technology

Lecture 1

Syada Tasmia Alvi
Lecturer, Dept.of CSE
Daffodil International University

The Ubiquity of Computers and the Rapid Pace of Change

Have you used a computer today?

You have if you used...

- an ATM,
- a cell phone,
- music player (iPod)
- a modern appliance, or
- an electronic device.



Negative impacts of new technology



Unemployment



Poor customer service



Crime



Alienation



Loss of Privacy

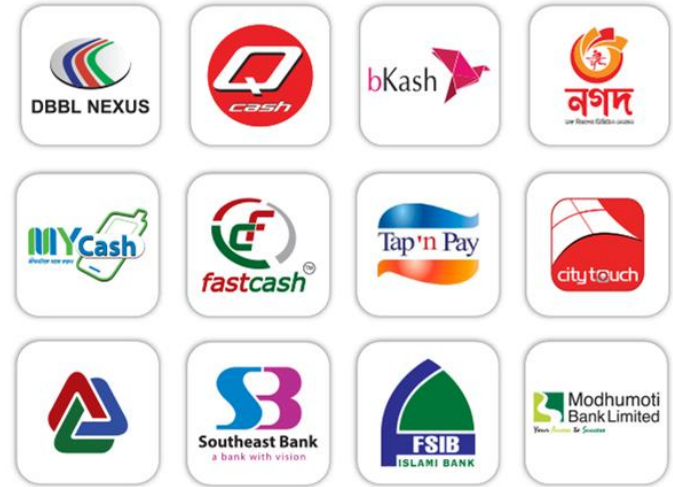
Some Positive Impacts of New



Convenience/ less effort



New types of jobs.



More options for transactions

Appreciating the Benefits

1. Benefits computers bring to communication.
2. Benefits computers bring to transportation, especially within our communities.
3. Benefits computers bring to education and training as they aid local schools.
4. Benefits of computers in fighting crime.
5. Benefits computers bring to medicine.
6. Benefits computers bring to the disabled in schools and the workplace.

Benefits to communication



- ☐ Non-invasive; read at recipients' convenience .
- ☐ Time-saving
- ☐ Access to immense (often timely) amounts of information
- ☐ Information on WWW can be in different formats such as text, graphics, video or sound.
- ☐ Much information on the web is “free” and can be accessed by all.
- ☐ Public forums not limited to geographic boundaries
 - Traditional boundaries are no obstacle
- ☐ More independence - people can do more themselves –
- ☐ Remote access available

Benefits to Transportation



- ❑ Navigation : Prevent from getting lost
- ❑ Diagnostics - e.g. chips that collect information about engine behaviour
- ❑ Safety : Anti-lock Braking System and Truck roll-over prevention
- ❑ Hybrid engines-New technologies, such as a warning system that alerts the driver he/she is veering off the road and at the same time corrects the drivers steering.
- ❑ Fuel efficiency
- ❑ Traffic pattern studies

Benefits to Education



- ❑ Language acquisition - Teach people to speak a foreign language
- ❑ Spelling
- ❑ People who are illiterate can learn in the comfort of their own home, without feeling embarrassed.
- ❑ Distance learning- . Students can learn and be taught from their own home without actually going to a school. (ex. Rural children unable to attend school due to distance or money)
- ❑ Simulations : Help train people to do a specific job.
- ❑ Speech recognition & synthesis (e.g. for learning a foreign language)

Benefits in fighting crime

- ❑ Improved crime reporting (e.g. information about stolen art may be posted)
- ❑ Faster search of arrest records and fingerprint files
- ❑ Remote access to records and reports (e.g., from patrol cars)
- ❑ Fingerprint matching has become easier with the introduction of there computer networks and databases. This task, as well as others, have become less time consuming and less painstaking
- ❑ Access to numerous databases
- ❑ Improved monitoring and surveillance equipment

Thank You

Unwrapping the Gift: Benefits of Computer Technology

Lecture 2

Syada Tasmia Alvi
Lecturer, Dept.of CSE
Daffodil International University

Benefits to medicine



- ❑ Sophisticated medical imaging
- ❑ Reduced surgery time
- ❑ Training surgeons
- ❑ Faster recovery time (e.g. minimally invasive surgery)
- ❑ Patient monitoring
- ❑ Improved treatments
- ❑ Patient records
 - More legible
 - Better organized
 - Accessible by many sites, simultaneously
 - Available immediately

Benefits to medicine



❑ Diagnosis

- Streamlined diagnosis
- Improved accuracy
- Reduced need for biopsy
- Improved screening and predictions

❑ Telemedicine

- Broad access to medical help
- Less expensive
- Expands consultations beyond rural/remote sites



Benefits to disabled



- ☐ Disability-specific computer applications (e.g., a braille printer, large size print).
- ☐ Control of household and workplace appliances via speech (people who can't use their hands.)
- ☐ Mobility
- ☐ Control of artificial limbs
- ☐ Improved vision (e.g. large font sizes)
- ☐ Access to adaptive educational equipment
- ☐ Improved communication (e.g. speech input for people who can't write, speech synthesis for those who can't talk)
- ☐ Opportunity to return to work

Benefits - Automation



- ☐ Organize and access large inventories
- ☐ Cost-saving
- ☐ Time-saving
- ☐ Improved customer satisfaction
- ☐ Tedious, monotonous work done by machines
- ☐ Improved workplace safety

Identifying people and products



- ☐ E.g. Bar codes & smart cards –
- ☐ Time-saving
- ☐ Accuracy
- ☐ Ease-of-use
- ☐ Cost-saving
- ☐ More secure
- ☐ Customizable
- ☐ Reliable

Monitoring Environment and Materials



- ☐ E.g. Bar codes & smart cards –
- ☐ Time-saving
- ☐ Accuracy
- ☐ Ease-of-use
- ☐ Cost-saving
- ☐ More secure
- ☐ Customizable
- ☐ Reliable

Benefits of Computers in Reducing Paper Use and Garbage



- ☐ Send/receive digital documents instead of hardcopy.
- ☐ Read, write, and edit online.
- ☐ Cost-saving; smaller storage needed.
- ☐ Creation of toxic wastes at paper mills reduced.

Other Benefits

- ☐ Tracking
- ☐ Rare plants and animals
- ☐ Migration habits
- ☐ Stolen goods
- ☐ Lost pets or people

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE

CHAPTER 2: PRIVACY

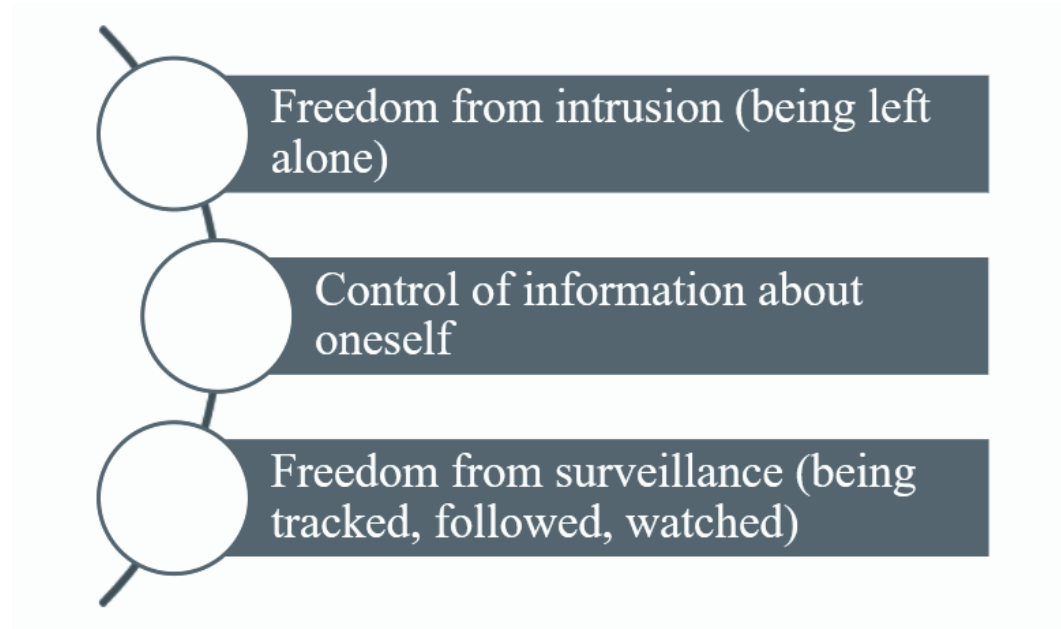
Lecture 1

Privacy and Computer Technology

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Lecturer, Dept.of CSE
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PRIVACY AND COMPUTER TECHNOLOGY

❑ Three Key Aspects of Privacy:



PRIVACY RISKS AND PRINCIPLES

Privacy threats come in several categories:

- ☐ Intentional, institutional uses of personal information
- ☐ Unauthorized use or release by “insiders”
- ☐ Theft of information
- ☐ Inadvertent leakage of information
- ☐ Our own actions

PRIVACY AND COMPUTER TECHNOLOGY

New Technology, New Risks:

- ❑ Government and private databases
 - Thousands of databases containing personal information about us.
 - Profiles of our personal information could be created easily.
- ❑ Sophisticated tools for surveillance and data analysis
 - Cameras, GPS, cell phones
- ❑ Vulnerability of data
 - Leaks of data happens, existence of data presents a risk.

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Examples:

- ❑ Search query data
 - Search engines collect many terabytes of data daily.
 - Data is analyzed to target advertising and develop new services.
 - Who gets to see this data? Why should we care?
- ❑ Smartphones
- ❑ Location apps
- ❑ Data sometimes stored and sent without user's knowledge

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- ❑ Anything we do in cyberspace is recorded.
- ❑ Huge amounts of data are stored.
- ❑ People are not aware of collection of data. Software is complex.
- ❑ Leaks happen.
- ❑ A collection of small items can provide a detailed picture.
- ❑ Re-identification has become much easier due to the quantity of information and power of data search and analysis tools.

PRIVACY RISKS AND PRINCIPLES

New Technology, New Risks – Summary of Risks:

- ❑ If information is on a public Web site, it is available to everyone.
- ❑ Information on the Internet seems to last forever.
- ❑ Data collected for one purpose will find other uses.
- ❑ Government can request sensitive personal data held by businesses or organizations.
- ❑ We cannot directly protect information about ourselves. We depend upon businesses and organizations to protect it.

PRIVACY AND COMPUTER TECHNOLOGY (CONT.)

Terminology and principles for data collection and use:

- ❑ Invisible information gathering
 - ❑ Collection of personal information about someone without the person's knowledge
 - ❑ Unauthorized software, Cookies, ISP providers •
- ❑ Secondary use
 - ❑ Use of personal information for a purpose other than the one it was provided for .
 - ❑ Data mining
 - ❑ Searching and analyzing masses of data to find patterns and develop new information or knowledge
 - ❑ Computer matching
 - ❑ Combining and comparing information from different databases (using social security number, for example, to match records)

Computer profiling

- ❑ Using data in computer files to predict likely behaviors of people
- ❑ Analyzing data in computer files to determine characteristics of people most likely to engage in certain behavior
- ❑ Businesses find new consumers
- ❑ Government detects fraud and crime activities
- ❑ Businesses engage in profiling to determine consumer propensity toward a product or service.
- ❑ Government agencies use profiling to create descriptions of possible terrorist

Principles for Data Collection and Use:

- ☐ Informed consent
- ☐ Opt-in and opt-out policies
- ☐ Fair Information Principles (or Practices)
- ☐ Data retention - Data retention is the storing of information for a specified period.

Principles for Data Collection and Use:

☐ Informed consent :

- People should be informed about the data collection and use policies of a business or organization,
- They can then decide whether or not to interact with that business or organization.

☐ Opt-in and opt-out policies :

- Opt-in :
 - Opting in means that a user will take an affirmative action to offer their consent.
 - The most common way businesses implement opt-in methods is through checkboxes. When presented with a checkbox, the user must take action to check the box, which denotes their consent.

Principles for Data Collection and Use:

- Opt-in policies:
 - Opting in can be used in a variety of situations, including subscribing to email and newsletter mailing lists, accepting cookie use, and agreeing to legal policies.

Email

Enter the E-mail

Create a password

At least 8 characters long

Confirm password

Enter your password again

☐ I want to receive news, feature updates, discounts, and offers from Termly.

☐ I acknowledge that I agree to the [Terms of Use](#) and have read the [Privacy Policy](#).

CREATE ACCOUNT

Cookie consent

By clicking "Accept All", you consent to our website's use of cookies to give you the most relevant experience by remembering your preferences and repeat visits. However, you may visit "Cookie Settings" to provide controlled consent. [Read More](#)

[Cookie Settings](#) [Reject All](#) [Accept All](#)

Advertisement ☐ Opt-in/Opt-out

Advertisement cookies are used to provide visitors with relevant ads and marketing campaigns. These cookies track visitors across websites and collect information to provide customized ads.

Cookie	Type	Duration	Description
_hjTLDTest	http	session	When the Hotjar script executes we try to determine the most generic cookie path we should use, instead of the page hostname. This is done so that cookies can be shared across subdomains (where applicable). To determine this, we try to store the _hjTLDTest cookie for different URL substrings alternatives until it fails. After this check, the cookie is removed.
uuid	https	6 months	To optimize ad relevance by collecting visitor data from multiple websites such as what pages have been loaded.

[Save Settings](#)

Powered by [CookieYes](#)

Principles for Data Collection and Use:

- ❑ Opt-out policies :
 - Opting out means a user takes action to withdraw their consent.
 - Not choosing to subscribe to newsletters, unticking a previously ticked checkbox, not consenting to save personal details, rejecting the use of cookies, etc. are some examples of opt-out.



Principles for Data Collection and Use:

❑ Fair Information Principles (or Practices)

- Inform people when you collect information about them, what you collect, and how you use it.
- Collect only data needed
- Offer a way for people to opt out from mailing lists, advertising, and other secondary uses.
- Offer a way for people to opt out from features and services that expose personal information.
- Keep data only as long as needed
- Protect security of data (from theft and from accidental leaks). Provide stronger protection for sensitive data.
- Policies for responding to law enforcement

Thank You

A GIFT OF FIRE THIRD EDITION - SARA BAASE

CHAPTER 2: PRIVACY

Lecture 2

"Big Brother is Watching You"

Syada Tasmia Alvi
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"BIG BROTHER IS WATCHING YOU"

Databases:

- Government agencies
 - Collect many types of information
 - Ask business to report about consumers
 - Buy personal information from sellers
- Main publicized reason: data mining and computer matching to fight terrorism
- Private information can be used to:
 - Arrest people
 - Jail people
 - Seize assets

"BIG BROTHER IS WATCHING YOU"

❏ Databases:

- Millions of crime suspects are searched in government databases
- Shift from presumption of innocence to presumption of guilt
- Computer software characterizes suspects
 - Innocent people are sometimes subject to embarrassing searches and expensive investigations and to arrest and jail.

"BIG BROTHER IS WATCHING YOU"

Small Sampling of Government Databases with Personal Information

❑ What data does the government have about you?

- Tax records
- Medical records
- Marriage and divorce records
- Property ownership
- Welfare records
- School records
- Motor vehicle records
- Voter registration records
- Books checked out of public libraries
- People with permits to carry firearms
- Applications for government grant and loan programs
- Professional and trade licenses
- Bankruptcy records
- Arrest records

"BIG BROTHER IS WATCHING YOU"

- ❑ Some constitution articles (laws) protect privacy.
- ❑ Modern surveillance techniques are redefining expectation of privacy.
- ❑ In some countries:
 - No court order or court oversight needed to get one's private information.
 - 2003-2005 report found "widespread and serious misuse" of the FBIs (**Federal Bureau of Investigation**)national security letter authorities.

"BIG BROTHER IS WATCHING YOU"

Two key problems arise from new technologies:

- ❑ Much of our personal information is no longer safe in our homes; it resides in huge databases outside our control.
- ❑ New technologies allow the government to search our homes without entering them and search our persons from a distance without our knowledge

"BIG BROTHER IS WATCHING YOU"

Video Surveillance:

- ❑ Security cameras in Shopping centers, malls, banks, etc.
- ❑ Cameras alone raise some privacy issues.
- ❑ When being combined with face recognition systems, they raise more privacy issues.
 - Increased security
 - Decreased privacy

Thank You

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CHAPTER 2: PRIVACY

Lecture 3

"DIVERSE PRIVACY TOPICS "

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Marketing, Personalization and Consumer

- Targeted and personalized marketing (business, political parties, etc)
- Paying for consumer information Examples -
 - Targeting ads to users by scanning their emails! Like Gmail.
- Data firms and consumer profiles
 - Companies (firms) that collect information about individuals
 - These firms sell data to other companies for marketing purposes.
- Data mining is being used
- Credit records might be sold to different parties

Social Networks

- ❑ What we do
 - Post opinions, gossip, pictures, “away from home” status
- ❑ What they do
 - New services with unexpected privacy settings

Location Tracking:

- ❑ Global Positioning Systems (GPS) -
 - computer or communication services that know exactly where a person is at a particular time
- ❑ Cell phones and other devices are used for location tracking

Stolen and Lost Data:

- ❑ Hackers
- ❑ Physical theft (laptops, thumb-drives, etc.)
- ❑ Requesting information under false pretenses
- ❑ Bribery of employees who have access

What We Do Ourselves

- ❑ Some people do not know or understand enough how the web works in order to make good decisions about what to put there.
- ❑ Some people do not think carefully.
- ❑ People often want a lot of information about others but do not want others to have access to the same kind of information about themselves.
- ❑ Our cell phone and email messages reside on computers outside our home or office.
- ❑ We have no direct control over such files.

What We Do Ourselves

- ❑ There have been many incidents of exposure of emails for politicians, businessmen, etc
- ❑ Personal information in blogs and online profiles
- ❑ Pictures of ourselves and our families
- ❑ File sharing and storing
- ❑ Is privacy old-fashioned?
 - Young people put less value on privacy than previous generations.
 - May not understand the risks.

Public Records: Access vs. Privacy:

- Public Records -
 - Governments maintain “public records,”
 - Records available to general public (bankruptcy, property, and arrest records, salaries of government employees, etc.)
- Identity theft can arise when public records are accesses.
- How should we control access to sensitive public records?

Children (privacy and safety)

➤ The Internet

- Not able to make decisions on when to provide information
- Vulnerable to online predators

➤ Parental monitoring

- Software to monitor Web usage
- Web cams to monitor children while parents are at work
- GPS tracking via cell phones or RFID

Children (privacy and safety)

- At what age does web monitoring become an invasion of the child's privacy?
- Should parents tell children about the tracking devices and services they are using?
- Informed consent is a basic principle for adults.
- At what age does it apply to children? Will intense tracking and monitoring slow the development of a child's responsible independence? Will parents rely more on gadgets than on talking to their children?

National ID System:

- Social Security Numbers
 - Too widely used
 - Easy to falsify
- A new national ID system - Pros
 - would require the card
 - harder to forge
 - have to carry only one card
- A new national ID system - Cons
 - Threat to freedom and privacy
 - Increased potential for abuse

Thank You

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CHAPTER 2: PRIVACY

Lecture 4

"PROTECTING PRIVACY"

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Technology and Markets:

Awareness:

Most people have figured out now you can't do anything on the web without leaving a record.

- ☐ We can decide to what extent we wish to use privacy-protecting tools.
- ☐ We can be more careful about the information we give out, and the privacy policies of sites we use or visit.

Technology and Markets:

Privacy-enhancing technologies for consumers

- ❑ New applications of technology often can solve problems that arise as side effects of technology.
- ❑ Example: cookie disablers, blocking pop-up ads, scanning PCs for spyware, etc.
- ❑ Using usernames and passwords for Blogs visitors (family , friends, etc.)

Technology and Markets:

Encryption

- ❑ Information sent to and from websites can be intercepted.
- ❑ Someone who steals a computer or hacks into it can view files on it .
- ❑ Encryption is a technology that transforms data into a form that that is meaningless to anyone who might intercept or view it.
- ❑ Encryption generally includes a coding scheme, or cryptography algorithm, and specific sequences of characters (digits or letters).

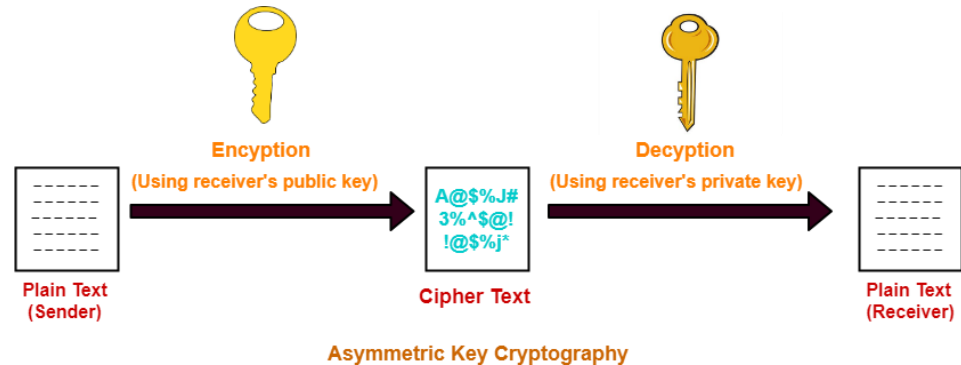
Cryptography is the art and science of hiding data in plain sight.
—Larry Loen

Technology and Markets:

Encryption

- ❑ Public-key cryptography: An encryption scheme, where two related keys are used.

- I. A key to encrypt messages.
- II. A key to decrypt messages.



Technology and Markets:

Business tools and policies for protecting privacy

- A well-designed database for sensitive information includes several features to protect against leaks, intruders, and unauthorized employee access.
 - Each person with authorized access to the system should have a unique identifier and a password.
 - A system can restrict users from performing certain operations, such as writing or deleting, on some files. User IDs can be coded so that they give access to only specific parts of a record.
 - For example, a billing clerk in a hospital does not need access to the results of a patient's lab tests.

Technology and Markets:

Business tools and policies for protecting privacy

- ❑ The computer system keeps track of information about each access, including the ID of the person looking at a record and the particular information viewed or modified. This is an audit trail that can later help trace unauthorized activity.
- ❑ Website operators pay thousands, sometimes millions, of dollars to companies that do privacy audits. Privacy auditors check for leaks of information, review the company's privacy policy and its compliance with that policy, evaluate warnings and explanations on its website that alert visitors when the site requests sensitive data, and so forth.
- ❑ Some large companies like IBM and Microsoft, use their economic influence to improve consumer privacy on the web, by removing ads from their web sites.

Rights and laws:

Warren and Brandeis: The inviolate personality -

- ❑ Warren and Brandeis criticized newspapers especially for the gossip columns.
- ❑ People have the right to prohibit publications of facts (and photos) about themselves.
- ❑ The kinds of information of most concern to them are personal appearance, statements, acts, and interpersonal relationships (marital, family, and others)
- ❑ Warren and Brandeis say privacy is distinct and needs its own protection.

Rights and laws:

Thomson: Is there a right to privacy? -

- ❑ Thomson argues the opposite point of view.
- ❑ There is no violation of privacy without violation of some other right, such as the right to control our property or our person, the right to be free from violent attacks, the right to form contracts (and expect them to be enforced).

Rights and laws:

Applying the theories:

- ❑ Many court decisions since Warren and Brandeis article, have taken their point of view.
- ❑ A person may win a case if someone published his/her consumer profile.
- ❑ Warren and Brandeis (and court decisions) allow disclosure of personal information to people who have an interest in it.
- ❑ An important aspect: consent

Rights and laws:

Transactions

- ❑ Privacy includes control of information about oneself.
- ❑ How to apply privacy notions to transactions, Which involve more than one person?

Rights and laws:

Ownership of personal data

- ❑ People should be given property rights in information about themselves.
- ❑ But some activities and transactions involve at least two people, each of whom would have claims to own the information about the activity.
- ❑ Can we own our profiles (collection of data describing our activities, purchase, interests, etc.) ?
- ❑ We cannot own the fact that our eyes have a certain color

Rights and laws:

Regulation

- ❑ Technical tools for privacy protection, market mechanisms, and business policies are not perfect.
- ❑ Regulation is not perfect either.
- ❑ Some Regulations may be so expensive and difficult to apply.
- ❑ Example: Health Insurance Portability and Accountability Act (HIPAA)- protect sensitive patient health information from being disclosed without the patient's consent or knowledge.

Rights and laws: Contrasting Viewpoints:

Free Market View

- ❑ Freedom of consumers to make voluntary agreements
- ❑ Diversity of individual tastes and values
- ❑ Response of the market to consumer preferences
- ❑ Usefulness of contracts

Rights and laws: Contrasting Viewpoints:

Consumer Protection View

- ❑ Advocates of strong privacy regulation emphasize the unsettling uses of personal information
- ❑ Costly and disruptive results of errors in databases
- ❑ Ease with which personal information leaks out, via loss, theft, and carelessness
- ❑ Consumers need protection from their own lack of knowledge, judgment, or interest

Communications

Wiretapping and Email Protection:

Telephone

- ❑ 1934 Communications Act prohibited interception of messages
- ❑ 1968 Omnibus Crime Control and Safe Streets Act allowed wiretapping and electronic surveillance by law-enforcement (with court order)

Communications

Wiretapping and Email Protection:

- ❑ Electronic Communications Privacy Act of 1986 (ECPA) extended the 1968 wiretapping laws to include electronic communications, restricts government access to e-mail

Thank You

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CHAPTER 3: Technology & Free Speech

Lecture 1

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Changing paradigms

❑ Regulatory paradigms

- Communications technologies differ with respect to constitutional protections
- Also with respect to government regulation

❑ The traditional three-part framework

- Print media (Newspapers, magazines)
- Broadcast media (Television, radio)
- Common carriers (Telephone, telegraph, and the postal system)

Changing paradigms

- ❑ Print media (Newspapers, magazines)
 - Has the strongest protections for free speech
- ❑ Broadcast media (Television, radio)
 - Has protections, but govt controls structure of industry + content
 - E.g. Canadian content regulations, restriction of some words
- ❑ Common carriers (Telephone, ISPs?)
 - Content not controlled and carrier not responsible for content

Changing paradigms

- ❑ Consider new technologies of the past 20 years
 - Bulletin-board systems (BBS)
 - World Wide Web
- ❑ How do these fit into the “print-broadcast-carrier” framework?
 - Not exactly print media
 - Not exactly broadcast media
 - Not exactly common carrier

Offensive Speech

Examples of such speech can include

- ☐ Political or religious speech
- ☐ Pornography
- ☐ Sexual or racial slurs
- ☐ Nazi (or White Supremacist) materials
- ☐ Libellous (false and damaging) statements
- ☐ Abortion information
- ☐ Alcohol advertisement

Censorship

- ❑ Censorship is the suppression of speech, public communication, or other information.
- ❑ This may be done on the basis that such material is considered objectionable, harmful, sensitive, or "inconvenient".
- ❑ Censorship can be conducted by governments, private institutions and other controlling bodies.
- ❑ Governments and private organizations may engage in censorship. Other groups or institutions may propose and petition for censorship.
- ❑ General censorship occurs in a variety of different media, including speech, books, music, films, and other arts, the press, radio, television, and the Internet for a variety of claimed reasons including national security, to control obscenity, pornography, and hate speech, to protect children or other vulnerable groups, to promote or restrict political or religious views, and to prevent slander and libel.

Offensive speech & censorship

- ❑ One approach used in the US
 - Limit Internet access in libraries and schools.
 - Accomplish this via “filtering software”
 - “Funding” control (any school or library receiving US federal funds must install such software on internet terminals)
- ❑ Example: Canadian company • Useful.com (Calgary)
- ❑ software does much, much more than just web filtering
- ❑ Is CIPA compliant
- ❑ Filters block sites containing child pornography, obscene material, anything deemed “harmful to minors”

Hate Speech

- ❑ Racist and hateful comments are offensive...
 - but not necessarily illegal.
- ❑ “hate speech” loosely refer to offensive discourse targeting a group or an individual based on inherent characteristics - such as
 - race,
 - religion or
 - gender - and that may threaten social peace.
- ❑ A judge has the authority to order removal of hate propaganda from a computer system available to public.

Hate Speech

- ❑ Canadian Human Rights Act
- ❑ Section 13: Applies to e-mail, web sites, and any other telecommunications activity.
- ❑ Prohibits messages likely to expose a person to hatred or contempt on any one of several categories.
 - Race
 - national or ethnic origin
 - Colour
 - Religion
 - Age
 - sexual orientation
 - marital status
 - family status
 - Disability
 - conviction for which a pardon has been granted

Challenges to Regulation

- ❑ For certain kinds of activity, a licence is required:
 - To practice law
 - To practice medicine (e.g., prescribe medication)
 - To practice engineering
 - To offer financial advice (e.g., investing in equities, futures, commodities)
- ❑ For other kinds of activity, no licence is required
 - To create downloadable, self-help legal software.
 - To publish newsletter / website about, or develop software for commodities and futures investing.

Censorship & the global net

- ❑ Internet technology has a global impact on free speech
- ❑ Can be used to avoid censorship
 - Global nature of Internet allows restrictions in one country to be circumvented by using networks in other, less restrictive countries. Such sites are usually posted in the US.
- ❑ Can be used to establish censorship
 - Global nature of Internet makes it easier for one nation to impose restrictive standards on others.
- ❑ eBay bans some items for sale (i.e., hate-group memorabilia) because of different cultural standards in different countries.

Thank You

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CHAPTER 3: Technology & Free Speech

Lecture 2

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Anonymity

- ❑ True anonymity
 - “Unlinkability”
 - Inability of anyone to determine the true author's identity
 - Legally problematic



Anonymity

❑ Pseudonymity:

- Pseudonymity is the near-anonymous state in which a user has a consistent identifier that is not their real name: a pseudonym.
- public pseudonyms: link between pseudonym and human being is easy for the public to know or discover
- non-public pseudonyms: link known by sysadmins, but no one else
- unlinkable pseudonyms: link not known and not discoverable by service operators

Anonymous Re-mailers and Web Browsers

- ❑ Johan Helsingius set up the first well-known anonymous email service in Finland in 1993.
 - Users were not entirely anonymous
 - the system retained identifying information.
 - Return address stripped off and then forwarded to recipient
- ❑ To send anonymous email using a “re-mailer” service, one sends the message to the remailer, where the return address is stripped off and the message is re-sent to the intended recipient.
- ❑ Anonymous web browsers - hide user’s IP address, do not track or store activity

Anonymity vs. Community?

❏ Supporters of anonymity

- Claim that it is necessary to protect privacy and free speech.
- Helps those who are concerned about political or economic retribution.
 - Human rights workers
 - Victims of domestic violence

Anonymity vs. Community?

❑ Opponents of anonymity

- Claim that it is anti-social and allows criminals to hide from law enforcement.
- Threatens civil discourse (“marketplace of ideas”)
- Examples:
 - Online harassment
 - Fraud

Spam

- ❑ Unsolicited, mass e-mail
 - Is cheap to senders but may impose costs on recipient's time or the recipient's online account (or both).
 - May contain objectionable content (political, commercial ads, solicitations for funds, pornography, etc.)
 - May contain a disguised return address.
 - May pass through filters.
- ❑ – Invades privacy.
- ❑ – Creates a financial and managerial burden on ISPs.

Spam: solutions (?!)

- ❑ Technology: Filters that screen out spam.
- ❑ Market pressure: Services that list spammers
- ❑ Small “microfee” for sending an email?
- ❑ requires consent for email marketing
- ❑ Vigilantism: Punish spammers

Thank You

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CHAPTER 2: Computer Crime

Lecture 1

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Challenges wrt Computer Crime

- ❑ There are at least three:
- ❑ Prevention
- ❑ Detection
- ❑ Prosecution

Defining “Hacking”: Phase one

Phase 1: Joy of Programming

- ❑ The early years (1960s and 1970s), when hacking was a positive term
- ❑ Hacker was a creative programmer who wrote clever code.
- ❑ First OSes and computer games were written by such hackers.
- ❑ Hackers were usually – but not always – high-school and college students.

Phase 2: From the 1970s to the mid-1990s

- ❑ Negative overtones appear in 1970s-90s
- ❑ Popular authors and media caused this use of term:
 - Described someone who used computers without authorization.
 - Sometimes to commit crimes.
- ❑ Early computer crimes were launched against business and government computers.
- ❑ Adult criminals began using computers
 - “White collar” crime
 - Organized crime increasingly responsible for computer break-ins

Phase 3: The growth of the Web and mobile devices

- ❑ Web era (mid 90s)
- ❑ Increased use of Internet by average people
- ❑ Attractive to criminals with basic computer skills.
- ❑ Crimes included the release of malicious code
- ❑ Unprotected computers are especially vulnerable
- ❑ Hacking for political motives increased

Phase 3: The growth of the Web and mobile devices

- ❑ Unsuspecting users may have their computers utilized to take part in a DDoS or fraud
- ❑ Minimal computer skills needed to create havoc
 - “Script kiddies”
 - Attackers who use tools / code written by others

Hacktivism

- ❑ Hacktivism is the use of hacking to promote a political cause
- ❑ Degree can range from mild to destructive
 - ❑ Defacing websites
 - ❑ Destroying data
 - ❑ Denial of service
- ❑ People who agree with the political or social position of the hackers will tend to see an act as “activism,” while those who disagree will tend to see it as ordinary crime (or worse).
- ❑ Some academic writers and political groups argue that hacktivism is ethical, that it is a modern form of civil disobedience.

Hacktivism

- ❑ Others argue that the political motive is irrelevant, or at the other extreme, that political hacking is a form of cyberterrorism.
- ❑ They think so:
 - ❑ Denies others their own freedom of speech.
 - ❑ Violates property rights.
 - ❑ Even some hacktivists reject web-site defacing as legitimate activity.

Thank You

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CHAPTER 2: Computer Crime

Lecture 2

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Cyber Crimes

- ❑ Cybercrime is any criminal activity that involves a computer, networked device or a network.
- ❑ Cyber crime is a subset of computer crime.
- ❑ Definitions :
 - Offences that are committed against individuals or group of individuals with a criminal motive
 - to intentionally harm the reputation of the victim or
 - cause physical or mental harm to the victim directly or indirectly, using modern telecommunication networks such as internet.

TARGETS OF CYBER CRIMES

❑ Against individual property

- Transmitting virus
- Un-authorized control/access over computer
- Intellectual property crimes
- Internet time thefts

❑ Against Organization

- Possession of un-authorized information
- Cyber terrorism against the government organization
- Distribution of pirated software,etc

TARGETS OF CYBER CRIMES

- ❑ Against Social at Large
 - Pornography (basically child pornography)
 - Trafficking
 - Financial crimes
 - Online gambling
 - Forgery
 - Sale of illegal articles

SOME COMMON CYBER CRIMES

- ❑ Computer Virus: A computer virus is a computer program or attaches itself to application programs or other executable system software causing damage to the files.
- ❑ Phishing: Phishing occurs when the perpetrator sends fictitious e-mails to individuals with links to fraudulent websites that appear official and thereby cause the victim to release personal information to the perpetrator.



SOME COMMON CYBER CRIMES

- ❑ Hacking: The act of penetrating or gaining unauthorized access to or use of data unavailable in a computer system or a computer network for the purpose of gaining knowledge, stealing or making unauthorized use of the data.
- ❑ Spoofing : Spoofing, as it pertains to cybersecurity, is when someone or something pretends to be something else in an attempt to gain our confidence, get access to our systems, steal data, steal money, or spread malware

SOME COMMON CYBER CRIMES

- ❑ Netsplonage: Netsplonage occurs when perpetrators back into online systems or individual PCs to obtain confidential information for the purpose of selling it to other parties.
- ❑ Cyber stalking: Cyber stalking refers to the use of the internet, email or other electronic communications device to stalk another person. It is an electronic harassment that involves harassing or threatening over a period of time.
- ❑ Cyber Terrorism: Cyberterrorism is the convergence of cyberspace and terrorism. It refers to unlawful attacks and threats of attacks against computers, networks and the information stored therein when done to intimidate or coerce a government or its people in furtherance of political or social objectives.

MOTIVES OF CYBER CRIMINALS

- ☐ Desire for entertainment
- ☐ Profit
- ☐ Infuriation or revenge
- ☐ Political agenda
- ☐ Sexual motivations
- ☐ Psychiatric illness

PREVENTION

- ☐ Use hard to guess passwords
- ☐ Use anti-virus software and firewalls-keep them up to date
- ☐ Don't open email or attachments from unknown sources
- ☐ Backup your data

Thank You





Ethics, also known as ***moral philosophy***, is a branch of philosophy that involves systematizing, defending, and recommending concepts of right and wrong behavior.

ANCIENT ETHICS



- Mesopotamian Epic of Gilgamesh
- Homer's Iliad
- The Icelandic Eddas
- The Sumerian Farmer's Almanac
- The Egyptian Instruction of Amenemope

Ethical Theories

An introduction to three main branches of normative ethics

Normative vs. descriptive Ethics

Descriptive Ethics: seeks to describe how people behave and/or what sorts of moral standards they claim to follow. (Think Anthropology or sociology)

- non-evaluative

Normative ethics: seeks to create or evaluate moral standards. it is an attempt to figure out what people **should** do or whether their current moral behavior is reasonable.

Three types of normative ethics

Teleological -- look at “ends” or consequences of what we do

Deontological – adheres to independent moral rules or duties; motives behind certain actions are right or wrong instead of focusing on the results

(Duty-based)

- The Golden rule
- Religion (ex. The ten commandments)

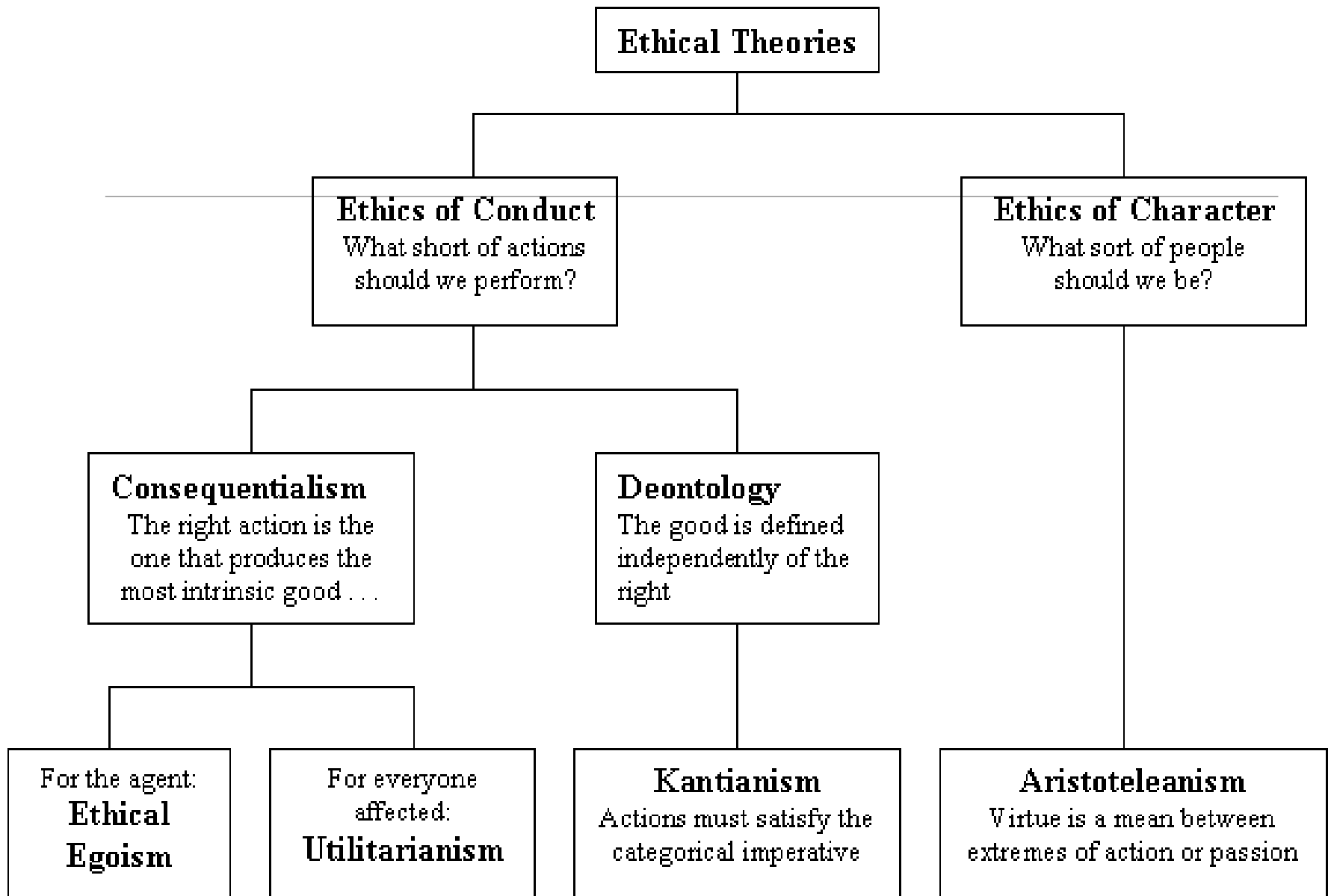
Virtue ethics- Character-based Theories that look for the role one’s character plays in the decision making and how that decision will affect the character of others

Three Main Ethical Approaches

Three main approaches to normative ethics that we will discuss:

- Virtue ethics (ethics of character)
- Utilitarianism (teleological)
- Deontology

The difference between these three approaches to morality tends to lie more in the way moral dilemmas are approached, rather than in the moral conclusions reached.



Virtue Theory (ethics of Character)

- **Virtue ethics** emphasizes the role of one's character and the virtues that one's character embodies for determining or evaluating ethical behavior.
- The roots of this theory lie in the works of Plato and Aristotle.
- Virtue ethics place emphasizes **being** rather than **doing**. In virtue ethics, morality stems from the identity and/or character of the individual, rather than being a reflection of the actions (or consequences) of the individual.
- A virtue ethics philosopher will identify virtues, desirable characteristics, that the moral or virtuous person embodies. Possessing these virtue in virtues, ethics, is what makes one moral, and one's actions are a mere reflection of one's inner morality.

Deontological Ethics (Kantian Ethics or Duty Ethics)

- places the emphasis on adhering to ethical principles or duties and fulfilling obligations
- How these duties are defined, however, is often a point of contention and debate in deontological ethics.
- Deontology also depends, at least partially, upon moral absolutes that make an action moral regardless of circumstances.
- There are multiple motivations for “duty” -
 - Immanuel Kant--our actions should be motivated by reason, rather than by emotions -you should not only do good things when you feel like it
 - Divine Command Theory– It has been decreed by God to be “good.”

Consequentialism

- Consequentialism bases the morality of an action upon the consequences of the outcome
- Instead of saying that one has a moral duty to abstain from murder, a consequentialist would say that we should abstain from murder because it causes undesirable effects.
- The main contention here is what outcomes should/ can be identified as objectively desirable.

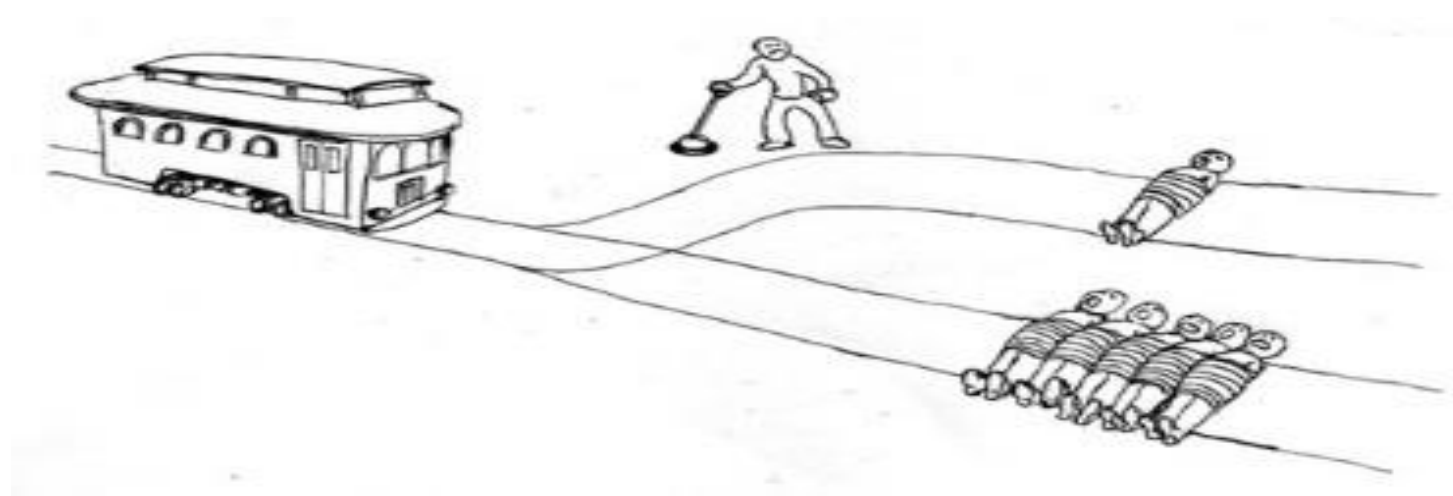
Utilitarianism

- Utilitarianism is one form of consequentialism which has as its main tenet that we should seek the greatest happiness of the greatest number.
 - Greatest Happiness Principle of John Stuart Mill
- Seeks utility over morality
- Our determinant of the desirability of an action is the net amount of happiness it brings, the number of people it brings it to, and the duration of the happiness.

UTILITARIANISM

Seeking utility can be problematic

Example: [The Trolley problem](#)

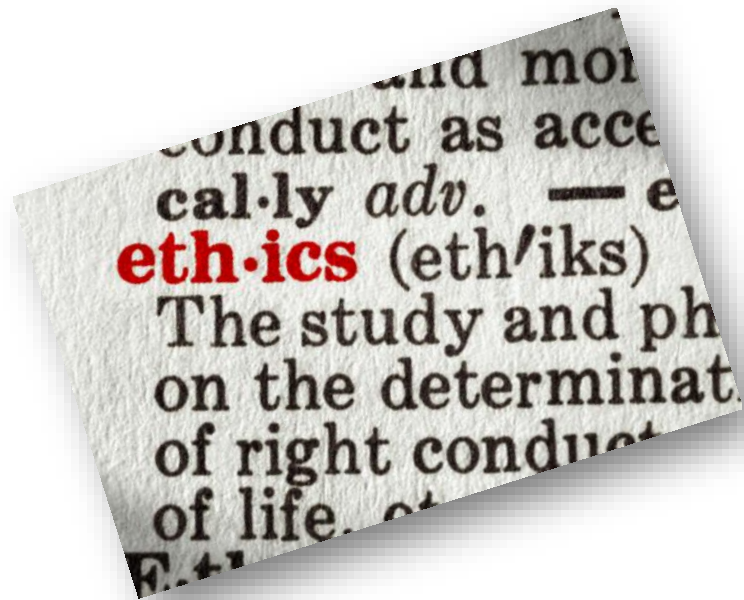


Comparing the Three Ethical Approaches

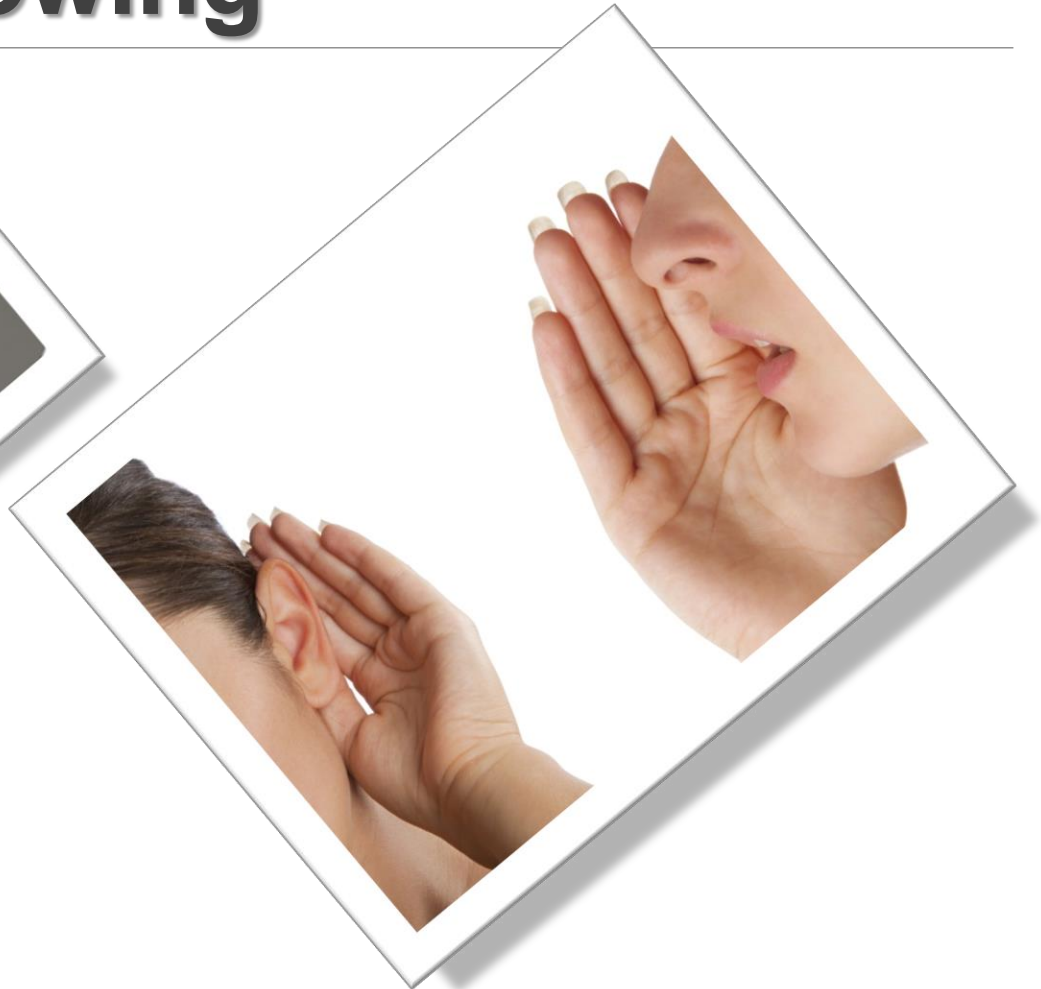
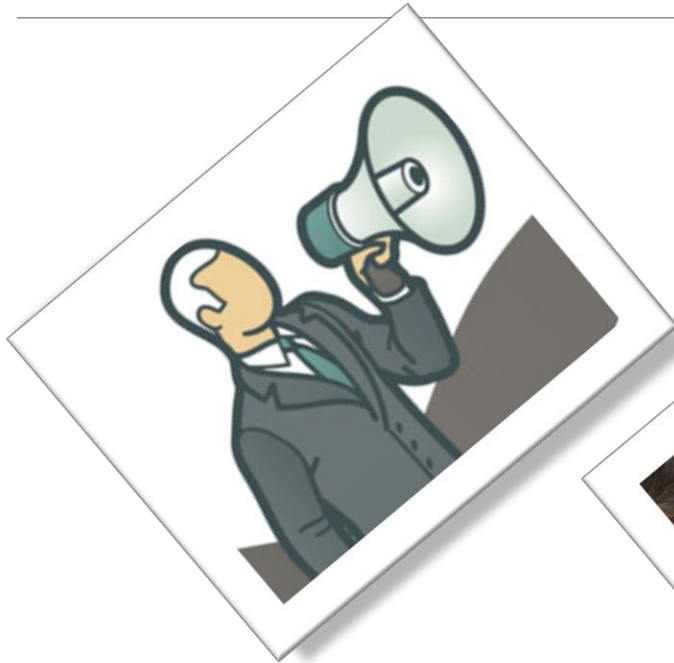
1. A consequentialist may argue that lying is wrong because of the negative consequences produced by lying – though a consequentialist may allow that certain foreseeable consequences might make lying acceptable.
2. A deontologist might argue that lying is *always* wrong, regardless of any potential "good" that might come from lying.
3. A virtue ethicist would focus less on lying in any particular instance and instead consider what a decision to tell a lie or not tell a lie said about one's character and moral behavior. Therefore, the decision to lie would be made in a case-by-case basis that weigh personal benefit, group benefit, and intentions.

COMPONENTS/QUALITIES

- Honesty
- Integrity
- Transparency
- Accountability
- Confidentiality
- Objectivity
- Respectfulness
- Obedience to the Law



“Whistleblowing”



“Whistleblowing”

A **whistleblower** is a person who tells the public or someone in authority about alleged dishonest or illegal activities occurring in a government department or private company or organization.

A **whistleblower** is a person who raises concern about frauds, corruptions, wrongdoings and mismanagement.

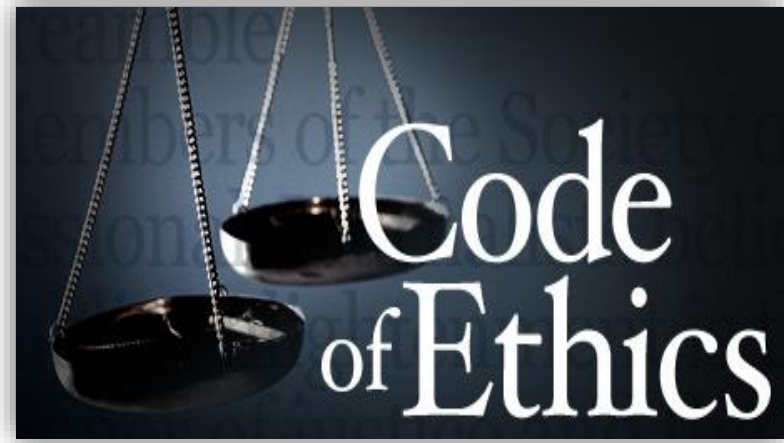
DIMENSIONS OF ETHICS (3 “R’s” of Ethics)

- RULES
- RESPONSIBILITY
- RESPECT



CODES OF ETHICS

The primary aspect of codes of ethics is to provide the basic framework for ethical judgment for a professional.



CODES OF ETHICS

The codes of ethics are guidelines for specific group of professionals to help them perform their roles, to know how to conduct themselves, and to know how to resolve various ethical issues.

The codes of ethics help the professionals to apply moral and ethical principles to the specific situations encountered in professional practice.

These codes convey the rights, duties, and obligations of the members of the profession.

POSITIVE ROLES OF CODES OF ETHICS

Inspiration

Guidance

Support for responsible conduct

Deterring and disciplining unethical professional conduct

Education and promoting of mutual understanding

Contributing to a positive public image of the profession

Protecting the status quo and suppressing dissent within the profession

Promoting business interests through restraint of trade

PROTECTING THE STATUS QUO

The codes institute ethical conventions. These ethical conventions can promote a minimum, acceptable level of ethical conduct.

The codes can also suppress the dispute within the profession.

**AT LEAST IT'S A STEP IN
THE RIGHT DIRECTION**



03-24-10

LIMITATIONS OF CODES OF ETHICS

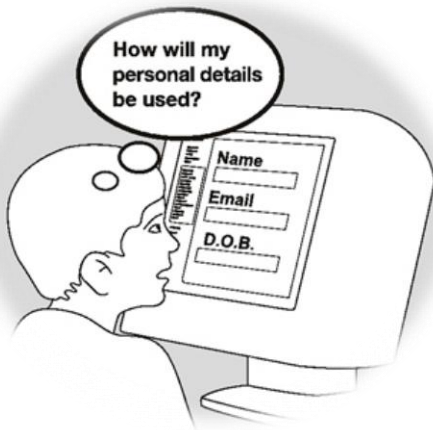
Codes of ethics are broad guidelines, restricted to general phrases. The codes cannot be applied directly to all situations.

Engineering codes often have internal conflicts, since several entries in codes overlap with each other, which may result in moral dilemmas.

The codes cannot serve as the final moral authority for professional conduct.

COMPUTER ETHICS

Computer ethics is the study of ethical issues that are associated primarily with computing machines and computing profession.



COMPUTER ETHICS

Don't use a computer to harm other people.

Don't interfere with other people's computer work.

Don't snoop around in other people's computer files.

Don't use a computer to steal.

Don't copy or use propriety software for which you have not paid.

Don't use other people's computer resources without authorization or proper compensation.

Don't appropriate other people's intellectual output.

COMPUTER ETHICS

Think about the social consequences of the program you are writing or the system you are designing.

Use a computer in ways that insure considerations and respect for your fellow humans.



Questions
are
guaranteed in
life;
Answers
aren't.

	Daffodil International University Department of Computer Science and Engineering (CSE) Course Outline			
Course Code:	CSE 498			
Course Title:	Social and Professional Issues in Computing			
Program:	B.Sc. in CSE			
Faculty:	Faculty of Science and Information Technology (FSIT)			
Semester:	Spring	Year:	2022	
Credit:	3.00	Contact Hour:	3:00 hr/Week	
Course Level:	L4T3	Prerequisite:	Null	
Course Category:	Core Engineering			
Instructor Name:	Dr. Fizar Ahmed			
Designation:	Assistant Professor			
Email:	fizar.cse@diu.edu.bd			
Office Address:	Room-411, CSE Building, Daffodil International University			

1. Course Rationale

Social and Professional Issues in Computing is a theory course which deals with different issues related to both social and professional life. Hence, this course will deal with different computing issues, i.e. privacy, social engineering, crime, hacking, and freedom of speech issue and so on. Furthermore, throughout the course we will focus on various professional issues like intellectual property, ethics and professional ethics, human vs computer, trusting computer etc. Besides, different social and international issues will also be discussed in this course.

1.1. Course Objective

- To see and understand the implications and impacts of the technology
- To know the ethics of leaking sensitive information
- Discussion of plagiarism
- Use of social media and personal devices at work
- To know hacking by governments to attack others

1.2. Course Outcome's (CO's)

CO1	Understanding the implications, impacts and benefits of the modern technology.
CO2	Understanding and application of freedom of speech, offensive or hate speech and censorship, spam, anonymity.
CO3	To know the different types of computer crime, the reason of computer crime and way of protecting them, privacy risks, privacy and computer technology, different privacy policies and to understand the code of ethics, different ethics type, professional ethics and their guidelines.

CO4	To learn about the intellectual property such as copyright, patent, trade mark and trade secret and how to use them.
CO5	To know the error, risks and failures occurred from the system development and to learn do's and don'ts in work life.

1.3. Program Outcomes (PO's)

Program Outcomes are reported in Appendix-I.

1.4. CO-PO Mapping:

Range: Weak = 1, Average = 2, Strong = 3.

PO's CO's	PLO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CLO1							2					
CLO2						3						
CLO3								3				
CLO4					3			2				
CLO5			3								2	2

1.5. CO Assessment Scheme

Assessment Task	CO's					Mark (Total = 100)
	CO1	CO2	CO3	CO4	CO5	
Attendance	-	--	--	--	--	7
Class Test (CT1, CT2, CT3)	-	--	--	--	--	15
Assignment	-	--	--	--	--	5
Presentation	-	--	--	--	--	8
Midterm Examination	5	10	10	--	--	25
Semester Final Examination	-	10	10	10	10	40
Total Mark	5	20	20	10	10	100

2. Strategies and approaches to learning

2.1. Teaching and Learning Activities (TLA)

TLA1	Lectures twice a week using whiteboard/multimedia of different topics.
TLA2	Active discussion in class regarding efficient solving of the logical and mathematical problems.
TLA3	Group discussion and presentation regarding diverse problems and corresponding lectures.
TLA4	Evaluation of class performances to reach each student in a class for every topic.

3. Course Schedule and Structure

3.1. Textbook

1. A Gift of Fire, 4th Edition Author: Sara Baase.

3.2. Reference Book

1. Ethics for the Information Age – Sixth Edition.

3.3. Course Plan/Lesson Plan

Week/Lesson (hour)	Discussion Topic & Book Reference	Student Activities during Online and Onsite and TLA	Mapping with CLO
Week 1 Lessen 1 & 2 (1.5 each)	Benefits (Unwrapping the Gift) 1. The Pace of Change 2. Change and Unexpected Developments 3. Themes 4. Ethics	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P;	CLO1

	Ref. A Gift of Fire – Fourth Edition (Chapter 1)	<u>TLA1, TLA3</u>	
Week 2 Lessen 1 & 2 (1.5 each)	Privacy 1. Privacy Risks and Principles 2. The Fourth Amendment, Expectation of Privacy, and Surveillance Technologies 3. The Business and Social Sectors. 4. Government Systems 5. Protecting Privacy 6. Technology, Markets, Rights, and Laws 7. Communications Ref. A Gift of Fire – Fourth Edition (Chapter 2)	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO2, CLO3
Week 3 Lessen 1 & 2 (1.5 each)	Freedom of Speech 1. Communications Paradigms 2. Controlling Speech 3. Posting, Selling, and Leaking Sensitive Material 4. Anonymity 5. The Global Net: Censorship and Political Freedom 6. Net Neutrality Regulations or the Market? Ref. A Gift of Fire – Fourth Edition (Chapter 3)	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO2
Week 4 Lessen 1 & 2 (1.5 each)	Computer Crime 1. Introduction 2. Hacking 3. Identity Theft and Credit Card Fraud 4. Whose Laws Rule the Web? Ref. A Gift of Fire – Fourth Edition (Chapter 5)	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO3 <u>Class Test# 1</u> (Either online or onsite based on Wk1-Wk3 discussion) based on CLO1, CLO2, CLO3

Week 5 Lessen 1 & 2 (1.5 each)	Ethics 1. Normative vs. descriptive Ethics 2. Three Main Ethical Approaches 3. Whistleblowing 4. Dimensions of Ethics (3 “R’s” of Ethics) 5. Codes of Ethics 6. Computer Ethics Ref. Handout	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO2, CLO3 <u>Assignment</u> : - Privacy - Computer Crime - Ethics Due: Week- 11
Week 6 Lessen 1 & 2 (1.5 each)	Review 1. Review for the MID Term Examinations 2. Problem solving Session Ref. Contents of Week 1 – Week 6	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA2, TLA4</u>	CLO1, CLO2, CLO3 <u>Class Test# 2</u> (Either online or onsite based on Wk4-Wk5 discussion) based on CLO2, CLO3
Week 7	Midterm Exam Week Topics: Week 1 – Week 5		
Week 8 Lessen 1 & 2 (1.5 each)	Intellectual Property 1. Principles, Laws, and Cases 2. Responses to Copyright Infringement 3. Search Engines and Online Libraries 4. Free Software 5. Patents for Inventions in Software Ref. A Gift of Fire – Fourth Edition (Chapter 4)	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO4
Week 9 Lessen 1 & 2 (1.5 each)	Errors, Failures, and Risks 1. Failures and Errors in Computer Systems 2. Case Study: The Therac-25 3. Increasing Reliability and Safety	Online/Onsite discussion; Review Feedback online; Using Interactive	CLO5

	4. Dependence, Risk, and Progress <i>Ref. A Gift of Fire – Fourth Edition (Chapter 8)</i>	content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3, TLA4</u>	
Week 10 Lessen 1 & 2 (1.5 each)	Professional Ethics and Responsibilities 1. What Is “Professional Ethics”? 2. Ethical Guidelines for Computer Professionals 3. Scenarios <i>Ref. A Gift of Fire – Fourth Edition (Chapter 9)</i>	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO3
Week 11 Lessen 1 & 2 (1.5 each)	Evaluating and Controlling Technology 1. Evaluating Information 2. The “Digital Divide” 3. Neo-Luddite Views of Computers, Technology, and Quality of Life 4. Making Decisions About Technology <i>Ref. A Gift of Fire – Fourth Edition (Chapter 7)</i>	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO5 <u>Class Test# 3</u> (Either online or onsite based on Wk8-Wk11 discussion) based on CLO3, CLO4, CLO5
Week 12 Lessen 1 & 2 (1.5 each)	Work and Wealth 1. Introduction 2. Automation and Employment 3. Workplace Changes 4. Globalization 5. The Digital Divide 6. The “Winner-Take-All Society” <i>Ref. Ethics for the Information Age – Sixth Edition (Chapter 10)</i>	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA1, TLA3</u>	CLO5 <u>Presentation</u>: Topics will be provided as Individual or Group.

Week 13 Lessen 1 & 2 (1.5 each)	Review 1. Review for the Final Examinations 2.Problem solving Session <i>Contents of Week 8 – Week 12</i>	Online/Onsite discussion; Review Feedback online; Using Interactive content e.g. Voice over PPT, PPT, Video, H5P; <u>TLA2, TLA4</u>	CLO3, CLO4, CLO5
Week 14	<i>Final Exam Week</i> <i>Topics: Week 8 – Week 13</i>		

4. Assessment Methods (Grading System)

Numerical Grade	Letter Grade	Grade Point
80-100	A+	4.00
75-79	A	3.75
70-74	A-	3.50
65-69	B+	3.25
60-64	B	3.00
55-59	B-	2.75
50-54	C+	2.50
45-49	C	2.25
40-44	D	2.00
Less than 40	F	0.00

5. Additional Support for Students

- Student Portal: <http://studentportal.diu.edu.bd/>
- Academic Guidelines <https://daffodilvarsity.edu.bd/article/academic-guidelines>
- Rules and Regulations of DIU <https://daffodilvarsity.edu.bd/article/rules-and-regulation>
- Career Development Center: <https://cdc.daffodilvarsity.edu.bd/>
- For general queries: <http://daffodilvarsity.edu.bd/>

Appendix-1: Program outcomes

POs	Category	Program Outcomes
PO1	Engineering Knowledge	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO2	Problem Analysis	Identify, formulate, research the literature and analyze complex engineering problems and reach substantiated conclusions using first principles of mathematics, the natural sciences and the engineering sciences.
PO3	Design/Development of Solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for public health and safety as well as cultural, societal and environmental concerns.
PO4	Investigations	Conduct investigations of complex problems, considering design of experiments, analysis and interpretation of data and synthesis of information to provide valid conclusions.
PO5	Modern tool usage	Create, select and apply appropriate techniques, resources and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society	Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to professional engineering practice.
PO7	Environment and sustainability	Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics	Apply ethical principles and commit to professional ethics, responsibilities and the norms of the engineering practice.
PO9	Individual work and teamwork	Function effectively as an individual and as a member or leader of diverse teams as well as in multidisciplinary settings.
PO10	Communication	Communicate effectively about complex engineering activities with the engineering community and with society at large. Be able to comprehend and write effective reports, design documentation, make effective presentations and give and receive clear instructions.

PO11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work as a member or a leader of a team to manage projects in multidisciplinary environments.
PO12	Life Long Learning	Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Representation and Computation

CSE 331

M. Arshad Khan

Lecturer(EFE)

Computer Science and Engineering(CSE)

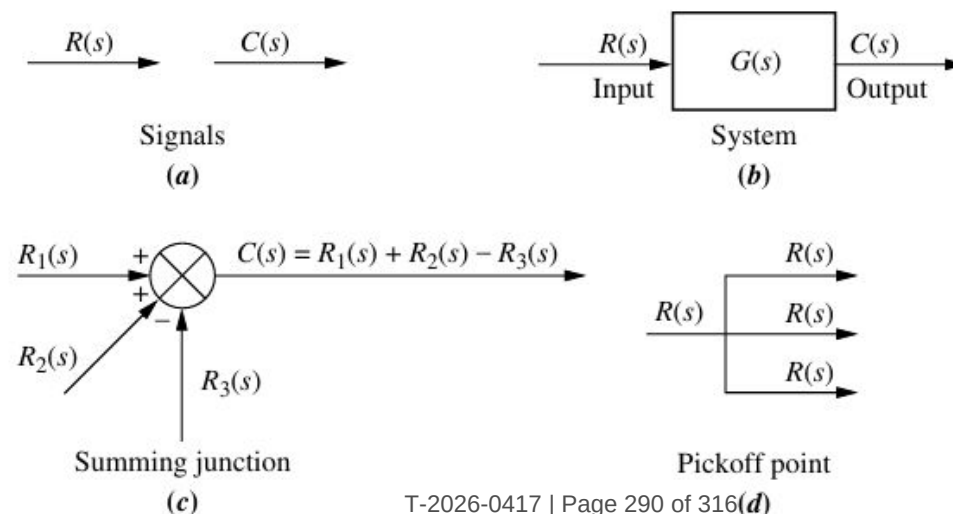
Difford International University(DIU)

Block Diagram

Any system can be described by a set of differential equations or can be represented by the schematic diagram containing all components and their connections.

Block diagrams use **single-line diagrams** to show the flow of system variables from one block to another. A **block** may represent a single component or a group of components, beside this there are two new elements. These new elements are summing junctions and pickoff points.

Overall, Block diagrams are single line diagram, that is the flow of system variables from one block to another block is represented by a single line.



Block Diagram from Electrical System

Procedure to Draw block Diagram From Electrical Circuit:

1. Apply circuit laws (Ohm's law, Kirchhoff's laws) to **each component with Laplace Transform (Assume Zero Initial Conditions)**.
2. Do circuit operation(Current and Voltage) **step by step from input to output**.
3. Now draw **block diagram of each operation**.
4. Now just **add each block one after another in serial** (Best possible way to arrange from last equation).

Block Diagram from Electrical System: Example 01

Laplace transform with initial condition zero

$$V_i(s) = I(s)R + sL I(s)$$

$$V_i(s) = I(s)(R + sL)$$

$$V_o(s) = sL I(s)$$

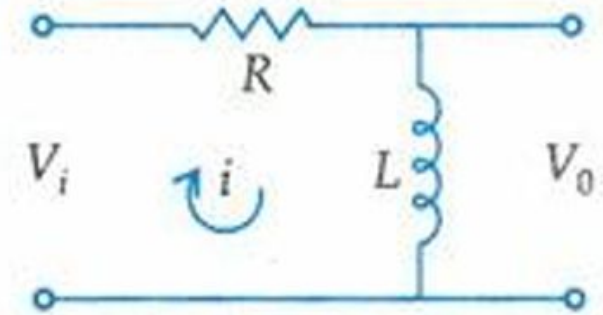
Now transfer function is,

$$\frac{V_o(s)}{V_i(s)} = \frac{sL}{R + sL}$$

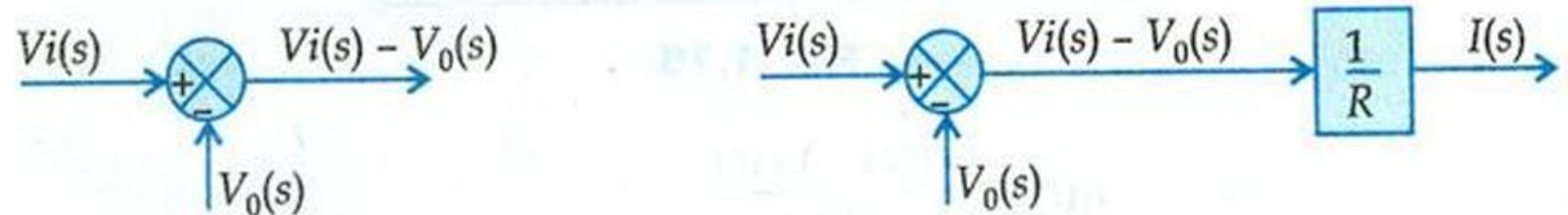
Now,

$$I(s) = \frac{1}{R} [V_i(s) - V_o(s)]$$

$$V_o(s) = sL I(s)$$



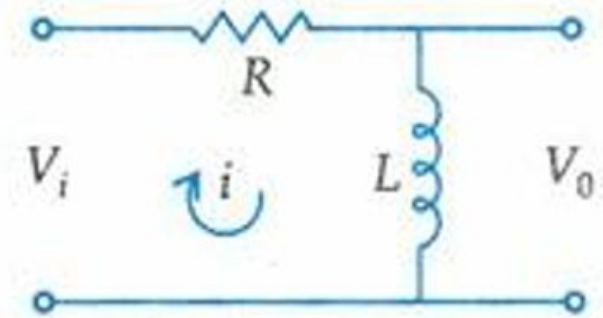
1st equation diagram:



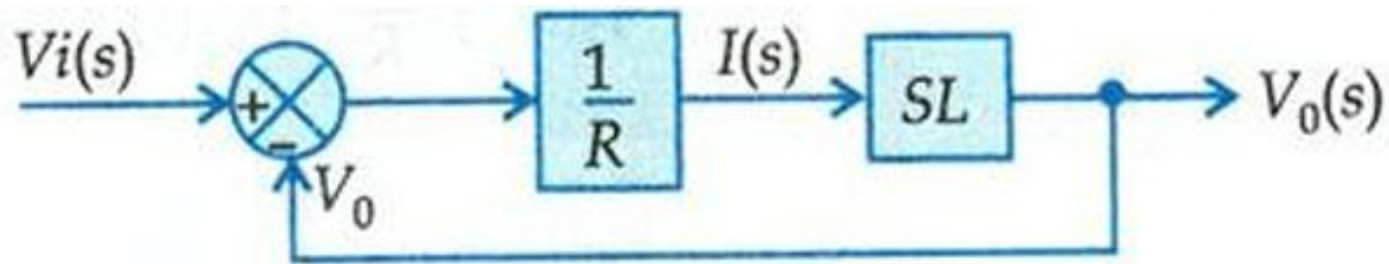
Block Diagram from Electrical System: Example 01

$$V_o(s) = sL I(s)$$

2nd equation diagram:



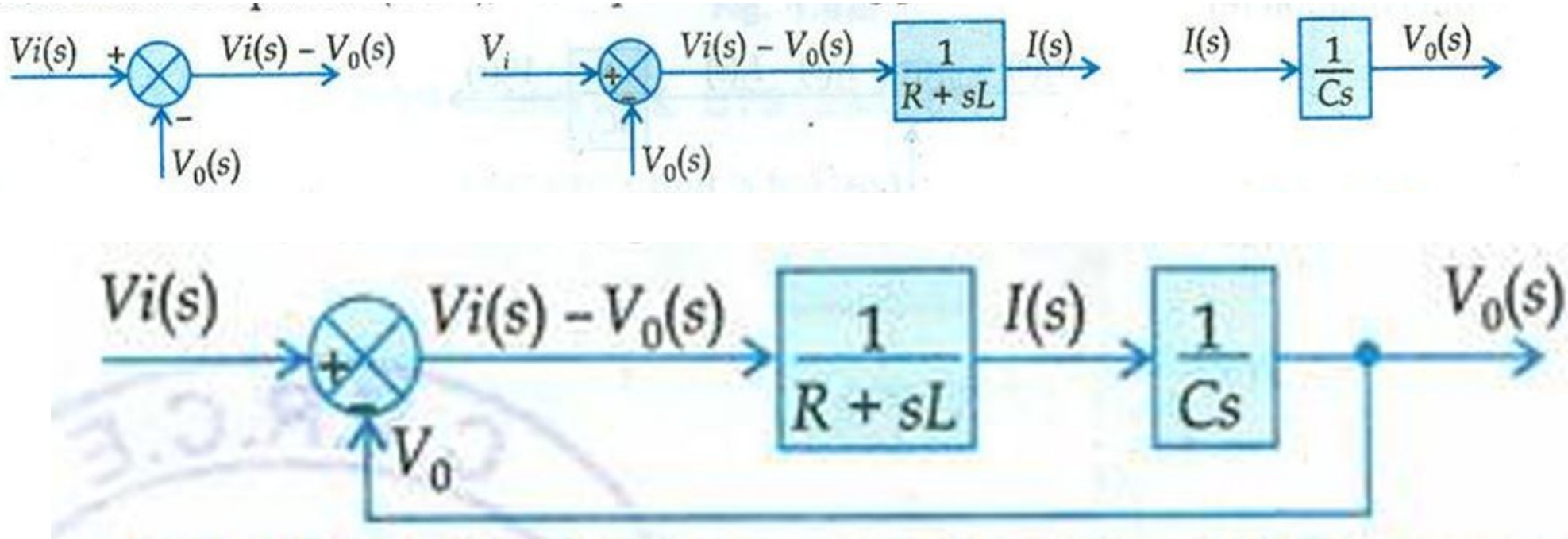
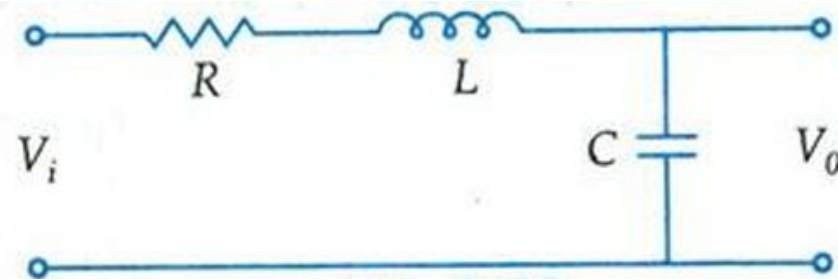
Complete Block Diagram:



Block Diagram from Electrical System: Example 02

Here,

$$I(s) = \frac{V_i(s) - V_o(s)}{R + sL}$$
$$V_o(s) = \frac{1}{Cs} I(s)$$



Block Diagram from Electrical System: Example 03

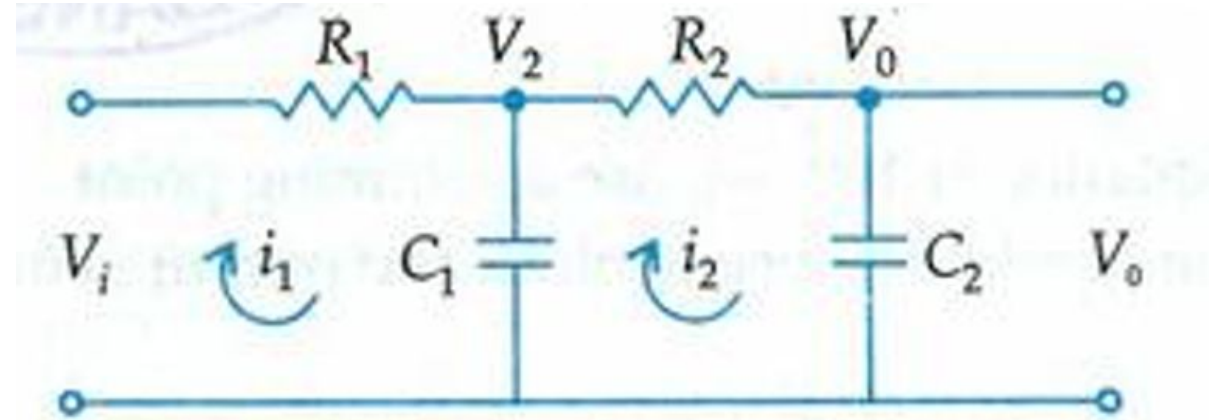
Take the Laplace transform from the circuit

$$I_1(s) = \frac{1}{R_1} [V_1(s) - V_2(s)]$$

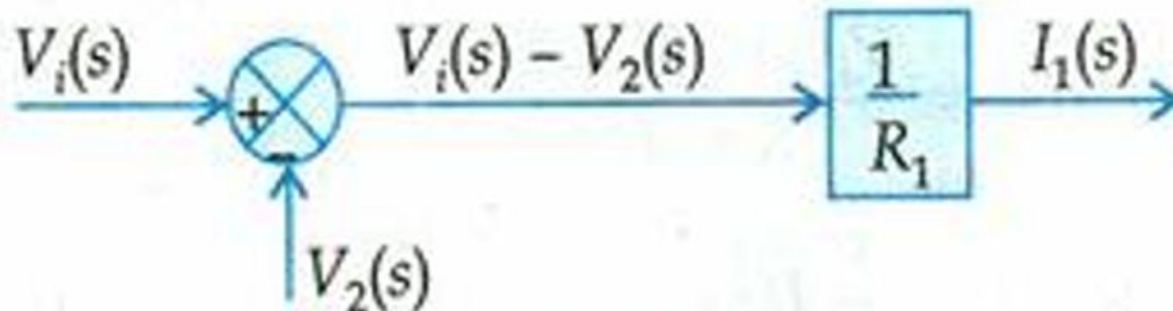
$$V_2(s) = \frac{1}{sC_1} [I_1(s) - I_2(s)]$$

$$I_2(s) = \frac{1}{R_2} [V_2(s) - V_o(s)]$$

$$V_o(s) = \frac{1}{sC_2} I_2(s)$$



From 1st Equation:



Block Diagram from Electrical System: Example 03

Take the Laplace transform from the circuit

From 2nd Equation:

$$V_2(s) = \frac{1}{sC_1} [I_1(s) - I_2(s)]$$

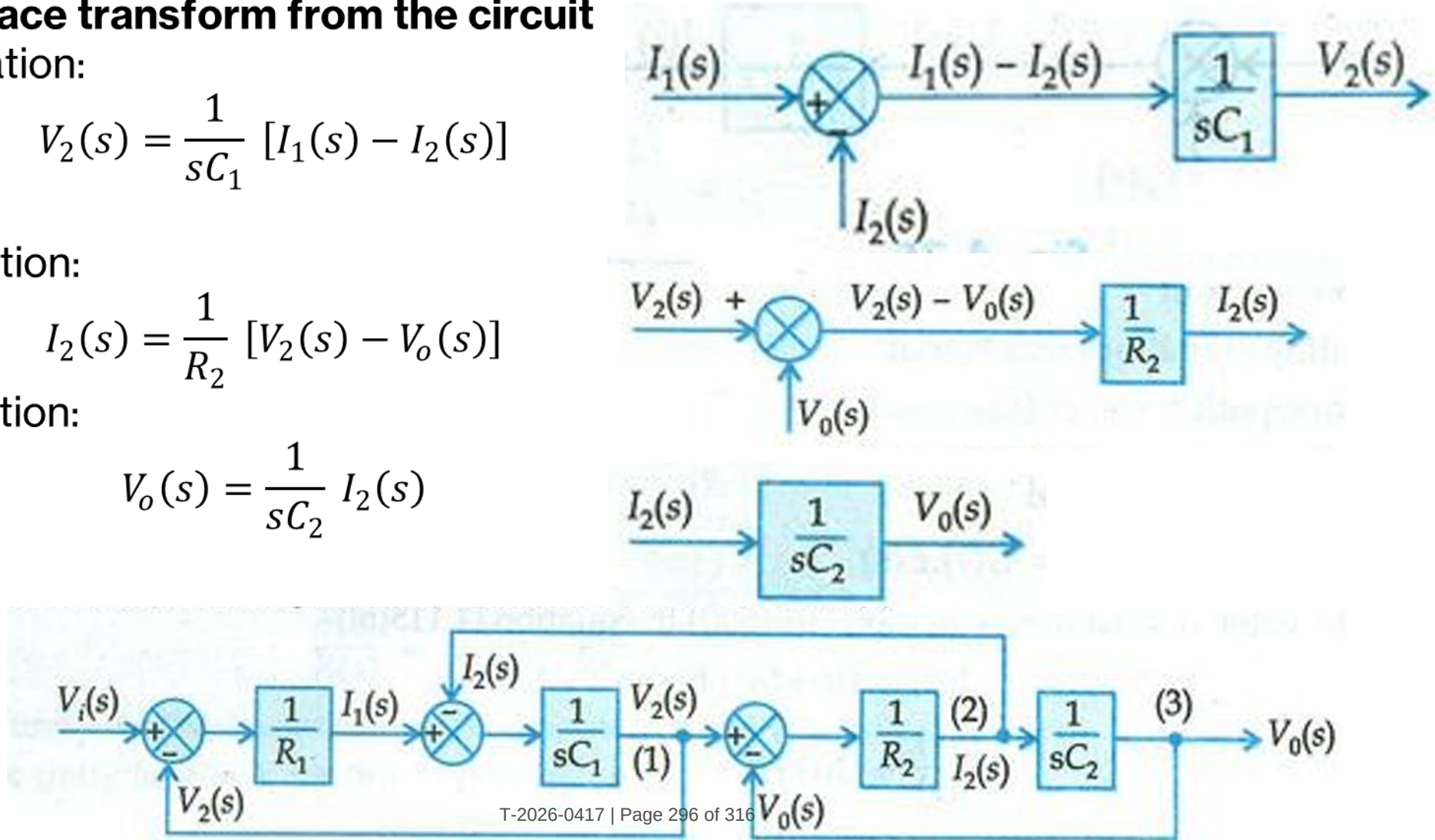
From 3rd Equation:

$$I_2(s) = \frac{1}{R_2} [V_2(s) - V_0(s)]$$

From 4th Equation:

$$V_0(s) = \frac{1}{sC_2} I_2(s)$$

Full Diagram:



Block Diagram from Electrical System: Example 04

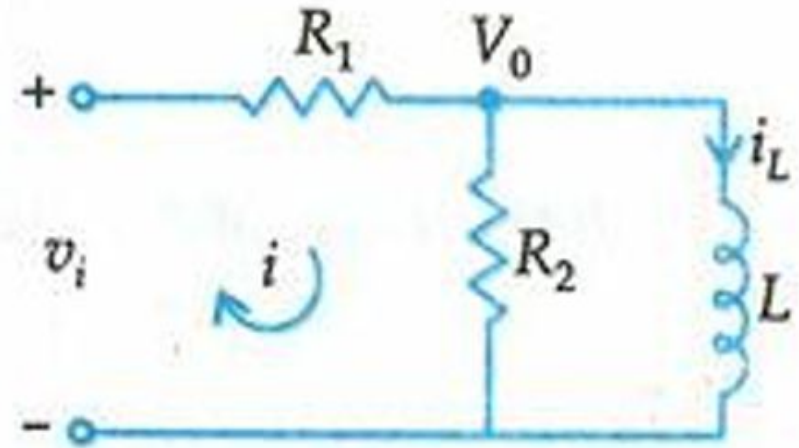
Laplace transform of the equations from the circuit:

$$I(s) = \frac{1}{R_1} [V_i(s) - V_o(s)]$$

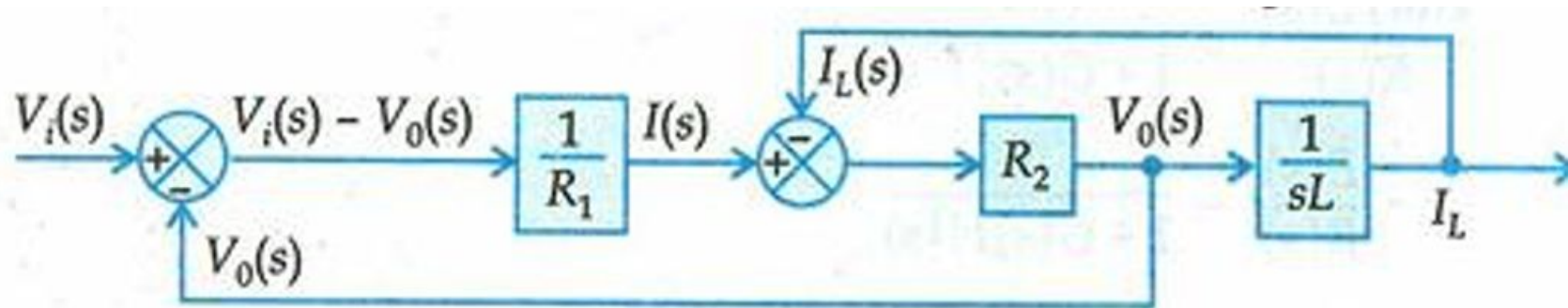
$$V_o(s) = R_2 [I(s) - I_L(s)]$$

$$V_o(s) = sL I_L(s)$$

$$I_L(s) = \frac{1}{sL} V_o(s)$$



Complete Block Diagram:



Representation and Computation

CSE 331

M. Asif Ullah

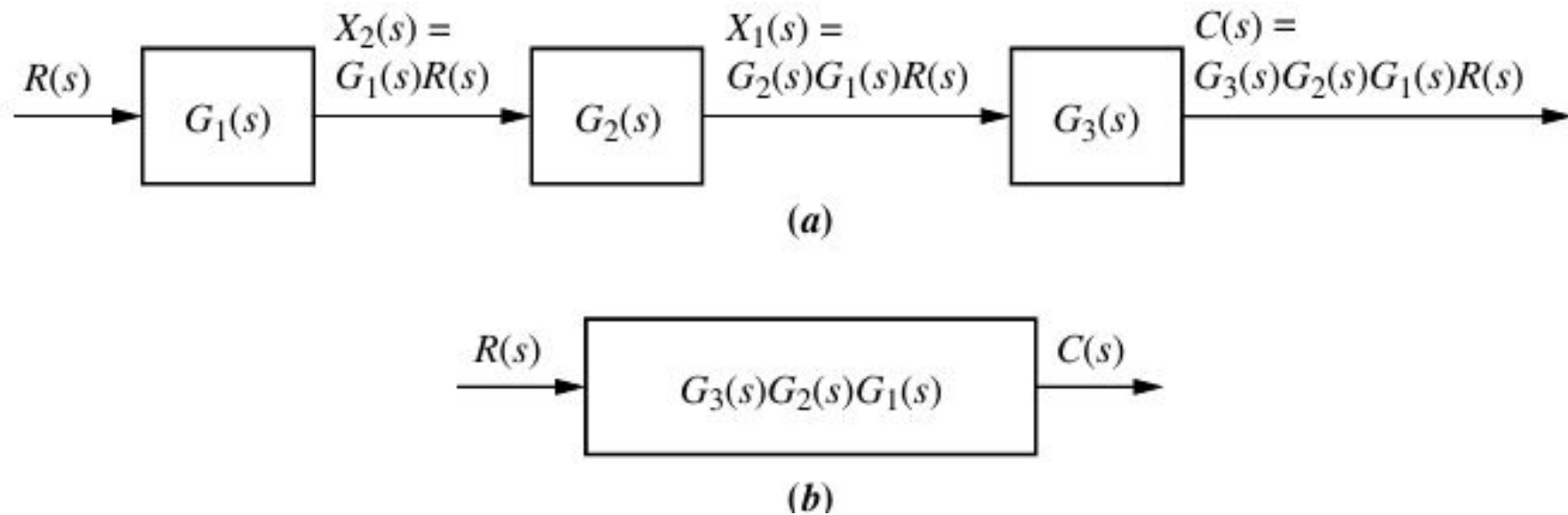
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Difford International University(DIU)

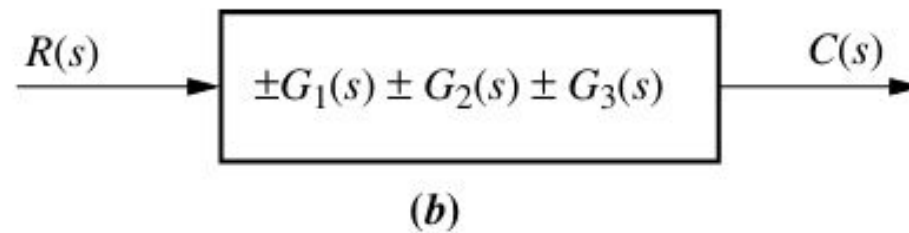
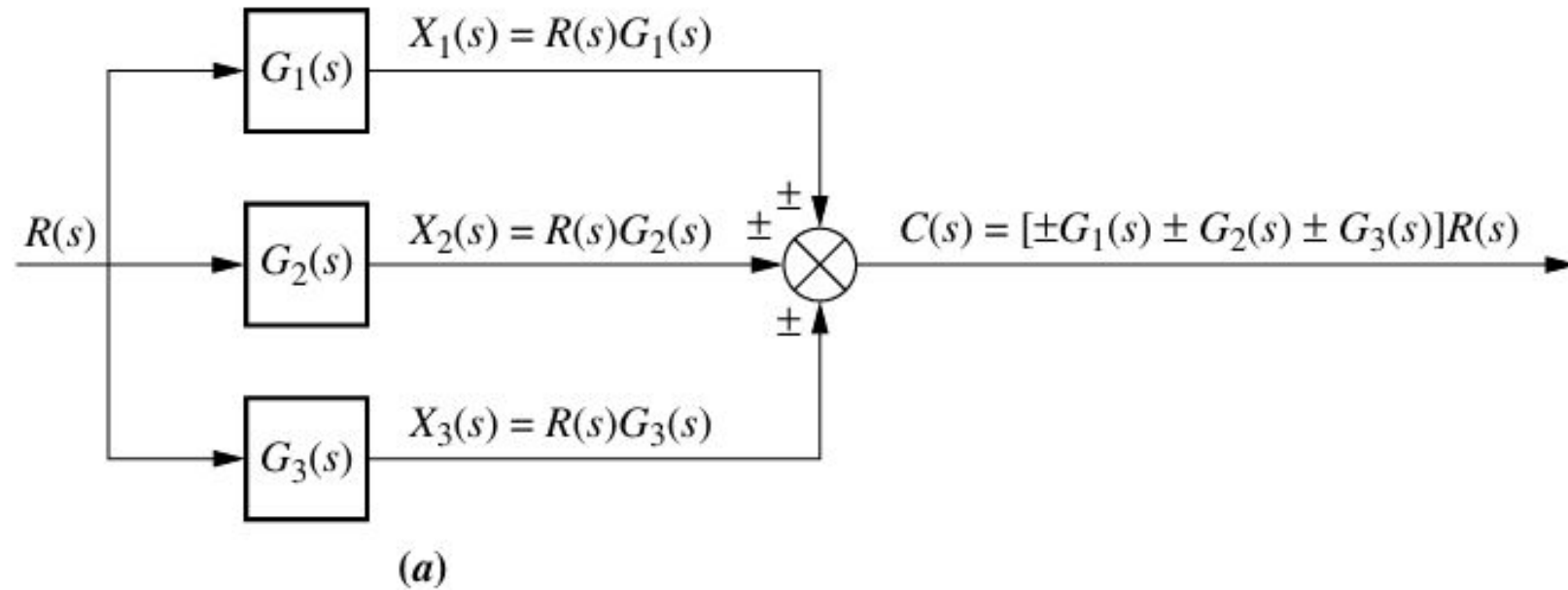
Block Diagram

Cascade Form



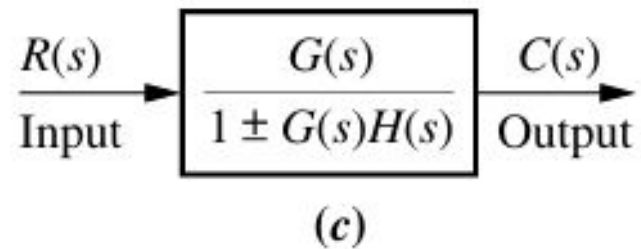
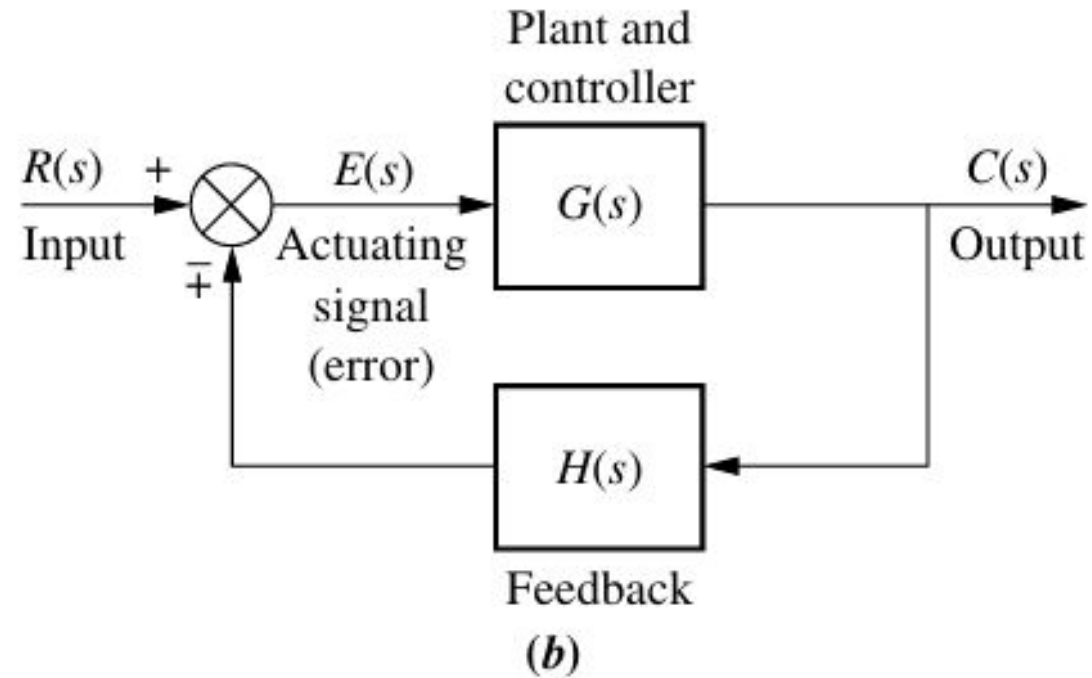
Block Diagram

Parallel Form



Block Diagram

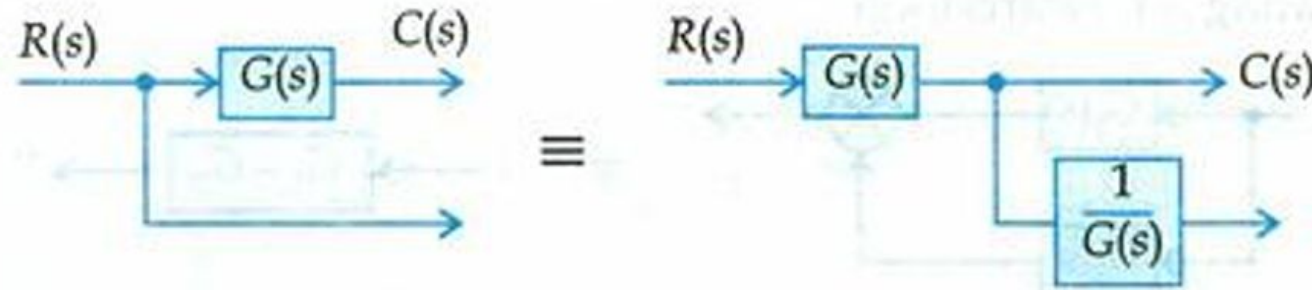
Feedback Form



Block Diagram

Rules to block reduction:

Rule No. 1: Moving a take off point after the block

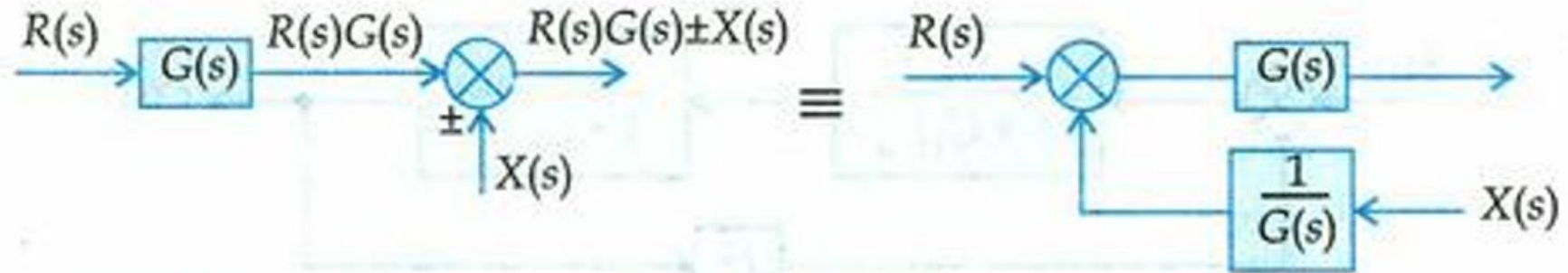


Rule No. 2: Moving a summing point beyond a block

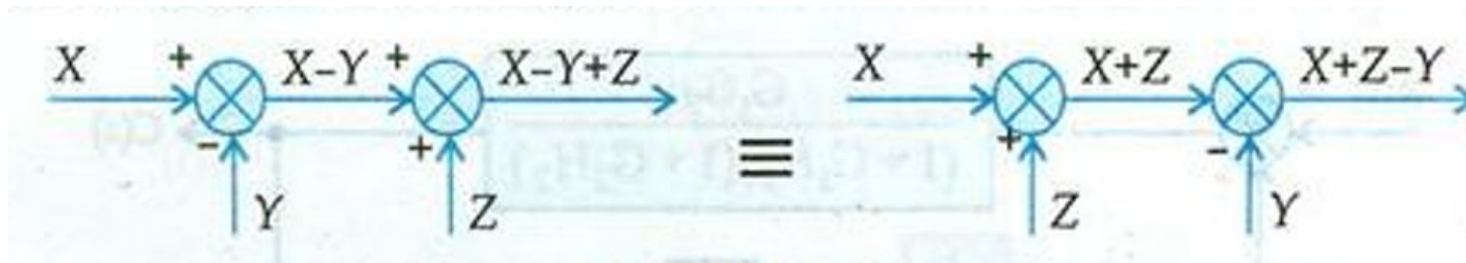


Block Diagram

Rule No. 3: Moving a summing point ahead of a block

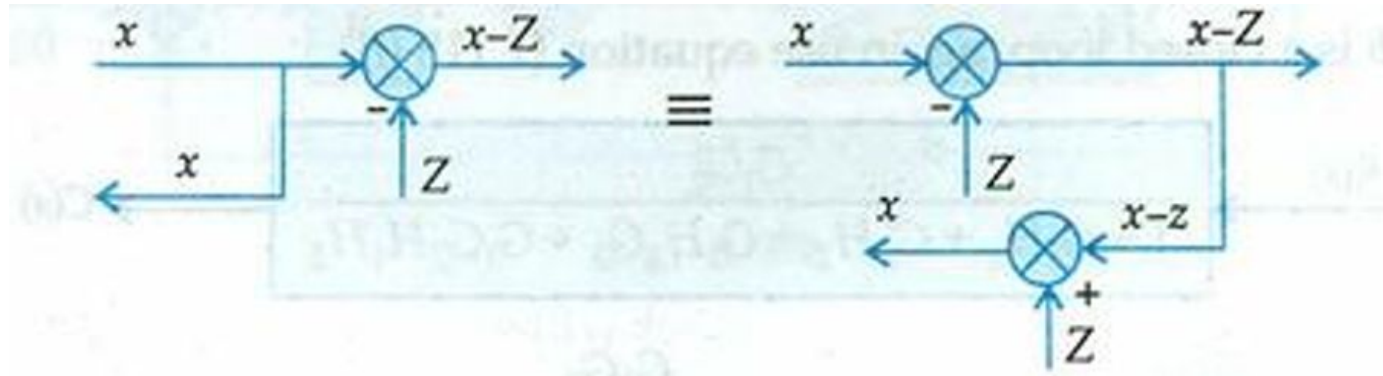


Rule No. 4: Interchanging two summing points



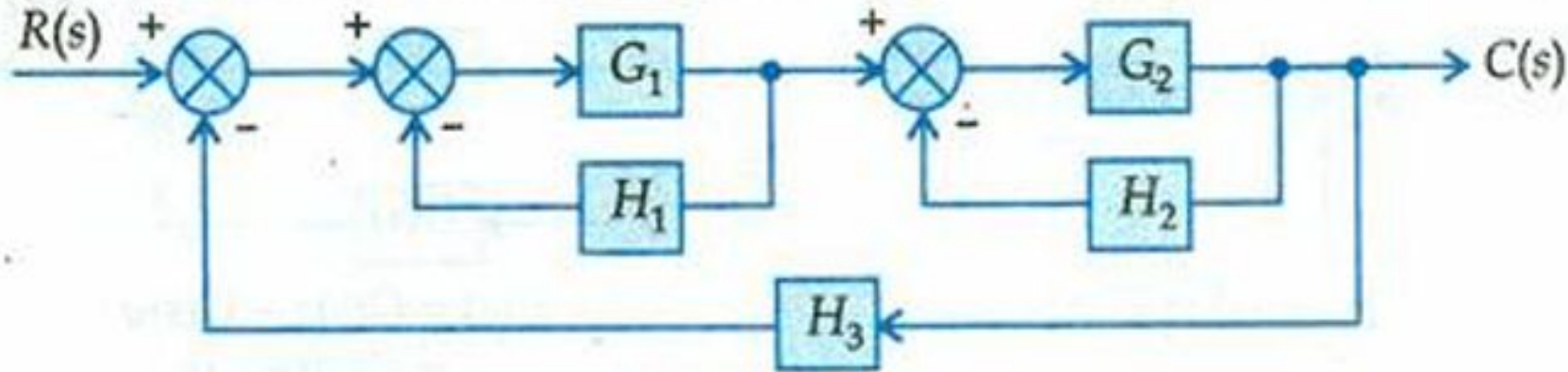
Block Diagram

Rule No. 5: Moving a take off point beyond a summing point

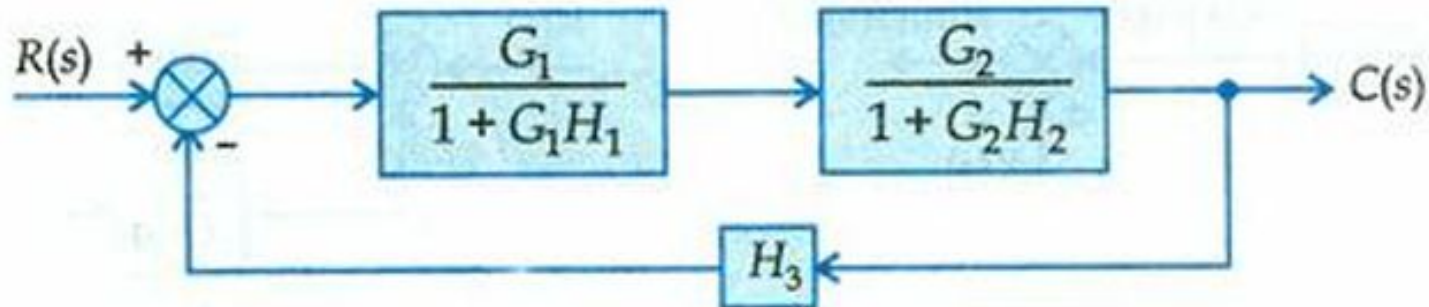


Block Diagram Reduction: Example 01

Derive the transfer function using block reduction technique.

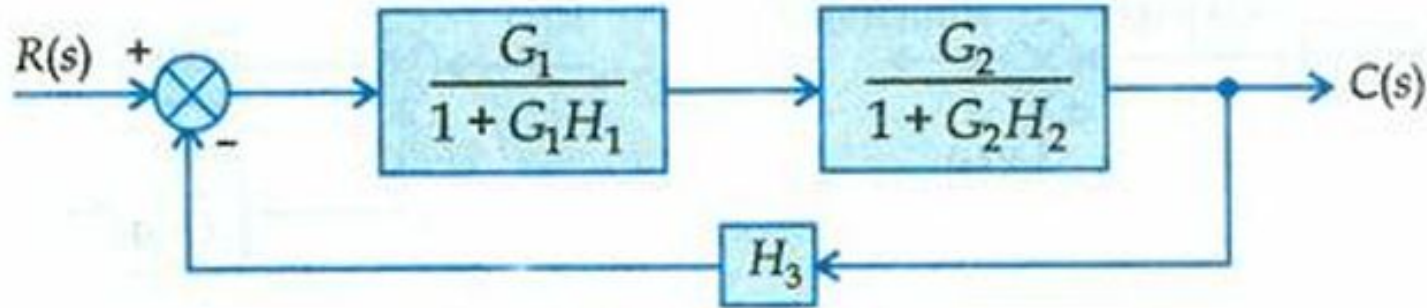


Step 01:

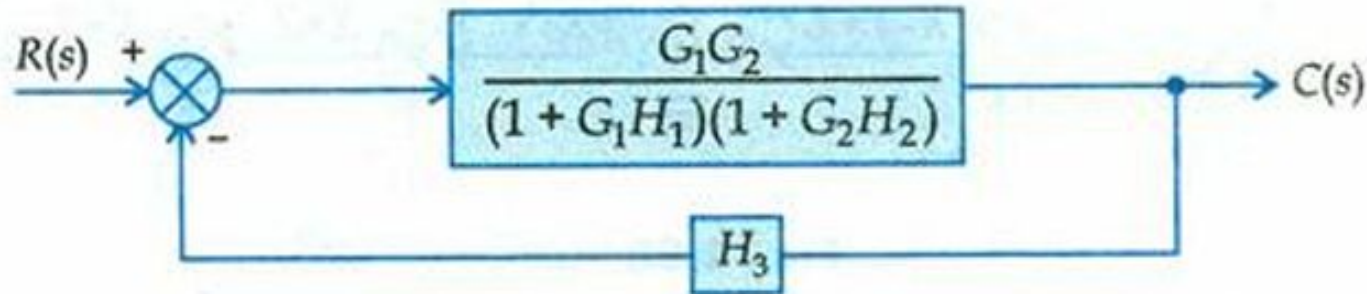


Block Diagram Reduction: Example 01

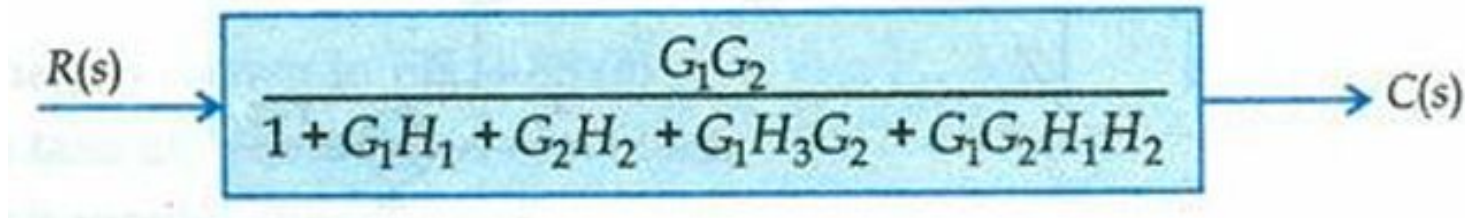
Step 01:



Step 02:



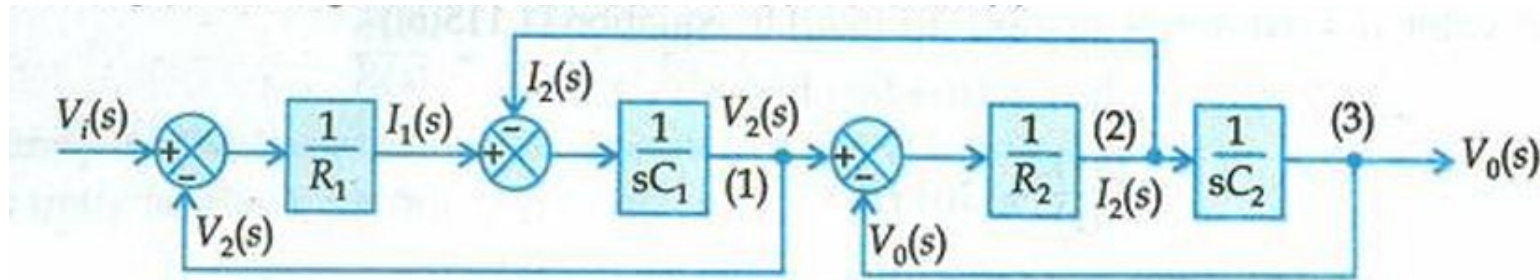
Step 03:



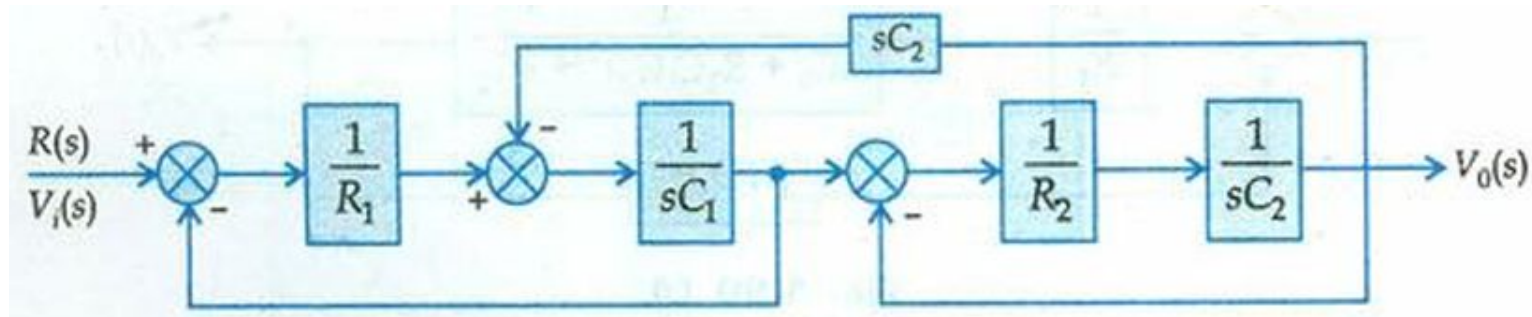
$$\frac{C(s)}{R(s)} = \frac{G_1 G_2}{1 + G_1 H_1 + G_2 H_2 + G_1 H_3 G_2 + G_1 G_2 H_1 H_2}$$

Block Diagram Reduction: Example 02

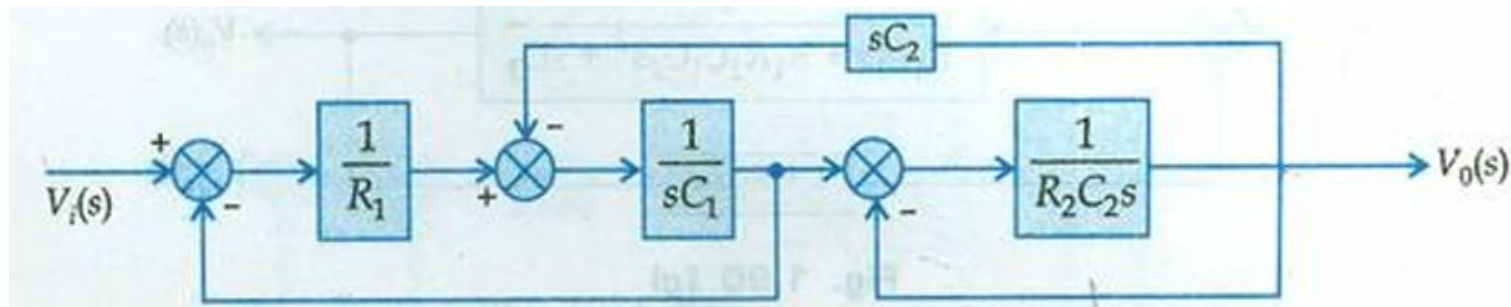
Derive the transfer function using block reduction technique.



Step 01:

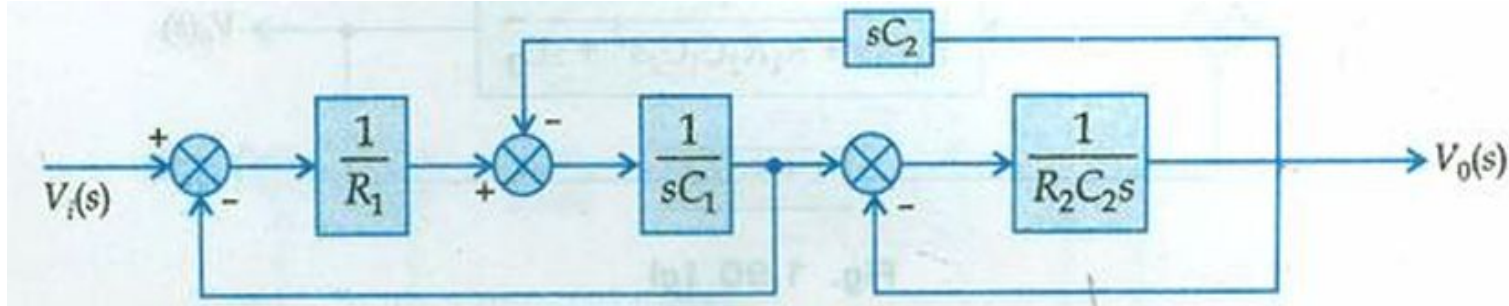


Step 02:

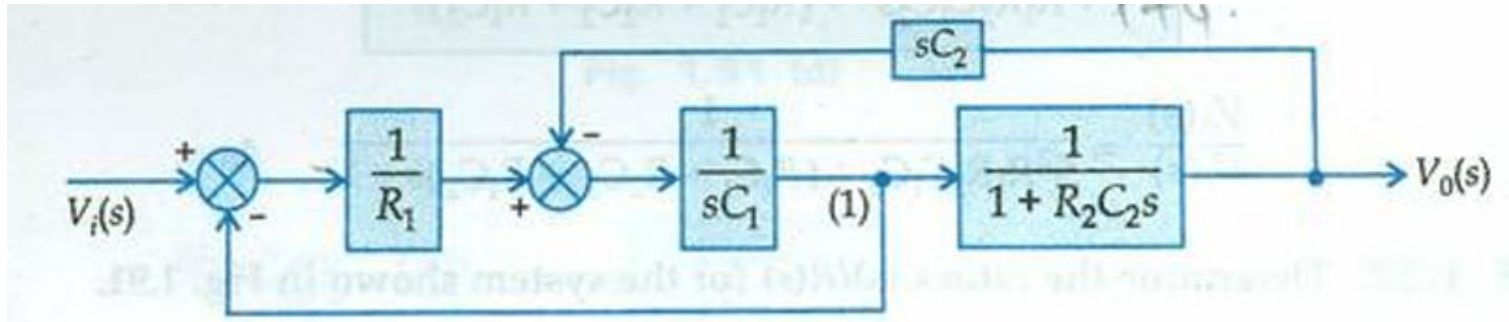


Block Diagram Reduction: Example 02

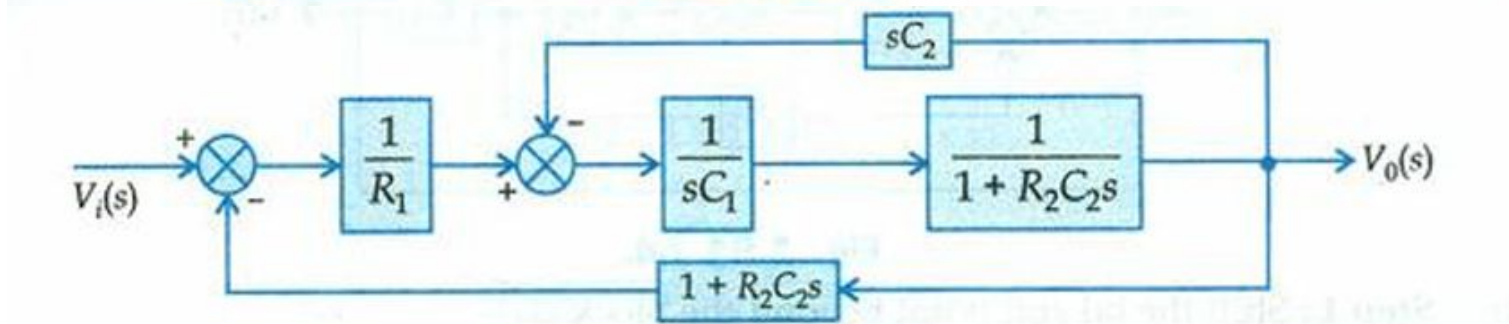
Step 02:



Step 03:

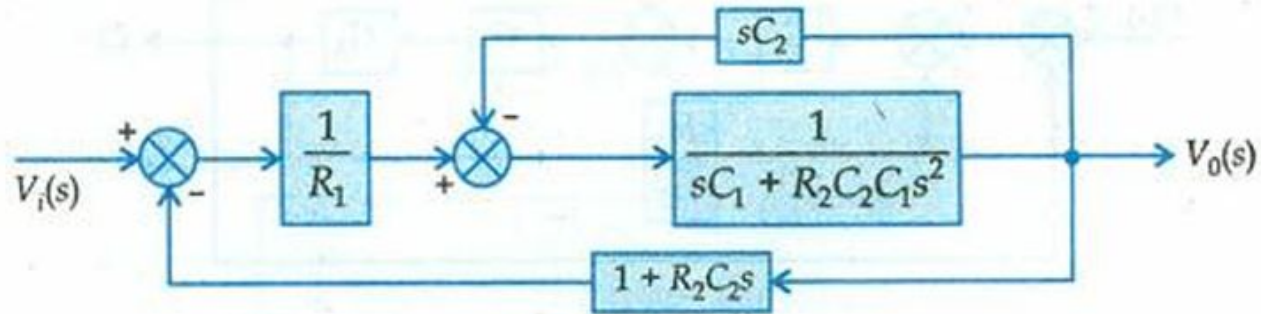


Step 04:

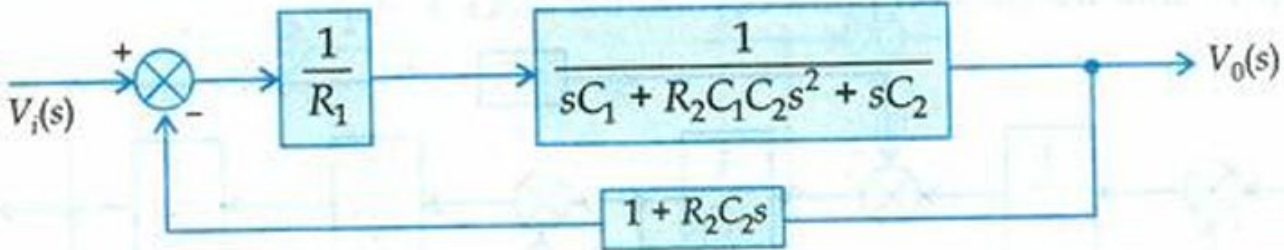


Block Diagram Reduction: Example 02

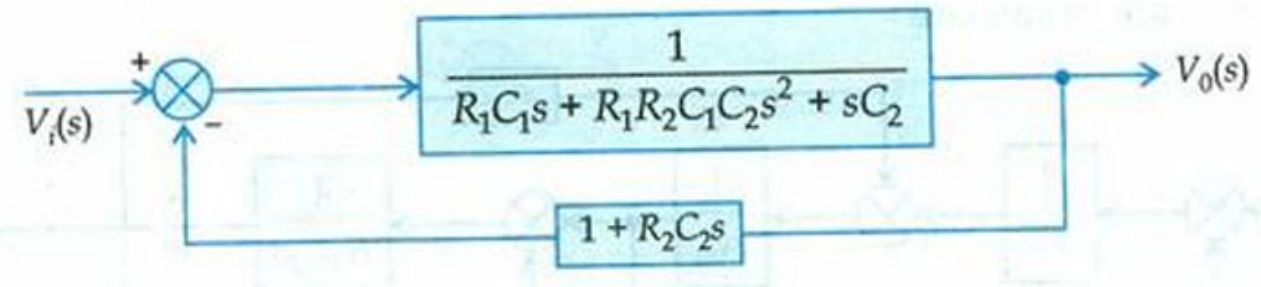
Step 05:



Step 06:

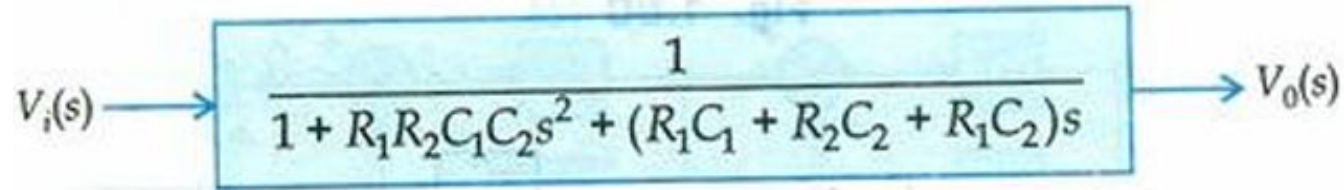


Step 07:



Block Diagram Reduction: Example 02

Step 08:

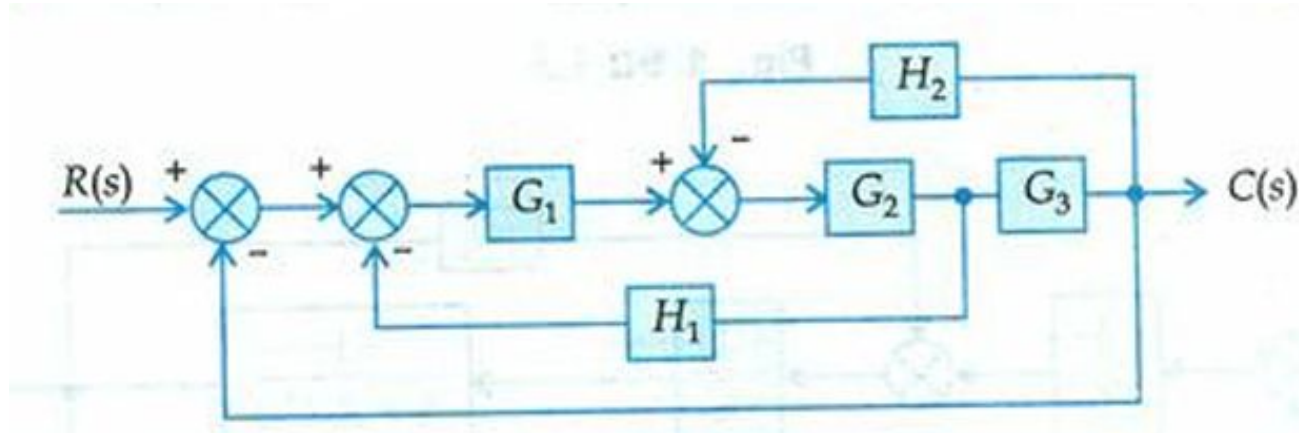


So, Transfer Function:

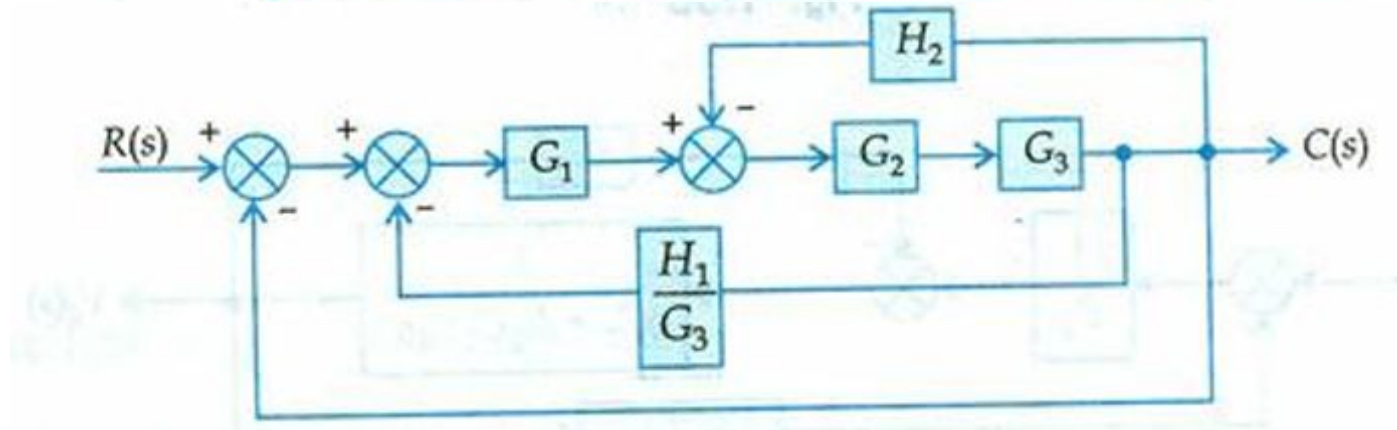
$$\frac{V_o(s)}{V_i(s)} = \frac{1}{s^2 R_1 R_2 C_1 C_2 + (R_1 C_1 + R_2 C_2 + R_1 C_2) s + 1}$$

Block Diagram Reduction: Example 03

Determine the ratio $\frac{C(s)}{R(s)}$ for the system shown in Fig.

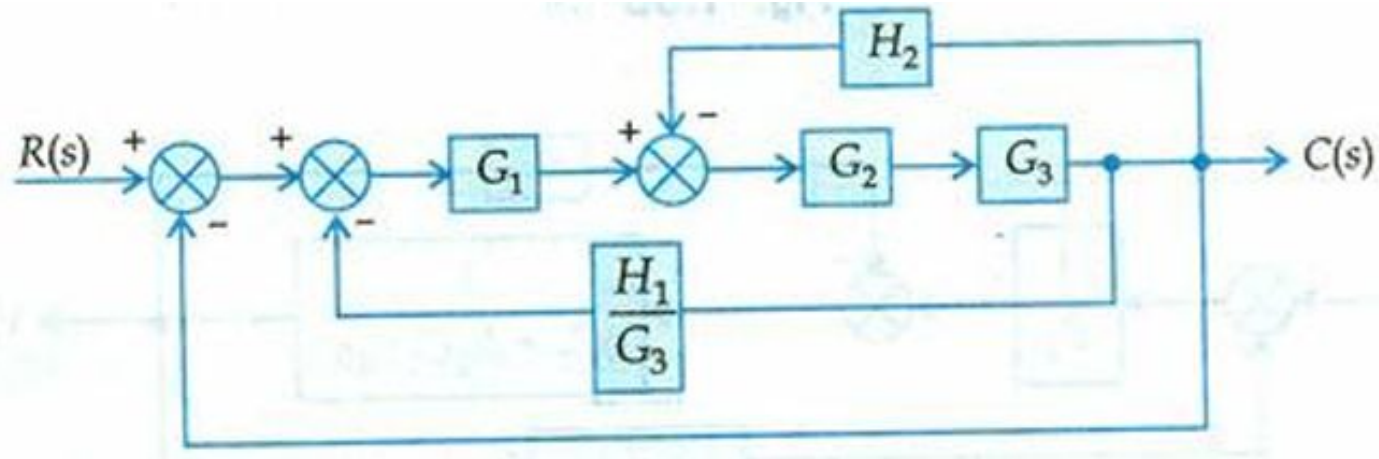


Step 01:

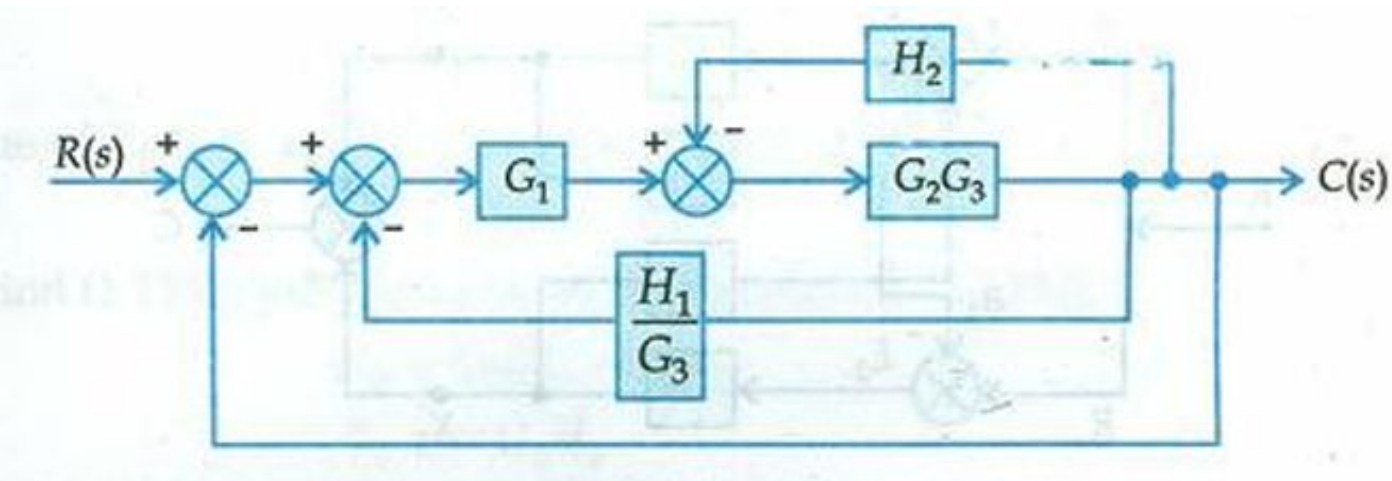


Block Diagram Reduction: Example 03

Step 01:

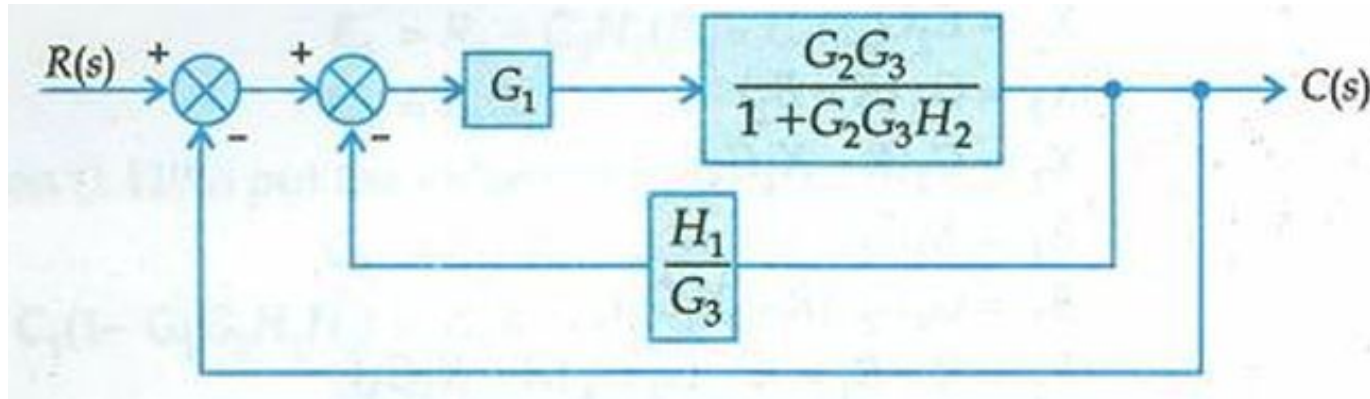


Step 2:

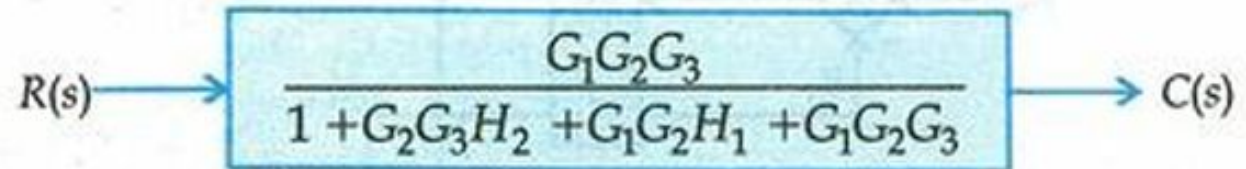
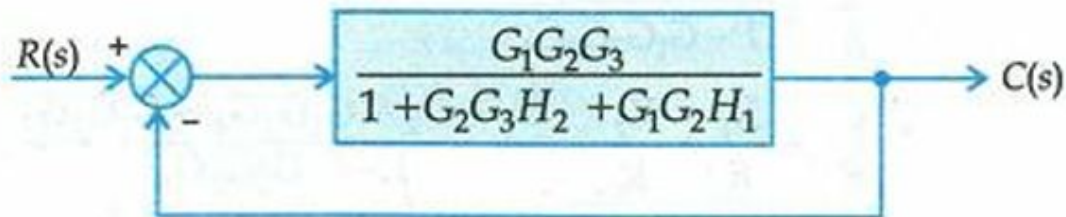


Block Diagram Reduction: Example 03

Step 03:



Step 04, 05 06 & 07 :



$$\frac{C(s)}{R(s)} = \frac{G_1 G_2 G_3}{1 + G_2 G_3 H_2 + G_1 G_2 H_1 + G_1 G_2 G_3}$$

Block Diagram Reduction: Example 04

Determine the ratio C/R .

Solution:

$$E_2 = R - B_2$$

$$B_2 = X_1 G_3$$

$$X_2 = E_2 G_2$$

$$X_2 = G_2(R - B_2)$$

$$X_2 = G_2(R - X_1 G_3)$$

$$B_1 = X_2 G_4$$

$$B_1 = G_2 G_4(R - X_1 G_3)$$

$$E_1 = R - B_1 = R - G_2 G_4(R - X_1 G_3)$$

$$X_1 = E_1 G_1$$

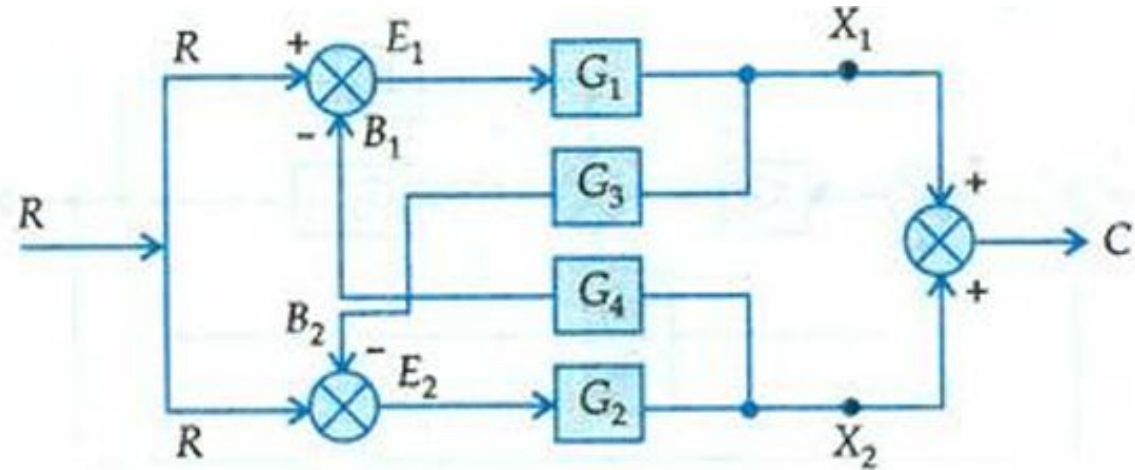
$$X_1 = G_1[R - G_2 G_4(R - X_1 G_3)]$$

$$X_1 = G_1 R - G_1 G_2 G_4 R + G_1 G_2 G_3 G_4 X_1$$

$$= R(G_1 - G_1 G_2 G_4) + G_1 G_2 G_3 G_4 X_1$$

$$\therefore X_1(1 - G_1 G_2 G_3 G_4) = R(G_1 - G_1 G_2 G_4)$$

$$\frac{X_1}{R} = \frac{G_1 - G_1 G_2 G_4}{1 - G_1 G_2 G_3 G_4}$$



put the value of x_1 in equation of x_2

$$X_2 = G_2 \left[R - \frac{G_1(1 - G_2 G_4)G_3 R}{1 - G_1 G_2 G_3 G_4} \right]$$

$$= \frac{G_2 R(1 - G_1 G_3)}{1 - G_1 G_2 G_3 G_4}$$

$$\therefore \frac{X_2}{R} = \frac{G_2(1 - G_1 G_3)}{1 - G_1 G_2 G_3 G_4}$$

Block Diagram Reduction: Example 05

Determine the transfer function C_1/R_1 , C_2/R_2 , C_1/R_2 and C_2/R_1 from the block diagram shown in Fig.

Solution:

$$E_2 = R_2 - B_2$$

$$B_2 = C_1 H_2$$

$$C_2 = E_2 G_2$$

Put the value of E_2 in equation of C_2

$$C_2 = G_2(R_2 - B_2)$$

put the value of B_2 in equation of C_2

$$C_2 = G_2(R_2 - C_1 H_2)$$

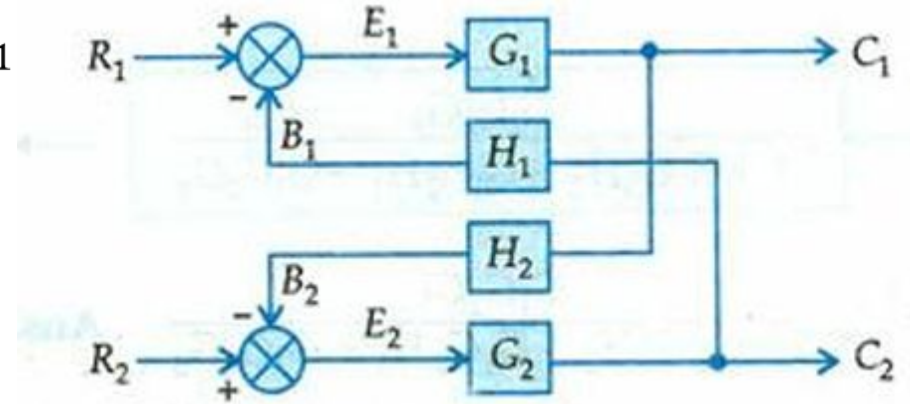
$$B_1 = C_2 H_1$$

Put the value of C_2 in equation of B_1

$$B_1 = G_2 H_1 (R_2 - C_1 H_2)$$

$$E_1 = R_1 - B_1$$

$$E_1 = R_1 - G_2 H_1 (R_2 - C_1 H_2)$$



Also,

$$C_1 = E_1 G_1$$

put the value of E_1 in equation C_1

$$C_1 = G_1 [R_1 - G_2 H_1 (R_2 - C_1 H_2)]$$

$$C_1 (1 - G_1 G_2 H_1 H_2) = G_1 R_1 - G_1 G_2 H_1 R_2$$

When Put $R_1 = 0$

$$\frac{C_1}{R_2} = \frac{-G_1 G_2 H_1}{1 - G_1 G_2 H_1 H_2}$$

Block Diagram Reduction: Example 05

Determine the transfer function C_1/R_1 , C_2/R_2 , C_1/R_2 and C_2/R_1 from the block diagram shown in Fig.

put the value of C_1 in equation of C_2

$$C_2 = G_2 \left[R_2 + \frac{G_1 G_2 H_1 H_2 R_2}{1 - G_1 G_2 H_1 H_2} \right]$$

$$\therefore \frac{C_2}{R_2} = \frac{G_2}{1 - G_1 G_2 H_1 H_2}$$

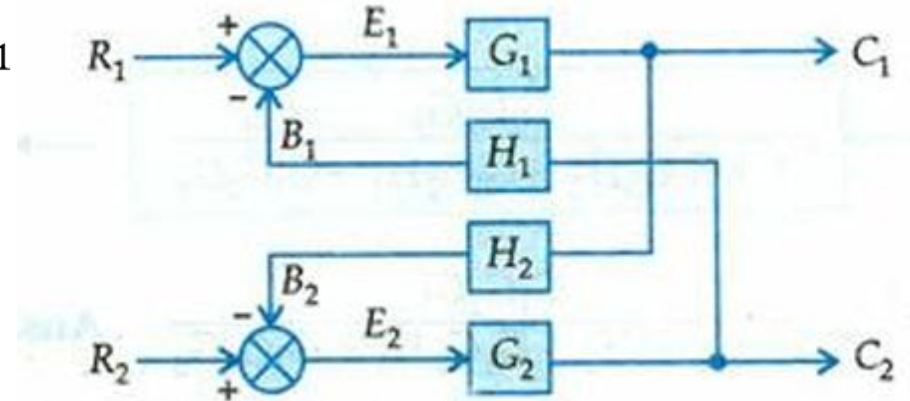
When

$$R_2 = 0, \text{ becomes}$$

$$C_1 = G_1 [R_1 + C_1 G_2 H_1 H_2]$$

$$C_1 = G_1 R_1 + C_1 G_1 G_2 H_1 H_2$$

$$\therefore \frac{C_1}{R_1} = \frac{G_1}{1 - G_1 G_2 H_1 H_2}$$



When

$$R_2 = 0, \text{ becomes}$$

$$C_2 = -G_2 C_1 H_2$$

put the value of C_1

$$C_2 = \frac{-G_1 G_2 R_1 H_2}{1 - G_1 G_2 H_1 H_2}$$

$$\therefore \frac{C_2}{R_1} = \frac{-G_1 G_2 H_2}{1 - G_1 G_2 H_1 H_2}$$